HUMAN ANATOMY AND PHYSIOLOGY

This unit specifies the competencies required to demonstrate the knowledge of human anatomy and physiology. It involves demonstrating the knowledge of organization of the human body, body fluids and their functions, body tissues and membranes and their functions, body cavities, human skeletal system, major muscles of the body and their functions, body joints, circulatory system, lymphatic system, endocrine system and its functions, nervous system and its function, digestive system and its function, urinary system and its function respiratory system and its function, special senses and their functions, reproductive system and its functions.

1 1.2 Summary of Learning Outcomes

1. Demonstrate knowledge of organization of the human body
2. Demonstrate knowledge of body fluids and their functions
3. Demonstrate knowledge of body tissues and membranes and their functions
4. Demonstrate knowledge of body cavities
5. Demonstrate knowledge of the human skeletal system
6. Demonstrate knowledge of the major muscles of the body and their functions
7. Demonstrate knowledge of the body joints
8. Demonstrate knowledge of circulatory system
9. Demonstrate knowledge of the lymphatic system
10. Demonstrate knowledge of the endocrine system and its functions
11. Demonstrate knowledge of the nervous system and its function
12. Demonstrate knowledge of the digestive system and its function
13. Demonstrate knowledge of the urinary system and its function
14. Demonstrate knowledge of the respiratory system and its function
15. Demonstrate knowledge of the special senses and their functions
16. Demonstrate knowledge of the reproductive system and its functions

1.2.1 Learning Outcome 1: Demonstrate knowledge of organization of the human body

1.2.1.1 Introduction to the learning outcome

This learning outcome specifies the content of competencies required to identify anatomical structures, outline anatomical position, planes and directions, and identify levels of human body organization, describe functions of the human body, human cell structure components are identified, human cell cycle outlined and homeostasis described.

1.2.1.2 Performance Standard

1.1 Anatomical structures are identified
1.2 Anatomical position, planes and directions are outlined
1.3 Levels of human body organization are identified
1.4 Functions of the human body are described
1.5 Human cell structure components are identified
1.6 Human cell cycle outlined

1.7 Homeostasis is described

1.2.1.3 Information Sheet

Definition of terms

Anatomical position *is the description of any region or part of the body in a specific stance. In the anatomical position, the body is upright, directly facing the observer, feet flat and directed forward. The upper limbs are at the body's sides with the palms facing forward.*

Anterior (or **ventral**) describes the front or direction toward the front of the body. The toes are anterior to the foot.

Posterior (or **dorsal**) Describes the back or direction toward the back of the body. The popliteus is posterior to the patella.

Superior (or **cranial**) describes a position above or higher than another part of the body proper. The orbits are superior to the oris.

Inferior (or **caudal**) describes a position below or lower than another part of the body proper; near or toward the tail (in humans, the coccyx, or lowest part of the spinal column). The pelvis is inferior to the abdomen.

Lateral describes the side or direction toward the side of the body. The thumb (pollex) is lateral to the digits.

Medial describes the middle or direction toward the middle of the body. The hallux is the medial toe.

Proximal describes a position in a limb that is nearer to the point of attachment or the trunk of the body. The brachium is proximal to the ante brachium.

Distal describes a position in a limb that is farther from the point of attachment or the trunk of the body. The crus is distal to the femur.

Superficial describes a position closer to the surface of the body. The skin is superficial to the bones.

Deep describes a position farther from the surface of the body. The brain is deep to the skull.

Branches of human anatomy

Gross Anatomy - the study of large body structures visible to the naked eye

Regional Anatomy - the study of all structures (blood vessels, nerves, muscles) located in a particular region of the body

Systemic Anatomy - the gross anatomy of the body studied system by system

Surface Anatomy - the study of internal body structures as they relate to the overlying skin.

Microscopic Anatomy - the study of structures too small to be seen without the aid of a microscope

Cellular Anatomy - the study of cells of the body

Histology - the study of the microscopic structure of tissues

Developmental Anatomy - the study of changes in an individual from conception to old age

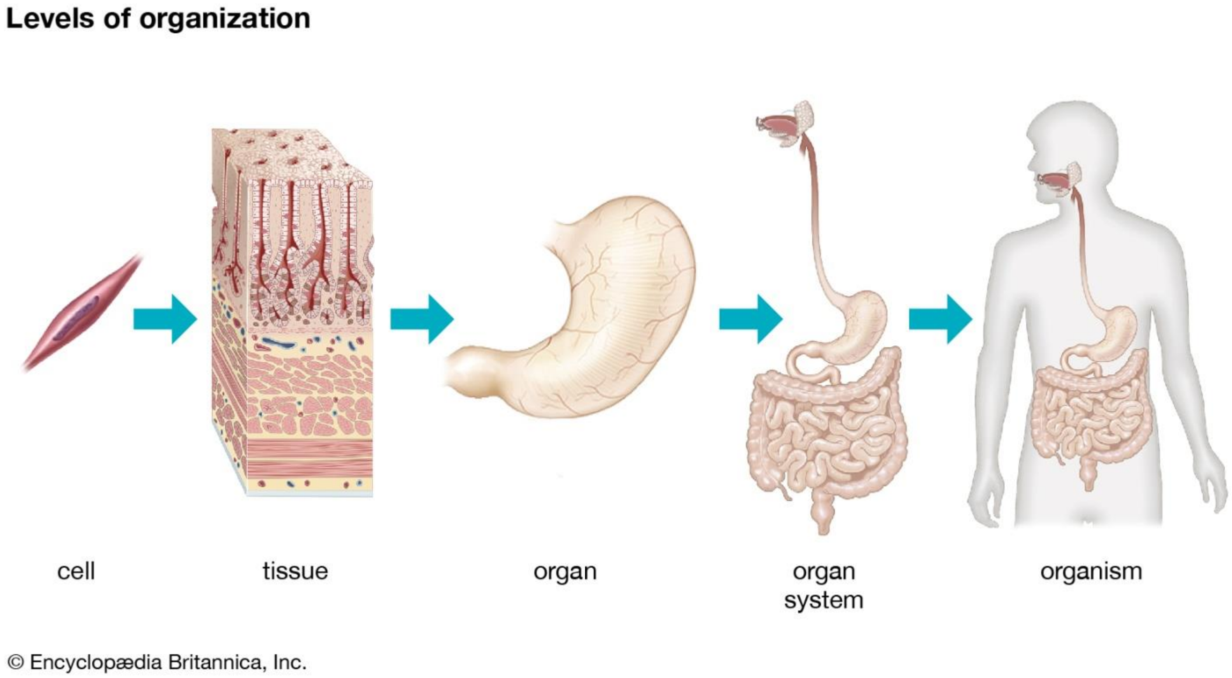
Specialized Branches of Anatomy

• Pathological anatomy - the study structural changes in cells, tissues, and organs caused by disease

• Radiographic anatomy - the study of internal body structures by means of x-rays and imaging techniques

• Functional morphology- the study of functional properties of body structures and efficiency of design.

Organization of the human body



Chemical Level

• At the chemical level atoms combine to form small molecules (CO_2 and H_2O) and larger macromolecules.

•Macromolecules of four classes are found in the body

• These macro molecules include carbohydrates (sugars), lipids (fats), proteins and nucleic acids
(DNA, RNA).

Cellular Level

• The smallest units of living tissue.

• Cells and their functional subunits called cellular organelles.

Tissue Level

• Consists of groups of similar cells that have a characteristic function
  - Epithelium
  - Muscle
  - Connective
  - Nervous

Organ Level

• A structure composed of at least two tissue types (with four the most common) that performs a specific physiological process or function.

Organ System Level

• Organs that cooperate with one another to perform a common function

• For example:- Cardiovascular system.

Functions of the human body

Metabolism

The first law of thermodynamics holds that energy can neither be created nor destroyed—it can only change form. Your basic function as an organism is to consume (ingest) energy and molecules in the foods you eat, convert some of it into fuel for movement, sustain your body functions, and build and maintain your body structures. There are two types of reactions that accomplish this: **anabolism** and **catabolism**.

- **Anabolism** is the process whereby smaller, simpler molecules are combined into larger, more complex substances. Your body can assemble, by utilizing energy, the complex chemicals it needs by combining small molecules derived from the foods you eat
- **Catabolism** is the process by which larger more complex substances are broken down into smaller simpler molecules. Catabolism releases energy. The complex molecules found in foods are broken down so the body can use their parts to assemble the structures and substances needed for life.

Responsiveness

Responsiveness is the ability of an organism to adjust to changes in its internal and external environments. An example of responsiveness to external stimuli could include moving toward sources of food and water and away from perceived dangers. Changes in an organism's internal environment, such as increased body temperature, can cause the responses of sweating and the dilation of blood vessels in the skin in order to decrease body temperature.

Movement

Human movement includes not only actions at the joints of the body, but also the motion of individual organs and even individual cells. As you read these words, red and white blood cells are moving throughout your body, muscle cells are contracting and relaxing to maintain your posture and to focus your vision, and glands are secreting chemicals to regulate body functions. Your body is coordinating the action of entire muscle groups to enable you to move air into and out of your lungs, to push blood throughout your body, and to propel the food you have eaten through your digestive tract. Consciously, of course, you contract your skeletal muscles to move the bones of your skeleton to get from one place to another and to carry out all of the activities of your daily life.

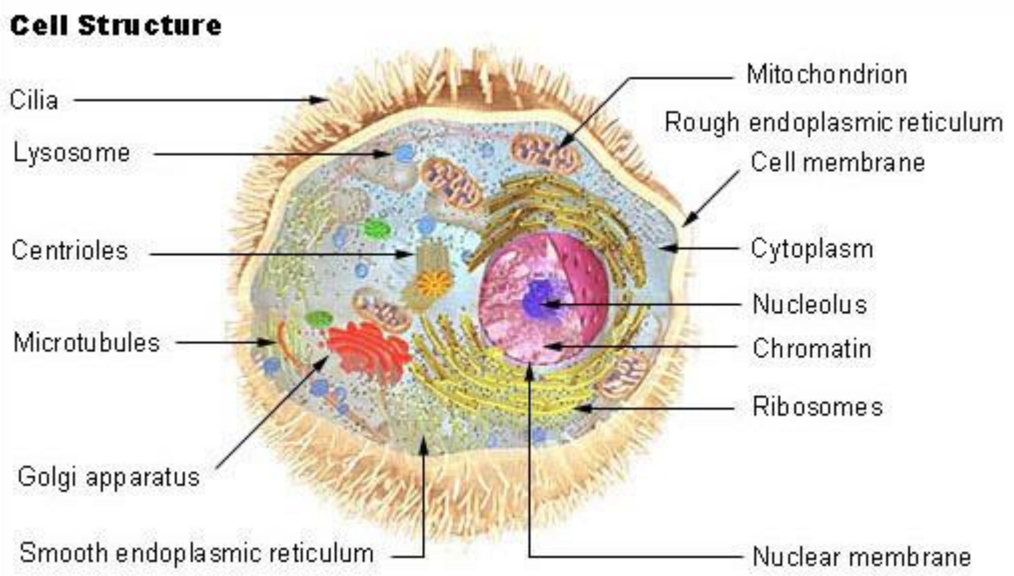
Development, growth and reproduction

Development is all of the changes the body goes through in life. Development includes the process of **differentiation**, in which unspecialized cells become specialized in structure and function to perform certain tasks in the body. Development also includes the processes of growth and repair, both of which involve cell differentiation.

Growth is the increase in body size. Humans, like all multicellular organisms, grow by increasing the number of existing cells, increasing the amount of non-cellular material around cells (such as mineral deposits in bone), and, within very narrow limits, increasing the size of existing cells.

Reproduction is the formation of a new organism from parent organisms. In humans, reproduction is carried out by the male and female reproductive systems. Because death will come to all complex organisms, without reproduction, the line of organisms would end.

The cell



All the tissues and organs of the body originate from a microscopic structure (the **fertilized ovum**), which consists of a soft jelly-like material enclosed in a membrane and containing a vesicle or small spherical body inside which are one or more denser spots. This may be regarded as a complete cell. All the solid tissues consist largely of cells essentially similar to it in nature but differing in external form.

Nucleolus, are not essential to the type cell, and in fact In the higher organisms a cell may be defined as "a nucleated mass of protoplasm of microscopic size." Its two essentials, therefore, are: a soft jelly-like material, similar to that found in the ovum, and usually styled cytoplasm, and a small spherical body imbedded in it, and termed a nucleus. Some of the unicellular
protozoa contain no nuclei but granular particles which, like true nuclei, stain with basic dyes. The other constituents of the ovum, viz., its limiting membrane and the denser spot contained in the nucleus, called the many cells exist without them.

A typical animal cell comprises the following cell organelles:

Cell Membrane – a thin semi permeable membrane layer of protein and fats surrounding the cell. Its primary role is to protect the cell from its surrounding. Also, it controls the entry and exit of nutrients and other microscopic entities into the cell.

Nuclear Membrane - is a double-membrane structure that surrounds the nucleus. It is also referred to as the nuclear envelope.

Nucleus - It is an organelle that contains several other sub-organelles such as nucleolus, nucleosomes and chromatins. It also contains DNA and other genetic materials.

Centrosome - is a small organelle found near to the nucleus which has a thick centre with radiating tubules. The centrosomes are where microtubules are produced.

Lysosome (Cell Vesicles) - They are round organelles surrounded by a membrane and comprising digestive enzymes which help in digestion, excretion and in the cell renewal process.

Cytoplasm - a jelly-like material which contains all the cell organelles, enclosed within the cell membrane. The substance found within the cell nucleus, contained by the nuclear membrane is called the nucleoplasm**.**

Golgi apparatus – is a flat, smooth layered, sac-like organelle which is located near the nucleus and involved in manufacturing, storing, packing and transporting the particles throughout the cell.

Mitochondrion - they are spherical or rod-shaped organelles with a double membrane. They are the powerhouse of a cell as they play an important role in releasing energy.

Ribosome - they are small organelles made up of RNA-rich cytoplasmic granules, and they are the sites of protein synthesis.

Endoplasmic Reticulum (ER) - This cellular organelle is composed of a thin, winding network of membranous sacs originating from the nucleus.

Vacuole – a membrane-bound organelle present inside a cell involved in maintaining shape and storing water, food, wastes, etc.

> ***Nucleopore*** *- they are tiny holes present in the nuclear membrane which are involved in the movement of nucleic acids and proteins within the cell.*

Cell cycle

The cell cycle is an ordered series of events involving cell growth and cell division that produces two new daughter cells. Cells on the path to cell division proceed through a series of precisely timed and carefully regulated stages of growth, DNA replication, and division that produces two identical (clone) cells.

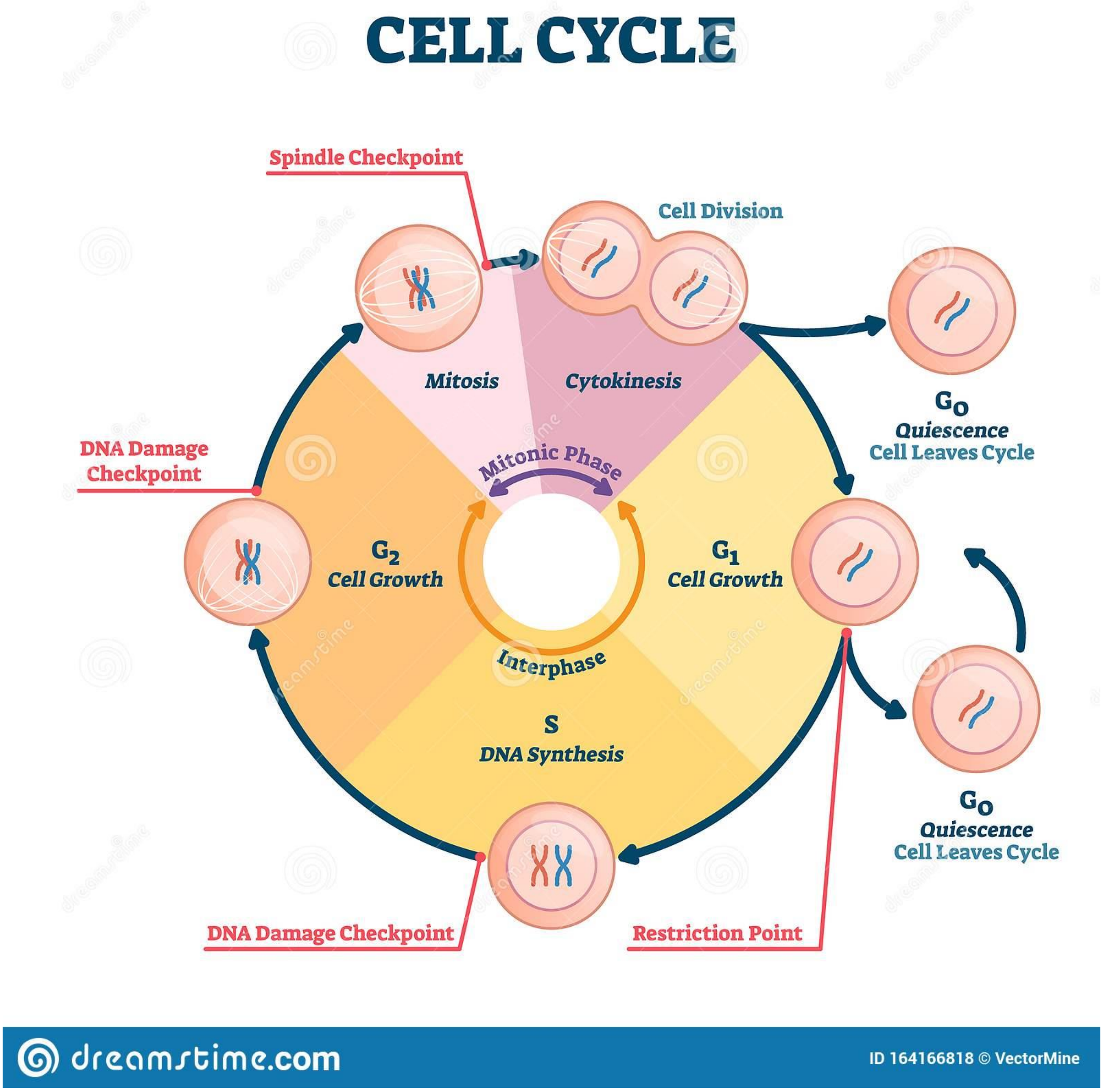


Fig: The cell cycle

The cell cycle has two major phases: interphase and the mitotic phase. During interphase, the cell grows and DNA is replicated. During the mitotic phase, the replicated DNA and cytoplasmic contents are separated, and the cell divides.

Interphase

During interphase, the cell undergoes normal growth processes while also preparing for cell division. In order for a cell to move from interphase into the mitotic phase, many internal and external conditions must be met. The three stages of interphase are called G_1, S, and G_2.

2

3 G1 Phase (First Gap)

The first stage of interphase is called the G1 phase (first gap) because, from a microscopic aspect, little change is visible. However, during the G1 stage, the cell is quite active at the biochemical level. The cell is accumulating the building blocks of chromosomal DNA and the associated proteins as well as accumulating sufficient energy reserves to complete the task of replicating each chromosome in the nucleus.

4 S Phase (Synthesis of DNA)

5 Throughout interphase, nuclear DNA remains in a semi-condensed chromatin configuration. In the S phase, DNA replication can proceed through the mechanisms that result in the formation of identical pairs of DNA molecules—sister chromatids—that are firmly attached to the centromeric region. The centrosome is duplicated during the S phase. The two centrosomes will give rise to the mitotic spindle, the apparatus that orchestrates the movement of chromosomes during mitosis. At the center of each animal cell, the centrosomes of animal cells are associated with a pair of rod-like objects, the centrioles, which are at right angles to each other. Centrioles help organize cell division. Centrioles are not present in the centrosomes of other eukaryotic species, such as plants and most fungi.

6 G2 Phase (Second Gap)

In the G2 phase, the cell replenishes its energy stores and synthesizes proteins necessary for chromosome manipulation. Some cell organelles are duplicated, and the cytoskeleton is dismantled to provide resources for the mitotic phase. There may be additional cell growth during G2. The final preparations for the mitotic phase must be completed before the cell is able to enter the first stage of mitosis.

The Mitotic Phase

The mitotic phase is a multistep processes during which the duplicated chromosomes are aligned, separated, and move into two new, identical daughter cells. The first portion of the mitotic phase is called karyokinesis, or nuclear division. The second portion of the mitotic phase, called cytokinesis, is the physical separation of the cytoplasmic components into the two daughter cells.

7 Karyokinesis (Mitosis)

Karyokinesis, also known as mitosis, is divided into a series of phases—prophase, prometaphase, metaphase, anaphase, and telophase—that result in the division of the cell nucleus (Figure 2). Karyokinesis is also called mitosis.

During prophase, the "first phase," the nuclear envelope starts to dissociate into small vesicles, and the membranous organelles (such as the Golgi complex or Golgi apparatus, and endoplasmic reticulum), fragment and disperse toward the periphery of the cell. The nucleolus disappears (disperses). The centrosomes begin to move to opposite poles of the cell. Microtubules that will form the mitotic spindle extend between the centrosomes, pushing them farther apart as the microtubule fibers lengthen. The sister chromatids begin to coil more tightly with the aid of condensin proteins and become visible under a light microscope.

During **prometaphase**, the "first change phase," many processes that were begun in prophase continue to advance. The remnants of the nuclear envelope fragment. The mitotic spindle continues to develop as more microtubules assemble and stretch across the length of the former nuclear area. Chromosomes become more condensed and discrete. Each sister chromatid develops a protein structure called a kinetochore in the centromeric region. The proteins of the kinetochore attract and bind mitotic spindle microtubules. As the spindle microtubules extend from the centrosomes, some of these microtubules come into contact with and firmly bind to the kinetochores. Once a mitotic fiber attaches to a chromosome, the chromosome will be oriented until the kinetochores of sister chromatids face the opposite poles. Eventually, all the sister chromatids will be attached via their kinetochores to microtubules from opposing poles. Spindle microtubules that do not engage the chromosomes are called polar microtubules. These microtubules overlap each other midway between the two poles and contribute to cell elongation. Astral microtubules are located near the poles, aid in spindle orientation, and are required for the regulation of mitosis.

During **metaphase**, the "change phase," all the chromosomes are aligned in a plane called the metaphase plate, or the equatorial plane, midway between the two poles of the cell. The sister chromatids are still tightly attached to each other by cohesin proteins. At this time, the chromosomes are maximally condensed.

During **anaphase**, the "upward phase," the cohesin proteins degrade, and the sister chromatids separate at the centromere. Each chromatid, now called a chromosome, is pulled rapidly toward the centrosome to which its microtubule is attached. The cell becomes visibly elongated (oval shaped) as the polar microtubules slide against each other at the metaphase plate where they overlap.

During **telophase,** the "distance phase," the chromosomes reach the opposite poles and begin to decondense (unravel), relaxing into a chromatin configuration. The mitotic spindles are depolymerized into tubulin monomers that will be used to assemble cytoskeletal components for each daughter cell. Nuclear envelopes form around the chromosomes, and nucleosomes appear within the nuclear area.

<u>**Homeostasis**</u>

8 Introduction to Homeostasis

Homeostasis refers to the body's ability to maintain a stable internal environment (regulating hormones, body temp., water balance, etc.). Maintaining homeostasis requires that the body continuously monitors its internal conditions. From body temperature to blood pressure to levels of certain nutrients, each physiological condition has a particular set point. A *set point* is the physiological value around which the normal range fluctuates. A *normal range* is the restricted set of values that is optimally healthful and stable. For example, the set point for normal human body temperature is approximately 37°C (98.6°F) Physiological parameters, such as body temperature and blood pressure, tend to fluctuate within a normal range a few degrees above and below that point. Control centers in the brain play roles in regulating physiological parameters and keeping them within the normal range. As the body works to maintain homeostasis, any significant deviation from the normal range will be resisted and homeostasis restored through a process called a **feedback loop**.

A feedback loop has three basic components (Figure 1.10a). A *sensor*, also known as a receptor, is a component of a feedback system that monitors a physiological value. It is responsible for detecting a change in the environment. This value is reported to the control center. The *control center* is the component in a feedback system that compares the value to the normal range. If the value deviates too much from the set point, then the control center activates an effector. An *effector* is the component in a feedback system that causes a change to reverse the situation and return the value to the normal range. Effectors are muscles and glands.

9 Two Types of Feedback Loops: Negative and Positive

Negative feedback is a mechanism in which the effect of the response to the stimulus is to shut off the original stimulus or reduce its intensity. Negative feedback loops are the body's most common mechanisms used to maintain homeostasis.

The maintenance of homeostasis by negative feedback goes on throughout the body at all times, and an understanding of negative feedback is thus fundamental to an understanding of human physiology.

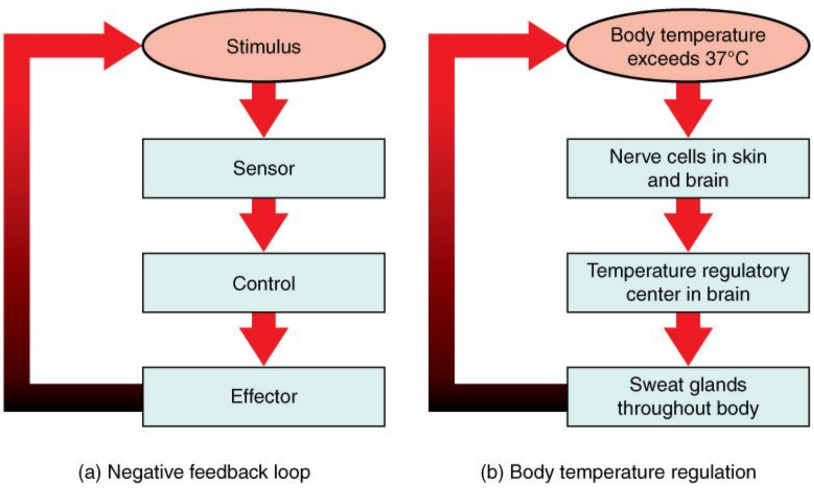


Figure 1: Feedback loop

In order to set the system in motion, a stimulus change an internal environment beyond its normal range (that is, beyond homeostasis). This stimulus is detected by a specific receptor. For example, in the control of blood glucose, specific endocrine cells in the pancreas detect excess glucose (the stimulus) in the bloodstream. These pancreatic beta cells respond to the increased level of blood glucose by releasing the hormone insulin into the bloodstream. The insulin signals skeletal muscle fibers, fat cells (adipocytes), and liver cells to take up the excess glucose, removing it from the bloodstream. As glucose concentration in the bloodstream drops, the decrease in concentration—the actual negative feedback—is detected by pancreatic alpha cells, and insulin release stops. This prevents blood sugar levels from continuing to drop below the normal range.

Humans have a similar temperature regulation feedback system that works by promoting either heat loss or heat gain. When the brain's temperature regulation center receives data from the sensors indicating that the body's temperature exceeds its normal range, it stimulates a cluster of brain cells referred to as the "heat-loss center." This stimulation has three major effects:

- Blood vessels in the skin begin to dilate allowing more blood from the body core to flow to the surface of the skin allowing the heat to radiate into the environment.
- As blood flow to the skin increases, sweat glands are activated to increase their output. As the sweat evaporates from the skin surface into the surrounding air, it takes heat with it.
- The depth of respiration increases, and a person may breathe through an open mouth instead of through the nasal passageways. This further increases heat loss from the lungs.

In contrast, activation of the brain's heat-gain center by exposure to cold reduces blood flow to the skin, and blood returning from the limbs is diverted into a network of deep veins. This arrangement traps heat closer to the body core and restricts heat loss. If heat loss is severe, the brain triggers an increase in random signals to skeletal muscles, causing them to contract and producing shivering. The muscle contractions of shivering release heat while using up ATP. The brain triggers the thyroid gland in the endocrine system to release thyroid hormone, which increases metabolic activity and heat production in cells throughout the body. The brain also signals the adrenal glands to release epinephrine (adrenaline), a hormone that causes the breakdown of glycogen into glucose, which can be used as an energy source. The breakdown of glycogen into glucose also results in increased metabolism and heat production.

Positive feedback intensifies a change in the body's physiological condition rather than reversing it. A deviation from the normal range results in more change, and the system moves farther away from the normal range. Positive feedback in the body is normal only when there is a definite end point. Childbirth and the body's response to blood loss are two examples of positive feedback loops that are normal but are activated only when needed.

Childbirth at full term is an example of a situation in which the maintenance of the existing body state is not desired. Enormous changes in the mother's body are required to expel the baby at the end of pregnancy. And the events of childbirth, once begun, must progress rapidly to a conclusion or the life of the mother and the baby are at risk. The extreme muscular work of labor and delivery are the result of a positive feedback system.

The first contractions of labor (the stimulus) push the baby toward the cervix (the lowest part of the uterus). The cervix contains stretch-sensitive nerve cells that monitor the degree of stretching (the sensors). These nerve cells send messages to the brain, which in turn causes the pituitary gland at the base of the brain to release the hormone oxytocin into the bloodstream. Oxytocin causes stronger contractions of the smooth muscles in of the uterus (the effectors), pushing the baby further down the birth canal. This causes even greater stretching of the cervix. The cycle of stretching, oxytocin release, and increasingly more forceful contractions stops only when the baby is born. At this point, the stretching of the cervix halts, stopping the release of oxytocin.

A second example of positive feedback centers on reversing extreme damage to the body. Following a penetrating wound, the most immediate threat is excessive blood loss. Less blood circulating means reduced blood pressure and reduced perfusion (penetration of blood) to the brain and other vital organs. If perfusion is severely reduced, vital organs will shut down and the person will die. The body responds to this potential catastrophe by releasing substances in the injured blood vessel wall that begin the process of blood clotting. As each step of clotting occurs, it stimulates the release of more clotting substances. This accelerates the processes of clotting and sealing off the damaged area. Clotting is contained in a local area based on the tightly controlled availability of clotting proteins. This is an adaptive, life-saving cascade of events.

1.2.1.5 Learning activities

1.2.1.5.1: Practical activity

- **a.** Choose one volunteer to complete jumping jacks at a pace that can be maintained for eight minutes. Make sure the volunteer is inactive for a few minutes before the experiment begins.
- **b.** Measure the heart rate by taking his or her pulse; you can do this one of two ways:
  - **i.** Radial Pulse
  - **ii.** Carotid Pulse
- **c.** Multiply the number of beats in 15 seconds by 4 to calculate the beats per minute. Record the data in the table below the 0 minutes box.
- **d.** Measure the person's perspiration level from 0 to 5: (0 = none; 5 = droplets dripping down the face). Note this observation in the table.
- **e.** Have the volunteer do jumping jacks for 2 minutes. After 2 minutes, measure the heart rate, breathing rate, and perspiration level (refer to steps 2 through 5), and record the data. Measure the pulse, breathing and perspiration levels as quickly as you can so that the volunteer can resume exercise. Do not have volunteer wait while you do the calculations and enter the data.
- **f.** Repeat step 6 three more times (remember, 2 minutes each at a time, totaling 8 minutes) and record your data at each point.

After the final recording of the dependent variables, wait one minute with the volunteer at reset. Then measure all of the variables again. Record this data under the "9 Time (Min.)

1.2.1.5.2Self-Assessment

a). Define the following terms

- Anterior
- Posterior
- Lateral

- Medial

b).Which are the two main types of reactions that accomplish metabolism?

c).What is the G_1 Phase of the cell cycle?

d).What is positive feedback?

e).What events precede the onset of first labor pains?

1.2.1.5.1 Model answers

Self assessment

a). Define the following terms

Anterior - (or **ventral**) describes the front or direction toward the front of the body. The toes are anterior to the foot.

Posterior - (or **dorsal**) Describes the back or direction toward the back of the body. The popliteus is posterior to the patella.

Lateral - describes the side or direction toward the side of the body. The thumb (pollex) is lateral to the digits

Medial - describes the middle or direction toward the middle of the body. The hallux is the medial toe.

b). What are the two main types of reactions that accomplish metabolism.

- **Anabolism** is the process whereby smaller, simpler molecules are combined into larger, more complex substances. Your body can assemble, by utilizing energy, the complex chemicals it needs by combining small molecules derived from the foods you eat
- **Catabolism** is the process by which larger more complex substances are broken down into smaller simpler molecules. Catabolism releases energy. The complex molecules found in foods are broken down so the body can use their parts to assemble the structures and substances needed for life.

c). Highlight the G_1 Phase of the cell cycle.

The first stage of interphase is called the G1 phase (first gap) because, from a microscopic aspect, little change is visible. However, during the G1 stage, the cell is quite active at the biochemical level. The cell is accumulating the building blocks of chromosomal DNA and the associated proteins as well as accumulating sufficient energy reserves to complete the task of replicating each chromosome in the nucleus.

d).Describe positive feedback

Intensifies a change in the body's physiological condition rather than reversing it. A deviation from the normal range results in more change, and the system moves farther away from the normal range. Positive feedback in the body is normal only when there is a definite end point. Childbirth and the body's response to blood loss are two examples of positive feedback loops that are normal but are activated only when needed.

e). What events precede the onset of first labor pains?

The first contractions of labor (the stimulus) push the baby toward the cervix (the lowest part of the uterus). The cervix contains stretch-sensitive nerve cells that monitor the degree of stretching (the sensors). These nerve cells send messages to the brain, which in turn causes the pituitary gland at the base of the brain to release the hormone oxytocin into the bloodstream. Oxytocin causes stronger contractions of the smooth muscles in of the uterus (the effectors), pushing the baby further down the birth canal. This causes even greater stretching of the cervix. The cycle of stretching, oxytocin release, and increasingly more forceful contractions stops only when the baby is born. At this point, the stretching of the cervix halts, stopping the release of oxytocin.

Practical activity

Data and Observations:

Table 1: Effect of Exercise

Total Time (Minutes)	Heart Rate (Beats per Minute)	Breathing Rate (Breaths per Minute)	Perspiration Level (Scale of 0 to 5)
0 – Prior to Exercise	84	36	0
2 Minutes	128	52	0
6 Minutes	96	36	1
8 Minutes	120	40	2
9 Minutes	144	44	2

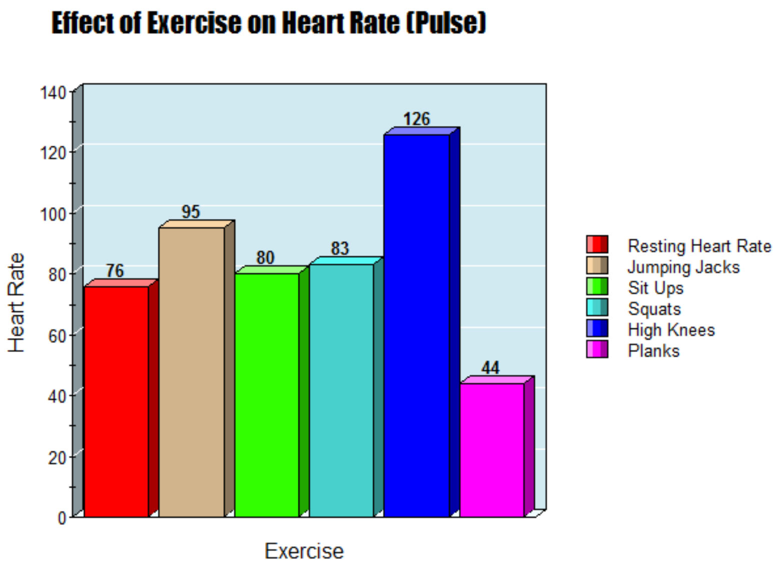
Table 1: shows the heart rate, breathing rate and perspiration level of exercisers over nine minutes

Table 2: Effects of Exercise: Class Averages

Variables Measured		Group 1	Group 2	Group 3	Group 4	Group 5	Averages:
Heart Rate	*Start:*	68	104	104	98	84	73
	Finish:	38	152	120	120	144	115
Breathing Rate	*Start:*	20	60	60	16	36	38
	Finish:	68	68	68	32	44	56
Perspiration Level	*Start:*	0	0	0	0	0	0
	Finish:	1	3	0	0	2	1

Table 2: shows the averages of the entire group's heart rate, breathing rate and perspiration levels at the start and finish of exercising

Graph 1: Effects of Exercise



Analysis:
There is an increase effect of exercise over time on the circulatory and respiratory systems, as shown in both Table 1 and Table 2. As the students began their exercise in two-minute intervals, their heart rates increased on average from 73 beats per minute to 115 beats per minute. Shawn's individual heart rate began at 84 beats per minute, and after lifting 5-pound weights for 8 minutes, his heart rate increased to 114 beats per minute. Also, his breathing rate showed an increase from 36 breaths per minute to 44 breaths per minute. Based on this data shown, the amount that you exercise does effect the circulatory and respiratory systems by causing a, on average, dramatic increase in beats/breath per minute in the measured variables.

Conclusion:
According to the data collected, the hypothesis that was developed at the beginning of the experiment: "If effects of exercise and exercise are related, then exercising will increase heart rate, breathing rate, and perspiration levels"; was accepted. Based on the data shown our individual group charts and the class averages, the effect that exercise has on the respiratory and circulatory systems is that it develops an increase in heart rate (Start: 73/ Finish: 115), breathing rate (Start: 38/ Finish: 56), and perspiration levels (Start: 0/ Finish: 1).

Another process that could have been used to measure the external and internal effects of exercise on the body could be to also measure the body temperature before and after the exercise is preformed. This could be a good variable to measure because often the amount of perspiration found on a body is a result of the internal and external body temperature. The relationship between the perspiration level and the body temperature is that they are both negative feedback. After exercise, the body tries to return to homeostasis; so when the body temperature increases due to exercise, the body perspires to cool off and return to its regular temperature, which is considered negative feedback.

1.2.1.7 Tools, equipment, supplies and materials

The following resources are provided;

A functional anatomy laboratory.

Stopwatches

Stethoscope

1.2.1.7 References

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1.2.2: Learning outcome 2: Demonstrate knowledge of human body fluids and their functions

1.2.2.1: Introduction to learning outcome

This learning outcome specifies the competencies required to identify **Components** of extracellular and intracellular fluids. Importance of body fluids is also explained.

1.2.2.2: Performance standard

1.2 Components of extracellular fluid are outlined
2.2 Components of intracellular fluid are outlined
3.2 Importance of body electrolytes is described

<u>1.2.2.3: Information Sheet</u>

<u>Introduction</u>

The chemical reactions of life take place in aqueous solutions. The dissolved substances in a solution are called solutes. In the human body, solutes vary in different parts of the body, but may include proteins—including those that transport lipids, carbohydrates, and, very importantly, electrolytes. Often in medicine, an electrolyte is referred to as a mineral dissociated from a salt that carries an electrical charge (an ion). For instance, sodium ions (Na^+) and chloride ions (Cl^-) are often referred to as electrolytes.

In the body, water moves through semi-permeable membranes of cells and from one compartment of the body to another by a process called osmosis. Osmosis is basically the diffusion of water from regions of higher concentration to regions of lower concentration, along an osmotic gradient across a semi-permeable membrane. As a result, water will move into and out of cells and tissues, depending on the relative concentrations of the water and solutes found there. An appropriate balance of solutes inside and outside of cells must be maintained to ensure normal function.

Body Water Content

Human beings are mostly water, ranging from about 75 percent of body mass in infants to about 50–60 percent in adult men and women, to as low as 45 percent in old age. The percent of body water changes with development, because the proportions of the body given over to each organ and to muscles, fat, bone, and other tissues change from infancy to adulthood. Your brain and kidneys have the highest proportions of water, which composes 80–85 percent of their masses. In contrast, teeth have the lowest proportion of water, at 8–10 percent.

Fluid Compartments

Body fluids can be discussed in terms of their specific fluid compartment, a location that is largely separate from another compartment by some form of a physical barrier. The intracellular fluid (ICF) compartment is the system that includes all fluid enclosed in cells by their plasma membranes. Extracellular fluid (ECF) surrounds all cells in the body. Extracellular fluid has two primary constituents: the fluid component of the blood (called plasma) and the interstitial fluid (IF) that surrounds all cells not in the blood.

Intracellular Fluid

The ICF lies within cells and is the principal component of the cytosol/cytoplasm. The ICF makes up about 60 percent of the total water in the human body, and in an average-size adult male, the ICF accounts for about 25 liters (seven gallons) of fluid. This fluid volume tends to be very stable, because the amount of water in living cells is closely regulated. If the amount of water inside a cell falls to a value that is too low, the cytosol becomes too concentrated with solutes to carry on normal cellular activities; if too much water enters a cell, the cell may burst and be destroyed.

10 Extracellular Fluid

11 The ECF accounts for the other one-third of the body’s water content. Approximately 20 percent of the ECF is found in plasma. Plasma travels through the body in blood vessels and transports a range of materials, including blood cells, proteins (including clotting factors and antibodies), electrolytes, nutrients, gases, and wastes. Gases, nutrients, and waste materials travel between capillaries and cells through the IF. Cells are separated from the IF by a selectively permeable cell membrane that helps regulate the passage of materials between the IF and the interior of the cell.

12 The body has other water-based ECF. These include the cerebrospinal fluid that bathes the brain and spinal cord, lymph, the synovial fluid in joints, the pleural fluid in the pleural cavities, the pericardial fluid in the cardiac sac, the peritoneal fluid in the peritoneal cavity, and the aqueous humor of the eye. Because these fluids are outside of cells, these fluids are also considered components of the ECF compartment.

Composition of Body Fluids

The compositions of the two components of the ECF—plasma and IF—are more similar to each other than either is to the ICF. Blood plasma has high concentrations of sodium, chloride, bicarbonate, and protein. The IF has high concentrations of sodium, chloride, and bicarbonate, but a relatively lower concentration of protein. In contrast, the ICF has elevated amounts of potassium, phosphate, magnesium, and protein. Overall, the ICF contains high concentrations of potassium and phosphate (HPO42−HPO42−), whereas both plasma and the ECF contain high concentrations of sodium and chloride.

Most body fluids are neutral in charge. Thus, cations, or positively charged ions, and anions, or negatively charged ions, are balanced in fluids. As seen in the previous graph, sodium (Na^+) ions and chloride ($Cl^–$) ions are concentrated in the ECF of the body, whereas potassium (K^+) ions are concentrated inside cells. Although sodium and potassium can “leak” through “pores” into and out of cells, respectively, the high levels of potassium and low levels of sodium in the ICF are maintained by sodium-potassium pumps in the cell membranes. These pumps use the energy supplied by ATP to pump sodium out of the cell and potassium into the cell.

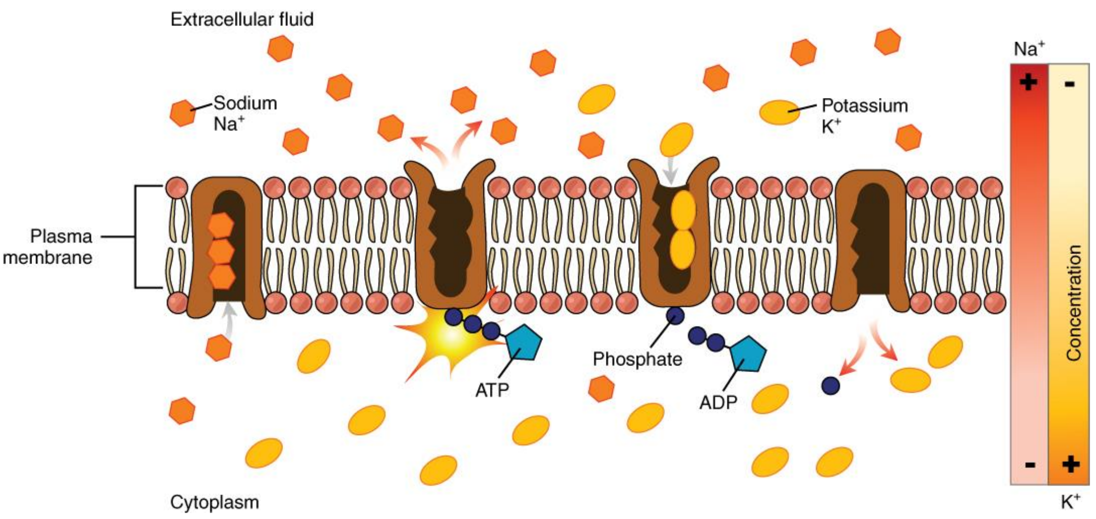

Figure 2” Sodium potassium pump

Examples of body fluids;

- **Amniotic fluid** - is the protective liquid contained by the amniotic sac of a gravid amniote. This fluid serves as a cushion for the growing fetus, but also serves to facilitate the exchange of nutrients, water, and biochemical products between mother and fetus.
- **Aqueous humour** - is the fluid produced by the eye. It provides nutrition to the eye, as well as maintains the eye in a pressurized state.
- **Bile** - is a fluid that is made and released by the liver and stored in the gallbladder. Bile helps with digestion. It breaks down fats into fatty acids, which can be taken into the body by the digestive tract.
- **Blood plasma** - is a yellowish liquid component of blood that holds the blood cells of whole blood in suspension. It is the liquid part of the blood that carries cells and proteins throughout the body. It makes up about 55% of the body's total blood volume.
- **Breast milk** - is milk produced by mammary glands, located in the breast of a human female. Breast milk is the primary source of nutrition for newborns, containing fat, protein, carbohydrates and variable minerals and vitamins.
- **Cerebrospinal fluid -** is a clear fluid that surrounds the brain and spinal cord. It cushions the brain and spinal cord from injury and also serves as a nutrient delivery and waste removal system for the brain.
- **Cerumen** - Earwax, also known by the medical term cerumen, is a brown, orange, red, yellowish or gray waxy substance secreted in the ear canal of humans and other mammals. It protects the skin of the human ear canal, assists in cleaning and lubrication, and provides protection against bacteria, fungi, and water.
- **Chyle -** Chyle is a milky bodily fluid consisting of lymph and emulsified fats, or free fatty acids. It is formed in the small intestine during digestion of fatty foods, and taken up by lymph vessels specifically known as lacteals. The lipids in the chyle are colloidally suspended in chylomicrons.
- **Exudates -** is any fluid that filters from the circulatory system into lesions or areas of inflammation. It can be a pus-like or clear fluid. When an injury occurs, leaving skin exposed, it leaks out of the blood vessels and into nearby tissues. The fluid is composed of serum, fibrin, and leukocytes.
- **Gastric juice -** is a unique combination of hydrochloric acid (HCl), lipase, and pepsin. Its main function is to inactivate swallowed microorganisms, thereby inhibiting infectious agents from reaching the intestine.
- **lymph -** is the fluid that flows through the lymphatic system, a system composed of lymph vessels (channels) and intervening lymph nodes whose function, like the venous system, is to return fluid from the tissues to the central circulation.
- **Mucus -** is a normal, slippery and stringy fluid substance produced by many lining tissues in the body. It is essential for body function and acts as a protective and moisturizing layer to keep critical organs from drying out. Mucus also acts as a trap for irritants like dust, smoke, or bacteria.
- **Pericardial fluid -** is the serous fluid secreted by the serous layer of the pericardium into the pericardial cavity. The pericardium consists of two layers, an outer fibrous layer and the inner serous layer.
- **Peritoneal fluid -** is a serous fluid made by the peritoneum in the abdominal cavity which lubricates the surface of tissue that lines the abdominal wall and pelvic cavity. It covers most of the organs in the abdomen. An increased volume of peritoneal fluid is called ascites.
- **Pleural fluid -** pleural effusion, sometimes referred to as “water on the lungs,” is the build-up of excess fluid between the layers of the pleura outside the lungs. The pleura are thin membranes that line the lungs and the inside of the chest cavity and act to lubricate and facilitate breathing.

- **Pus -** a thick yellowish or greenish opaque liquid produced in infected tissue, consisting of dead white blood cells and bacteria with tissue debris and serum.
- **Saliva** - is the watery and usually somewhat frothy substance produced in the mouths of some animals, including humans. Produced in salivary glands, saliva is 98% water, but it contains many important substances, including electrolytes, mucus, antibacterial compounds and various enzymes.
- **Sebum -** is an oily, waxy substance produced by your body's sebaceous glands. It coats, moisturizes, and protects your skin. It's also the main ingredient in what you might think of as your body's natural oils.
- **Serous fluid -** is any of various body fluids resembling serum, that are typically pale yellow and transparent and of a benign nature. The fluid fills the inside of body cavities.
- **Semen -** also known as seminal fluid, is an organic fluid created to contain spermatozoa. It is secreted by the gonads (sexual glands) and other sexual organs of male or hermaphroditic animals and can fertilize the female ovum.
- **Sputum -** a mixture of saliva and mucus coughed up from the respiratory tract, typically as a result of infection or other disease and often examined microscopically to aid medical diagnosis.
- **Synovial fluid -** also called synovia, is a viscous, non-Newtonian fluid found in the cavities of synovial joints. With its egg white–like consistency, the principal role of synovial fluid is to reduce friction between the articular cartilage of synovial joints during movement.
- **Sweat** - moisture exuded through the pores of the skin, typically in profuse quantities as a reaction to heat, physical exertion, fever, or fear.
- **Tears -** are a clear liquid secreted by the lacrimal glands found in the eyes of all land mammals. Their functions include lubricating the eyes, removing irritants, and aiding the immune system. Tears also occur as a part of the body's natural pain response.
- **Urine** - is a liquid by-product of metabolism in humans and in many other animals. Urine flows from the kidneys through the ureters to the urinary bladder. Urination results in urine being excreted from the body through the urethra.
- **Vomit -** matter vomited from the stomach.

Electrolytes in the human body

Electrolytes are elements and compounds that occur naturally in the body. They control important physiologic functions.

Examples of electrolytes include:

- calcium
- chloride
- magnesium
- phosphate
- potassium
- sodium

These substances are present in your blood, bodily fluids, and urine. They’re also ingested with food, drinks, and supplements.

An electrolyte disorder occurs when the levels of electrolytes in your body are either too high or too low. Electrolytes need to be maintained in an even balance for your body to function properly. Otherwise, vital body systems can be affected.
Severe electrolyte imbalances can cause serious problems such as coma, seizures, and cardiac arrest.

13 Symptoms of electrolyte disorders

Mild forms of electrolyte disorders may not cause any symptoms. Such disorders can go undetected until they're discovered during a routine blood test. Symptoms usually start to appear once a particular disorder becomes more severe.
Not all electrolyte imbalances cause the same symptoms, but many share similar symptoms.
Common symptoms of an electrolyte disorder include:

- irregular heartbeat
- fast heart rate
- fatigue
- lethargy
- convulsions or seizures
- nausea
- vomiting
- diarrhea or constipation
- abdominal cramping
- muscle cramping
- muscle weakness
- irritability
- confusion
- headaches
- numbness and tingling
- Call your doctor right away if you're experiencing any of these symptoms and suspect you might have an electrolyte disorder. Electrolyte disturbances can become life-threatening if left untreated.
- Causes of electrolyte disorders
- Electrolyte disorders are most often caused by a loss of bodily fluids through prolonged vomiting, diarrhea, or sweating. They may also develop due to fluid loss related to burns.
- Certain medications can cause electrolyte disorders as well. In some cases, underlying diseases, such as acute or chronic kidney disease, are to blame.
- The exact cause may vary depending on the specific type of electrolyte disorder.

14 Types of electrolyte disorders

Elevated levels of an electrolyte are indicated with the prefix “hyper-.” Depleted levels of an electrolyte are indicated with “hypo-.”

Conditions caused by electrolyte level imbalances include:

- calcium: hypercalcemia and hypocalcemia
- chloride: hyperchloremia and hypochloremia
- magnesium: hypomagnesaemia and hypomagnesemia
- phosphate: hyperphosphatemia or hypophosphatemia
- potassium: hyperkalemia and hypokalemia
- sodium: hypernatremia and hyponatremia

Calcium

Calcium is a vital mineral that your body uses to stabilize blood pressure and control skeletal muscle contraction. It’s also used to build strong bones and teeth.

Hypercalcemia occurs when you have too much calcium in the blood. This is usually caused by:

- kidney disease
- thyroid disorders, including hyperparathyroidism
- lung diseases, such as tuberculosis or sarcoidosis
- certain types of cancer, including lung and breast cancers
- excessive use of antacids and calcium or vitamin D supplements
- medications such as lithium, theophylline, or certain water pills

Hypocalcemia occurs due to a lack of adequate calcium in the bloodstream. Causes can include:

- kidney failure
- hypoparathyroidism
- vitamin D deficiency
- pancreatitis
- prostate cancer
- malabsorption
- certain medications, including heparin, osteoporosis drugs, and antiepileptic drugs

Chloride

Chloride is necessary for maintaining the proper balance of bodily fluids.

Hyperchloremia occurs when there’s too much chloride in the body. It can happen as a result of:

- severe dehydration

- kidney failure
- dialysis

Hypochloremia develops when there’s too little chloride in the body. It’s often caused by sodium or potassium problems.

Other causes can include:

- cystic fibrosis
- eating disorders, such as anorexia nervosa
- scorpion stings
- acute kidney failure

Magnesium

Magnesium is a critical mineral that regulates many important functions, such as:

- muscle contraction
- heart rhythm
- nerve function

Hypermagnesemia means excess amounts of magnesium. This disorder primarily affects people with Addison’s disease and end-stage kidney disease.

Hypomagnesemia means having too little magnesium in the body. Common causes include:

- alcohol use disorder
- malnutrition
- malabsorption
- chronic diarrhea
- excessive sweating
- heart failure
- certain medications, including some diuretics and antibiotics

Phosphate

The kidneys, bones, and intestines work to balance phosphate levels in the body. Phosphate is necessary for a wide variety of functions and interacts closely with calcium.

Hyperphosphatemia can occur due to:

- low calcium levels
- chronic kidney disease
- severe breathing difficulties
- underactive parathyroid glands
- severe muscle injury
- tumor lysis syndrome, a complication of cancer treatment

- excessive use of phosphate-containing laxatives

Low levels of phosphate, or hypophosphatemia, can be seen in:

- acute alcohol abuse
- severe burns
- starvation
- vitamin D deficiency
- overactive parathyroid glands
- certain medications, such as intravenous (IV) iron treatment, niacin (Niacor, Niaspan), and some antacids

Potassium

Potassium is particularly important for regulating heart function. It also helps maintain healthy nerves and muscles.

Hyperkalemia may develop due to high levels of potassium. This condition can be fatal if left undiagnosed and untreated. It's typically triggered by:

- severe dehydration
- kidney failure
- severe acidosis, including diabetic ketoacidosis
- certain medications, including some blood pressure medications and diuretics
- adrenal insufficiency, which is when your cortisol levels are too low

Hypokalemia occurs when potassium levels are too low. This often happens as a result of:

- eating disorders
- severe vomiting or diarrhea
- dehydration
- certain medications, including laxatives, diuretics, and corticosteroids.

Sodium

Sodium is necessary for the body to maintain fluid balance and is critical for normal body function. It also helps to regulate nerve function and muscle contraction.

Hypernatremia occurs when there's too much sodium in the blood. Abnormally high levels of sodium may be caused by:

- inadequate water consumption
- severe dehydration
- excessive loss of bodily fluids as a result of prolonged vomiting, diarrhea, sweating, or respiratory illness
- certain medications, including corticosteroids

Hyponatremia develops when there's too little sodium. Common causes of low sodium levels include:

- excessive fluid loss through the skin from sweating or burns
- vomiting or diarrhea
- poor nutrition
- alcohol use disorder
- overhydration
- thyroid, hypothalamic, or adrenal disorders
- liver, heart, or kidney failure
- certain medications, including diuretics and seizure medications
- syndrome of inappropriate secretion of antidiuretic hormone (SIADH)

1.2.2.4: Practical Activity

Using a black marker pen on a white manila paper, draw a well labeled diagram to illustrate the functioning of the sodium pump.

1.2.2.4: Self Assessment

a). High four causes of abnormally high sodium levels

b). Highlight five symptoms of electrolyte disorder.

c). Highlight five causes of hypercalcemia.

d). Highlight five causes of hypomagnesia

1.2.2.4.1: Responses

a). High four causes of abnormally high sodium levels

- inadequate water consumption
- severe dehydration
- excessive loss of bodily fluids as a result of prolonged vomiting, diarrhea, sweating, or respiratory illness
- certain medications, including corticosteroids

b). Highlight five symptoms of electrolyte disorder.

- irregular heartbeat
- fast heart rate
- fatigue
- lethargy
- convulsions or seizures
- nausea
- vomiting
- diarrhea or constipation

- abdominal cramping
- muscle cramping

c). Highlight five causes of hypercalcemia.

- kidney disease
- thyroid disorders, including hyperparathyroidism
- lung diseases, such as tuberculosis or sarcoidosis
- certain types of cancer, including lung and breast cancers
- excessive use of antacids and calcium or vitamin D supplements

d). Highlight five causes of hypomagnesia

- alcohol use disorder
- malnutrition
- malabsorption
- chronic diarrhea
- excessive sweating
- heart failure
- certain medications, including some diuretics and antibiotics

1.2.2.5: Tools, equipment, supplies and materials

Black marker pen

White Manila paper

Anatomy lab

1.2.2.5: References

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1.2.3: Learning Outcome 3: Demonstrate knowledge of body tissues and membranes and their functions

1.2.3.1: Introduction to learning outcome

This learning outcome specifies the knowledge and competencies required to describe type of tissues as well as their functions.

1.2.3.2: Performance standard

1.8 **Types and structure** of tissues are described
1.9 **Functions** of tissues are described

1.2.3.3: Information Sheet

Introduction

The term **tissue** is used to describe a group of cells that are similar in structure and perform a specific function. **Histology** is the field of study that involves the microscopic examination of tissue appearance, organization, and function.

Tissues are organized into four broad categories based on structural and functional similarities. These categories are epithelial, connective, muscle, and nervous. The primary tissue types work together to contribute to the overall health and maintenance of the human body. Thus, any disruption in the structure of a tissue can lead to injury or disease.

Types of tissues

Epithelial tissue refers to groups of cells that cover the exterior surfaces of the body, line internal cavities and passageways, and form certain glands.

Connective tissue, as its name implies, binds the cells and organs of the body together.

Muscle tissue contracts forcefully when excited, providing movement.

Nervous tissue is also excitable, allowing for the generation and propagation of electrochemical signals in the form of nerve impulses that communicate between different regions of the body.

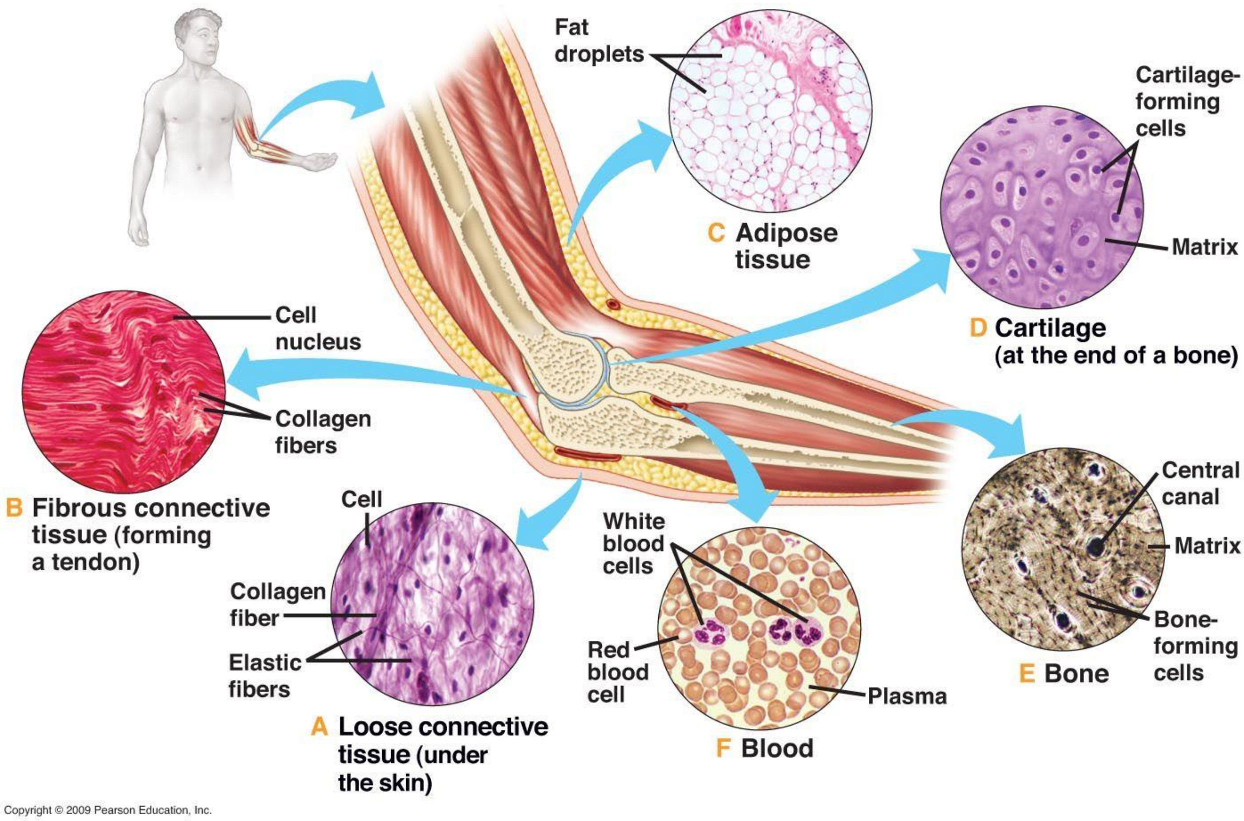

Fig: tissues

Connective Tissue Membranes

A **connective tissue membrane** is built entirely of connective tissue. This type of membrane may be found encapsulating an organ, such as the kidney, or lining the cavity of a freely movable joint (e.g., shoulder). When lining a joint, this membrane is referred to as a **synovial membrane**. Cells in the inner layer of the synovial membrane release synovial fluid, a natural lubricant that enables the bones of a joint to move freely against one another with reduced friction.

Epithelial Membranes

An **epithelial membrane** is composed of an epithelial layer attached to a layer of connective tissue. A **mucous membrane,** sometimes called a mucosa, lines a body cavity or hollow passageway that is open to the external environment. This type of membrane can be found lining portions of the digestive, respiratory, excretory, and reproductive tracts. Mucus, produced by uniglandular cells and glandular tissue, coats the epithelial layer. The underlying connective tissue, called the **lamina propria** (literally "own layer"), helps support the epithelial layer.

A **serous membrane** lines the cavities of the body that do not open to the external environment. Serous fluid secreted by the cells of the epithelium lubricates the membrane and reduces abrasion and friction between organs. Serous membranes are identified according to location. Three serous membranes are found lining the thoracic cavity; two membranes that cover the lungs (pleura) and one membrane that covers the heart (pericardium). A fourth serous membrane, the peritoneum, lines the peritoneal cavity, covering the abdominal organs and forming double sheets of mesenteries that suspend many of the digestive organs.

A **cutaneous membrane** is a multi-layered membrane composed of epithelial and connective tissues. The apical surface of this membrane exposed to the external environment and is covered with dead, keratinized cells that help protect the body from desiccation and pathogens. The skin is an example of a cutaneous membrane.

Muscle Tissue

Muscle tissue is a soft tissue that composes muscles in animal bodies, and gives rise to muscles' ability to contract. It is also referred to as myopropulsive tissue. This is opposed to other components or tissues in muscle such as tendons or perimysium. It is formed during embryonic development through a process known as myogenesis. Muscle tissue consists of elongated cells also called as muscle fibers. This tissue is responsible for movements in our body. Muscles contain special proteins called contractile protein which contract and relax to cause movement.

Nervous tissue

Nervous tissue, also called neural tissue, is the main tissue component of the nervous system. The nervous system regulates and controls bodily functions and activity and consists of two parts: the central nervous system (CNS) comprising the brain and spinal cord, and the peripheral nervous system (PNS) comprising the branching peripheral nerves. It is composed of neurons, also known as nerve cells, which receive and transmit impulses, and neuroglia, also known as glial cells or glia, which assist the propagation of the nerve impulse as well as provide nutrients to the neurons.

Nervous tissue is made up of different types of neurons, all have an axon. An axon is the long stem-like part of the cell that sends action potentials to the next cell. Bundles of axons make up the nerves in the PNS and tracts in the CNS.

Functions of the nervous system are sensory input, integration, control of muscles and glands, homeostasis, and mental activity.

1.2.3.4: Practical Activity

Using the anatomy mannequin provided in the skills lab, identify the various muscles and tissues in the human body.

1.2.3.4: Self assessment

a). Highlight four types of tissues

b). Describe the muscle tissue

c).What is a cutaneous membrane?

d). What is a serous membrane?

1.2.3.4.1: Model Answers

a). Highlight four types of tissues

Epithelial tissue refers to groups of cells that cover the exterior surfaces of the body, line internal cavities and passageways, and form certain glands.

Connective tissue, as its name implies, binds the cells and organs of the body together.

Muscle tissue contracts forcefully when excited, providing movement.

Nervous tissue is also excitable, allowing for the generation and propagation of electrochemical signals in the form of nerve impulses that communicate between different regions of the body.

b). Describe the muscle tissue

Muscle tissue is a soft tissue that composes muscles in animal bodies, and gives rise to muscles' ability to contract. It is also referred to as myopropulsive tissue. This is opposed to other components or tissues in muscle such as tendons or perimysium. It is formed during embryonic development through a process known as myogenesis. Muscle tissue consists of elongated cells also called as muscle fibers. This tissue is responsible for movements in our body. Muscles contain special proteins called contractile protein which contract and relax to cause movement.

c). what is a cutaneous membrane?

This is a multi-layered membrane composed of epithelial and connective tissues. The apical surface of this membrane exposed to the external environment and is covered with dead, keratinized cells that help protect the body from desiccation and pathogens. The skin is an example of a cutaneous membrane.

d). What is a serous membrane?

A **serous membrane** lines the cavities of the body that do not open to the external environment. Serous fluid secreted by the cells of the epithelium lubricates the membrane and reduces abrasion and friction between organs.

1.2.3.5: Tools, equipment, supplies and materials

Anatomy and physiology laboratory

Well labeled charts

Magnifying lens

Anatomy mannequins

1.2.3.6: References

1. Moore, K.; Agur, A.; Dalley, A. F. (2015). "Essesntial Clinical Anatomy". Nervous System (4th ed.). Inkling.
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1.2.4Learning outcome 4: Demonstrate knowledge of body cavities

1.2.4.1: Introduction to learning outcome

This learning outcome specifies the knowledge and competencies required to identify body cavities and describe their contents.

1.2.4.2: Performance standard
Body cavities are identified
Contents of body cavities are described

1.2.4.3: Information Sheet
Introduction

A body cavity is any space or compartment, or potential space in the animal body. Cavities accommodate organs and other structures; cavities as potential spaces contain fluid.

The two largest human body cavities are the ventral body cavity, and the dorsal body cavity. In the dorsal body cavity the brain and spinal cord are located.

The membranes that surround the central nervous system organs (the brain and the spinal cord, in the cranial and spinal cavities are the three meninges. The differently lined spaces contain different types of fluid. In the meninges for example the fluid is cerebrospinal fluid; in the abdominal cavity the fluid contained in the peritoneum is a serous fluid.

In amniotes and some invertebrates the peritoneum lines their largest body cavity called the coelom.

15 Human body cavities

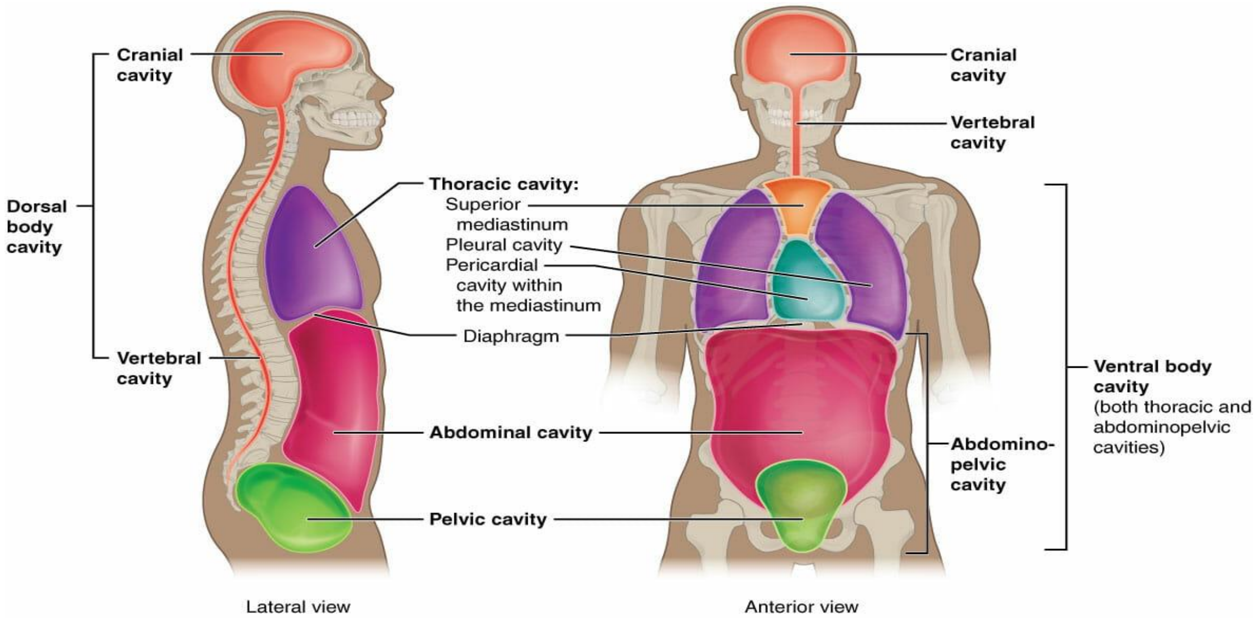

Fig: Human body cavities

16 The dorsal (posterior) cavity and the ventral (anterior) cavity are the largest body compartments.

The dorsal body cavity includes the cranial cavity, enclosed by the skull and contains the brain, and the spinal cavity, enclosed by the spine and contains the spinal cord.The ventral body cavity includes the thoracic cavity, enclosed by the ribcage and contains the lungs and heart; and the abdominopelvic cavity. The abdominopelvic cavity can be divided into the abdominal cavity, enclosed by the ribcage and pelvis and contains the kidneys, ureters, stomach, intestines, liver, gallbladder, and pancreas; and the pelvic cavity, enclosed by the pelvis and contains bladder, anus and reproductive system.

Table 4

Human body cavities and membranes				
Name of cavity			Principal contents	Membranous lining
Dorsal body cavity	Cranial cavity		Brain	Meninges
	Vertebral canal		Spinal cord	Meninges
Ventral body cavity	Thoracic cavity		Heart, Lungs	Pericardium Pleural cavity
	Abdominopelvic cavity	Abdominal cavity	Digestive organs, spleen, kidneys	Peritoneum
		Pelvic cavity	Bladder, reproductive organs	Peritoneum

Table 4: Human body cavities and membranes

Ventral body cavity

The thoracic cavity consists of three cavities that fill the interior area of the chest.

- The two pleural cavities are situated on both sides of the body, anterior to the spine and lateral to the breastbone.
- The superior mediastinum is a wedge-shaped cavity located between the superior regions of the two thoracic cavities.
- The pericardial cavity within the mediastinum is located at the center of the chest below the superior mediastinum. The pericardial cavity roughly outlines the shape of the heart.

The diaphragm divides the thoracic and the abdominal cavities. The abdominal cavity occupies the entire lower half of the trunk, anterior to the spine. Just under the abdominal cavity, anterior to the buttocks, is the pelvic cavity. The pelvic cavity is funnel shaped and is located inferior and anterior to the abdominal cavity.

Together the abdominal and pelvic cavity can be referred to as the abdominopelvic cavity while the thoracic, abdominal, and pelvic cavities together can be referred to as the ventral body cavity. Subdivisions of the Posterior (Dorsal) and Anterior (Ventral) Cavities The anterior (ventral) cavity has two main subdivisions: the thoracic cavity and the abdominopelvic cavity. The thoracic cavity is the more superior subdivision of the anterior cavity, and it is enclosed by the rib cage.

The thoracic cavity contains the lungs and the heart, which is located in the mediastinum. The diaphragm forms the floor of the thoracic cavity and separates it from the more inferior abdominopelvic cavity. The abdominopelvic cavity is the largest cavity in the body. Although no membrane physically divides the abdominopelvic cavity, it can be useful to distinguish between the abdominal cavity, the division that houses the digestive organs, and the pelvic cavity, the division that houses the organs of reproduction.

Dorsal body cavity

The cranial cavity is a large, bean-shaped cavity filling most of the upper skull where the brain is located. The spinal cavity is a very narrow, thread-like cavity running from the cranial cavity down the entire length of the spinal cord.

Together the cranial cavity and spinal (or vertebral) cavity can be referred to as the dorsal body cavity. In the posterior (dorsal) cavity, the cranial cavity houses the brain, and the spinal cavity encloses the spinal cord. Just as the brain and spinal cord make up a continuous, uninterrupted structure, the cranial and spinal cavities that house them are also continuous. The brain and spinal cord are protected by the bones of the skull and vertebral column and by cerebrospinal fluid, a colorless fluid produced by the brain, which cushions the brain and spinal cord within the posterior (dorsal) cavity.

17 **Development**

At the end of the third week, the neural tube, which is a fold of one of the layers of the trilaminar germ disc, called the ectoderm, appears. This layer elevates and closes dorsally, while the gut tube rolls up and closes ventrally to create a "tube on top of a tube."
The mesoderm, which is another layer of the trilaminar germ disc, holds the tubes together and the lateral plate mesoderm, the middle layer of the germ disc, splits to form a visceral layer associated with the gut and a parietal layer, which along with the overlying ectoderm, forms the lateral body wall. The space between the visceral and parietal layers of lateral plate mesoderm is the primitive body cavity. When the lateral body wall folds, it moves ventrally and fuses at the midline. The body cavity closes, except in the region of the connecting stalk. Here, the gut tube maintains an attachment to the yolk sac. The yolk sac is a membranous sac attached to the embryo, which provides nutrients and functions as the circulatory system of the very early embryo.

The lateral body wall folds, pulling the amnion in with it so that the amnion surrounds the embryo and extends over the connecting stalk, which becomes the umbilical cord, which connects the fetus with the placenta. If the ventral body wall fails to close, ventral body wall defects can result, such as ectopia cordis, a congenital malformation in which the heart is abnormally located outside the thorax. Another defect is gastroschisis, a congenital defect in the anterior abdominal wall through which the abdominal contents freely protrude. Another possibility is bladder exstrophy, in which part of the urinary bladder is present outside the body. In normal circumstances, the parietal mesoderm will form the parietal layer of serous membranes lining the outside (walls) of the peritoneal, pleural, and pericardial cavities. The visceral layer will form the visceral layer of the serous membranes covering the lungs, heart, and abdominal organs. These layers are continuous at the root of each organ as the organs lie in their respective cavities. The peritoneum, a serum membrane that forms the lining of the abdominal cavity, forms in the gut layers and in places mesenteries extend from the gut as double layers of peritoneum. Mesenteries provide a pathway for vessels, nerves, and lymphatics to the organs. Initially, the gut tube from the caudal end of the foregut to the end of the hindgut is suspended from the dorsal body wall by dorsal mesentery. Ventral mesentery, derived from the septum transversum, exists only in the region of the terminal part of the esophagus, the stomach, and the upper portion of the duodenum.

These cavities contain and protect delicate internal organs, and the ventral cavity allows for significant changes in the size and shape of the organs as they perform their functions.

Anatomical structures are often described in terms of the cavity in which they reside. The body maintains its internal organization by means of membranes, sheaths, and other structures that separate compartments.

The lungs, heart, stomach, and intestines, for example, can expand and contract without distorting other tissues or disrupting the activity of nearby organs. The ventral cavity includes the thoracic and abdominopelvic cavities and their subdivisions. The dorsal cavity includes the cranial and spinal cavities.

1.2.4.4: Practical activity

Using the mannequin provided, identify the various cavities in the human body.

1.2.4.5.: Self Assessment

a). Highlight three components of the thoracic cavity.
b). Describe the dorsal cavity.
c). Highlight the functions of the body cavities.
d). what is the function of the peritoneum?

1.2.4.5.1: Model Answers

a). Highlight three components of the thoracic cavity.

- The two pleural cavities are situated on both sides of the body, anterior to the spine and lateral to the breastbone.
- The superior mediastinum is a wedge-shaped cavity located between the superior regions of the two thoracic cavities.
- The pericardial cavity within the mediastinum is located at the center of the chest below the superior mediastinum. The pericardial cavity roughly outlines the shape of the heart.

b). Describe the dorsal cavity.

The cranial cavity is a large, bean-shaped cavity filling most of the upper skull where the brain is located. The spinal cavity is a very narrow, thread-like cavity running from the cranial cavity down the entire length of the spinal cord.

Together the cranial cavity and spinal (or vertebral) cavity can be referred to as the dorsal body cavity. In the posterior (dorsal) cavity, the cranial cavity houses the brain, and the spinal cavity

encloses the spinal cord. Just as the brain and spinal cord make up a continuous, uninterrupted structure, the cranial and spinal cavities that house them are also continuous. The brain and spinal cord are protected by the bones of the skull and vertebral column and by cerebrospinal fluid, a colorless fluid produced by the brain, which cushions the brain and spinal cord within the posterior (dorsal) cavity.

c). Highlight the functions of the body cavities.

These cavities contain and protect delicate internal organs, and the ventral cavity allows for significant changes in the size and shape of the organs as they perform their functions.

Anatomical structures are often described in terms of the cavity in which they reside. The body maintains its internal organization by means of membranes, sheaths, and other structures that separate compartments.

The lungs, heart, stomach, and intestines, for example, can expand and contract without distorting other tissues or disrupting the activity of nearby organs. The ventral cavity includes the thoracic and abdominopelvic cavities and their subdivisions. The dorsal cavity includes the cranial and spinal cavities.

d). what is the function of the peritoneum?

The peritoneum, a serum membrane that forms the lining of the abdominal cavity, forms in the gut layers and in places mesenteries extend from the gut as double layers of peritoneum.

1.2.4.5: Tools, equipment, supplies and materials

Anatomy and physiology lab
Anatomical models and mannequins
Magnifying lens

1.2.4.6: References

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1.2.5: Learning Outcome 5: Demonstrate knowledge of the human skeletal system

1.2.5.1: Introduction to the learning outcome.

This learning outcome specifies the competencies required to identify divisions of the human skeleton, outline components of the axial skeleton as well as describe components of the appendicular skeleton. Types of bones are also described.

1.2.5.2:Performance Standard

Divisions of the human skeleton outlined
Components of the axial skeleton are described
Components of the appendicular skeleton are described
Types of bones are described

1.2.5.3: Information Sheet

Introduction

Human skeleton, the internal skeleton that serves as a framework for the body. This framework consists of many individual bones and cartilages. There also are bands of fibrous connective tissue the ligaments and the tendons in intimate relationship with the parts of the skeleton. This article is concerned primarily with the gross structure and the function of the skeleton of the normal human adult.

Functions of the skeletal System

The primary organs of the human skeletal system – bones -lie buried within the muscles and other soft tissues. Bones are living organs and have the ability to grow.

The six main functions of the skeletal system

Support – Bones form the body's supporting framework. All of the softer tissues of the body hang from the skeletal system.

Protection – Hard, bony "boxes" protect delicate structures enclosed within them. For example,
our skull protects our brain. Our breastbone and ribs protect our heart and lungs. Our pelvis protects our intestines and bladder and, in females, our uterus and during pregnancy the developing foetus.

Movement - Muscles are anchored firmly to bones. As muscles contract and shorten, they pull on bones and thereby move them.

Mineral storage – Bones act as reserves of minerals important for our body, most notably calcium and phosphorus. Bones maintain homeostasis of blood calcium.

Growth factor storage - Bones store important growth factors, for example fibroblast growth factor, which acts on the kidney to reduce phosphate reabsorption. In this function, our bones are an endocrine organ.

Hemopoiesis – This is the process of blood formation. Blood cells are formed in red bone marrow.

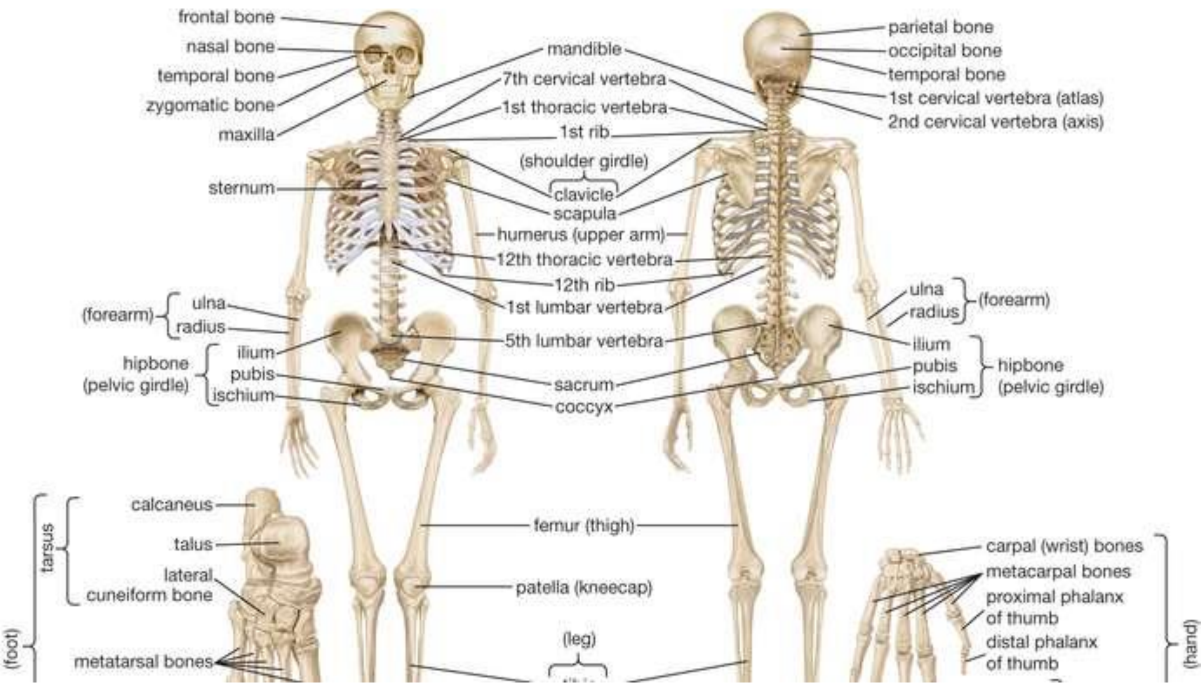


Figure 3: Skeletal system

The human skeleton, like that of other vertebrates, consists of two principal subdivisions, each with origins distinct from the others and each presenting certain individual features.

These are

(1) the axial, comprising the vertebral column the spine and much of the skull, and (2)the appendicular, to which the pelvic (hip) and pectoral (shoulder) girdles and the bones and cartilages of the limbs belong. Discussed in this article as part of the axial skeleton is a

third subdivision, the visceral, comprising the lower jaw, some elements of the upper jaw, and the branchial arches, including the hyoid bone.

When one considers the relation of these subdivisions of the skeleton to the soft parts of the human body such as the nervous system, the digestive system, the respiratory system, the cardiovascular system, and the voluntary muscles of the muscle system it is clear that the functions of the skeleton are of three different types: support, protection, and motion. Of these functions, support is the most primitive and the oldest; likewise, the axial part of the skeleton was the first to evolve. The vertebral column, corresponding to the notochord in lower organisms, is the main support of the trunk.

The central nervous system lies largely within the axial skeleton, the brain being well protected by the cranium and the spinal cord by the vertebral column, by means of the bony neural arches (the arches of bone that encircle the spinal cord) and the intervening ligaments.

Protection of the heart, lungs, and other organs and structures in the chest creates a problem somewhat different from that of the central nervous system. These organs, the function of which involves motion, expansion, and contraction, must have a flexible and elastic protective covering. Such a covering is provided by the bony thoracic basket, or rib cage, which forms the skeleton of the wall of the chest, or thorax. The connection of the ribs to the breastbone the sternum is in all cases a secondary one, brought about by the relatively pliable rib (costal) cartilages. The small joints between the ribs and the vertebrae permit a gliding motion of the ribs on the vertebrae during breathing and other activities. The motion is limited by the ligamentous attachments between ribs and vertebrae.

The third general function of the skeleton is that of motion. The great majority of the skeletal muscles are firmly anchored to the skeleton, usually to at least two bones and in some cases to many bones. Thus, the motions of the body and its parts, all the way from the lunge of the football player to the delicate manipulations of a handicraft artist or of the use of complicated instruments by a scientist, are made possible by separate and individual engineering arrangements between muscle and bone.

In this article the parts of the skeleton are described in terms of their sharing in these functions. The disorders and injuries that can affect the human skeleton are described in the article bone disease.

Axial And Visceral Skeleton

19 The cranium

20 The cranium is the part of the skull that encloses the brain is sometimes called the braincase, but its intimate relation to the sense organs for sight, sound, smell, and taste and to other structures makes such a designation somewhat misleading.

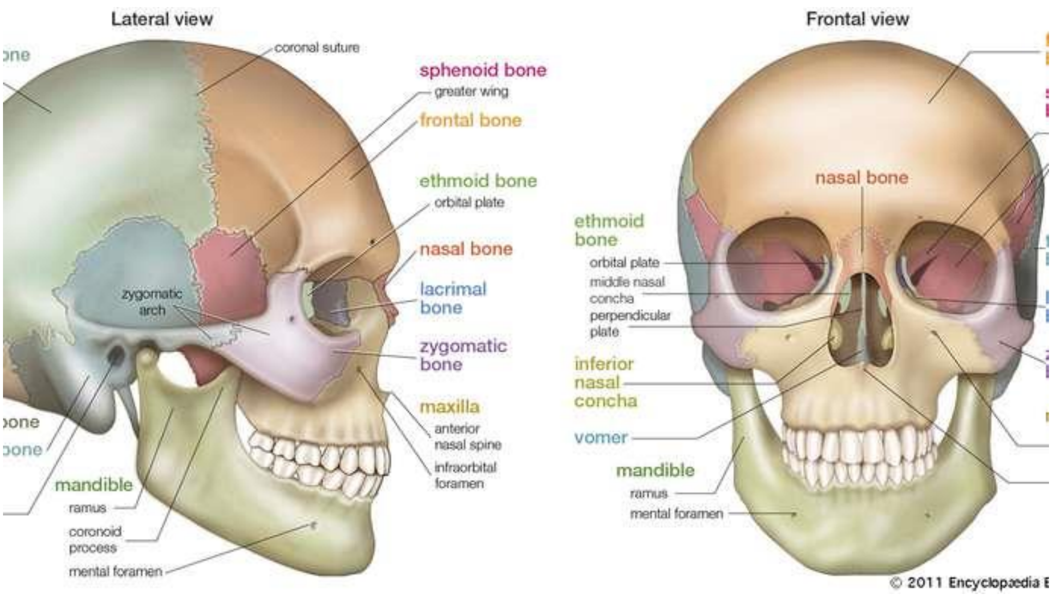


Figure 4: The cranium

21 Development of cranial bones

The cranium is formed of bones of two different types of developmental origin—the cartilaginous, or substitution, bones, which replace cartilages preformed in the general shape of the bone; and membrane bones, which are laid down within layers of connective tissue. For the most part, the substitution bones form the floor of the cranium, while membrane bones form the sides and roof.

The range in the capacity of the cranial cavity is wide but is not directly proportional to the size of the skull, because there are variations also in the thickness of the bones and in the size of the air pockets, or sinuses. The cranial cavity has a rough, uneven floor, but its landmarks and details of structure generally are consistent from one skull to another.

The cranium forms all the upper portion of the skull, with the bones of the face situated beneath its forward part. It consists of a relatively few large bones, the frontal bone, the sphenoid bone, two temporal bones, two parietal bones, and the occipital bone. The frontal bone underlies the forehead region and extends back to the coronal suture, an arching line that separates the frontal bone from the two parietal bones, on the sides of the cranium. In front, the frontal bone forms a joint with the two small bones of the bridge of the nose and with the zygomatic bone (which forms part of the cheekbone; The facial bones and their complex functions), the sphenoid, and the maxillary bones. Between the nasal and zygomatic bones, the horizontal portion of the frontal bone extends back to form a part of the roof of the eye socket, or orbit; it thus serves an important protective function for the eye and its accessory structures.

Each parietal bone has a generally four-sided outline. Together they form a large portion of the side walls of the cranium. Each adjoins the frontal, the sphenoid, the temporal, and the occipital bones and its fellow of the opposite side. They are almost exclusively cranial bones, having less relation to other structures than the other bones that help to form the cranium.

Interior of the cranium

The interior of the cranium shows a multitude of details, reflecting the shapes of the softer structures that are in contact with the bones.

The internal surface of the vault is relatively uncomplicated. In the midline front to back, along the sagittal suture, the seam between the two parietal bones, is a shallow depression—the groove for the superior longitudinal venous sinus, a large channel for venous blood. A number of depressions on either side of it mark the sites of the pacchionian bodies, structures that permit the venous system to absorb cerebrospinal fluid. The large thin-walled venous sinuses all lie within the cranial cavity. While they are thus protected by the cranium, in many places they are so close beneath the bones that a fracture or a penetrating wound may tear the sinus wall and lead to bleeding. The blood frequently is trapped beneath the outermost and toughest brain covering, the dura mater, in a mass called a subdural hematoma.

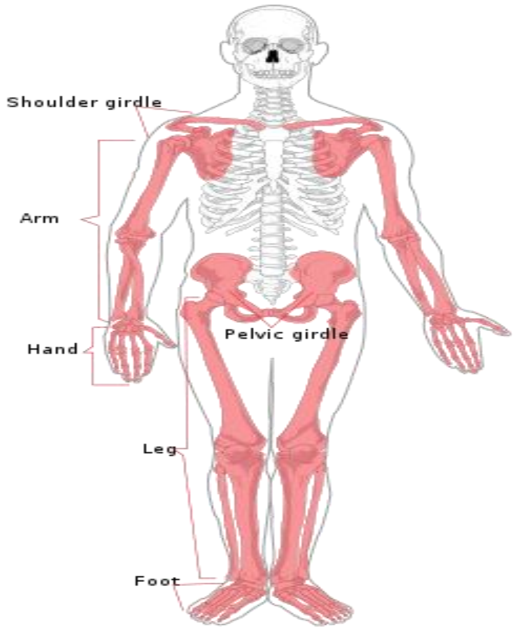
Conspicuous markings on the internal surface of the projection of the sphenoid, called the greater wing, and on the internal surfaces of the parietal and temporal bones are formed by the middle meningeal artery and its branches, which supply blood to the brain coverings. Injury to these vessels may lead to extradural hematoma, a mass of blood between the dura mater and the bone.

In contrast to the vault and sides of the cranium, the base presents an extremely complicated aspect. It is divided into three major depressions, or fossae, in a descending stair-step arrangement from front to back. The fossae are divided strictly according to the borders of the bones of the cranium but are related to major portions of the brain. The anterior cranial fossa serves as the bed in which rest the frontal lobes of the cerebrum, the large forward part of the brain. The middle cranial fossa, sharply divided into two lateral halves by a central eminence of bone, contains the temporal lobes of the cerebrum. The posterior cranial fossa serves as a bed for the hemispheres of the cerebellum (a mass of brain tissue behind the brain stem and beneath the rear portion of the cerebrum) and for the front and middle portion of the brain stem. Major portions of the brain are thus partially enfolded by the bones of the cranial wall.

There are openings in the three fossae for the passage of nerves and blood vessels, and the markings on the internal surface of the bones are from the attachments of the brain coverings—the meninges—and venous sinuses and other blood vessels.

Appendicular skeleton

The appendicular skeleton is the portion of the skeleton of vertebrates consisting of the bones that support the appendages. There are 126 bones. The appendicular skeleton includes the skeletal elements within the limbs, as well as supporting shoulder girdle pectoral and pelvic girdle. The word appendicular is the adjective of the noun appendage, which itself means a part that is joined to something larger.



Of the 206 bones in the human skeleton, the appendicular skeleton comprises 126. Functionally it is involved in locomotion (lower limbs) of the axial skeleton and manipulation of objects in the environment (upper limbs).

The appendicular skeleton forms during development from cartilage, by the process of endochondral ossification.

The appendicular skeleton is divided into six major regions:

1. Shoulder girdles (4 bones) - Left and right clavicle (2) and scapula (2).
2. Arms and forearms (6 bones) - Left and right humerus (2) (arm), ulna (2) and radius (2) (forearm).
3. Hands (54 bones) - Left and right carpals (16) (wrist), metacarpals (10), proximal phalanges (10), intermediate phalanges (8) and distal phalanges (10).
4. Pelvis (2 bones) - left hip bone and right hip bone (2).
5. Thighs and legs (8 bones) - Left and right femur (2) (thigh), patella (2) (knee), tibia (2) and fibula (2) (leg).
6. Feet and ankles (52 bones) - Left and right tarsals (14) (ankle), metatarsals (10), proximal phalanges (10), intermediate phalanges (8) and distal phalanges (10).

It is important to realize that through anatomical variation it is common for the skeleton to have many accessory bones (sutural bones in the skull, cervical ribs, lumbar ribs and even extra lumbar vertebrae).

The appendicular skeleton of 126 bones and the axial skeleton of 80 bones together form the complete skeleton of 206 bones in the human body. Unlike the axial skeleton, the appendicular skeleton is unfused. This allows for a much greater range of motion.

Types of bones

Bone, or osseous tissue, is a connective tissue that constitutes the endoskeleton. It contains specialized cells and a matrix of mineral salts and collagen fibers.

The mineral salts primarily include hydroxyapatite, a mineral formed from calcium phosphate. Calcification is the process of deposition of mineral salts on the collagen fiber matrix that crystallizes and hardens the tissue. The process of calcification only occurs in the presence of collagen fibers.

The bones of the human skeleton are classified by their shape: long bones, short bones, flat bones, sutural bones, sesamoid bones, and irregular bones.

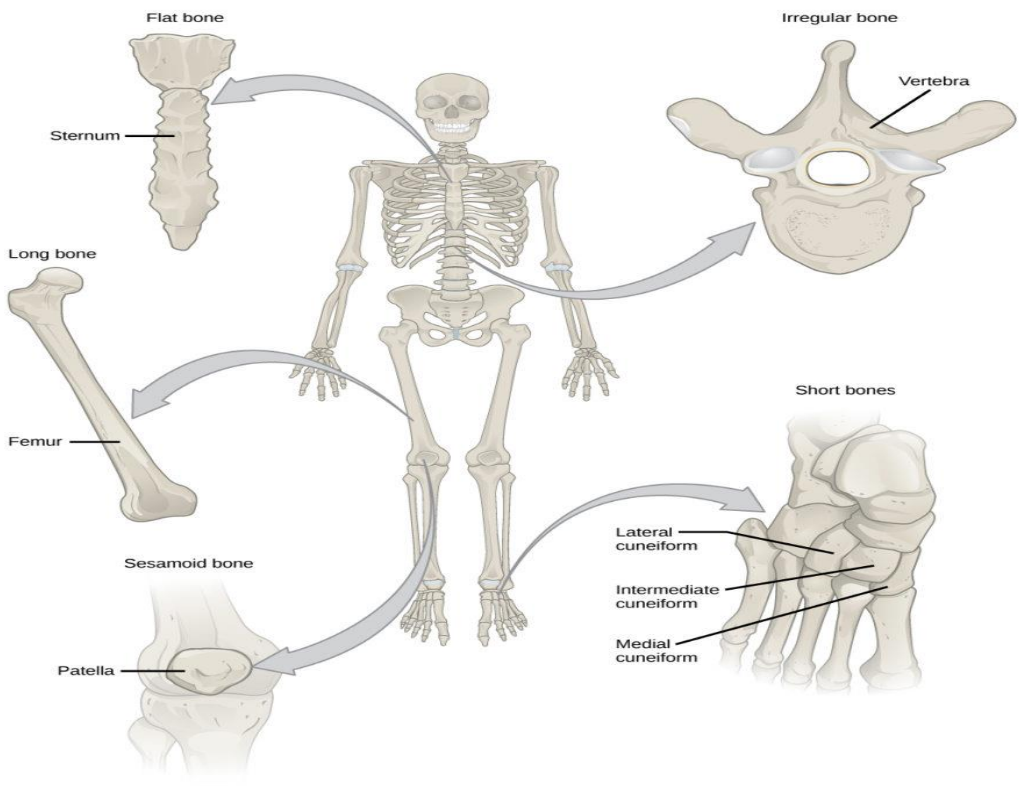


Figure 5:Types of bones

Long bones are longer than they are wide and have a shaft and two ends. The **diaphysis**, or central shaft, contains bone marrow in a marrow cavity. The rounded ends, the **epiphyses**, are covered with articular cartilage and are filled with red bone marrow, which produces blood cells. Most of the limb bones are long bones—for example, the femur, tibia, ulna, and radius. Exceptions to this include the patella and the bones of the wrist and ankle.

Short bones, or cuboidal bones, are bones that are the same width and length, giving them a cube-like shape. For example, the bones of the wrist (carpals) and ankle (tarsals) are short bones.

Flat bones are thin and relatively broad bones that are found where extensive protection of organs is required or where broad surfaces of muscle attachment are required. Examples of flat bones are the sternum (breast bone), ribs, scapulae (shoulder blades), and the roof of the skull.

Irregular bones are bones with complex shapes. These bones may have short, flat, notched, or ridged surfaces. Examples of irregular bones are the vertebrae, hip bones, and several skull bones.

Sesamoid bones are small, flat bones and are shaped similarly to a sesame seed. The patellae are sesamoid bones. Sesamoid bones develop inside tendons and may be found near joints at the knees, hands, and feet.

Sutural bones are small, flat, irregularly shaped bones. They may be found between the flat bones of the skull. They vary in number, shape, size, and position.

1.2.5.4: Practical Activity

Using the skeletal model provided in the anatomy lab, describe the various components of the human skeletal system.

1.2.5.5: Self Assessment

a). Highlight types of bones

b). Highlight the six major regions of the appendicular skeleton

c). What are the six functions of the skeletal system?

1.2.5.4.1: Model Answers

a). Highlight types of bones.

Long bones are longer than they are wide and have a shaft and two ends. The **diaphysis**, or central shaft, contains bone marrow in a marrow cavity. The rounded ends, the **epiphyses**, are covered with articular cartilage and are filled with red bone marrow, which produces blood cells. Most of the limb bones are long bones—for example, the femur, tibia, ulna, and radius. Exception sto this include the patella and the bones of the wrist and ankle.

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Sutural bones are small, flat, irregularly shaped bones. They may be found between the flat bones of the skull. They vary in number, shape, size, and position.

b). Highlight the six major regions of the appendicular skeleton

1. Shoulder girdles (4 bones) - Left and right clavicle (2) and scapula (2).
2. Arms and forearms (6 bones) - Left and right humerus (2) (arm), ulna (2) and radius (2) (forearm).
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c). what are the six functions of the skeletal system?

Support – Bones form the body's supporting framework. All of the softer tissues of the body hang from the skeletal system.
Protection – Hard, bony "boxes" protect delicate structures enclosed within them. For example,
our skull protects our brain. Our breastbone and ribs protect our heart and lungs. Our pelvis protects our intestines and bladder and, in females, our uterus and during pregnancy the developing foetus.
Movement - Muscles are anchored firmly to bones. As muscles contract and shorten, they pull on bones and thereby move them.
Mineral storage – Bones act as reserves of minerals important for our body, most notably calcium and phosphorus. Bones maintain homeostasis of blood calcium.
Growth factor storage - Bones store important growth factors, for example fibroblast growth factor, which acts on the kidney to reduce phosphate reabsorption. In this function, our bones are an endocrine organ.
Hemopoiesis – This is the process of blood formation. Blood cells are formed in red bone marrow.

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1.2.6: Learning outcome 6: Demonstrate knowledge of the major muscles of the body and their functions

1.2.6.1: Introduction to learning outcome

This learning outcome specifies the competencies required to identify the structure of muscles their types and functions.

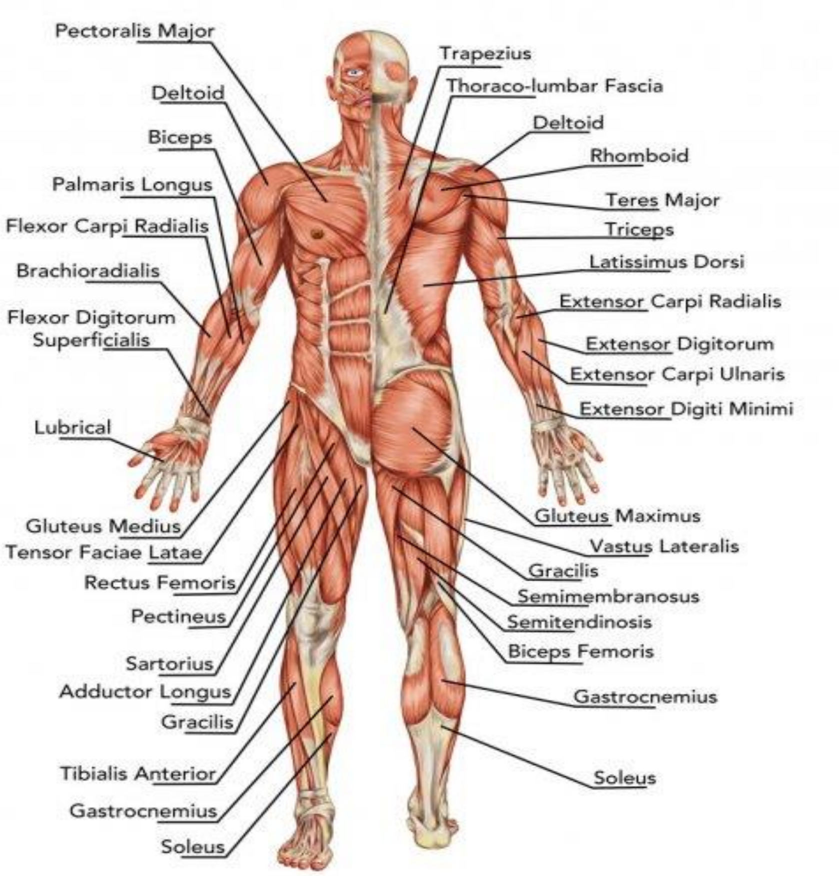
1.2.6.2: Performance Standards

1. **Structure** of a muscle outlined
2. **Types** of muscles outlined
3. **Functions** of muscles are listed

1.2.6.3:Information Sheet

The muscular system is an organ system consisting of skeletal, smooth and cardiac muscles. It permits movement of the body, maintains posture and circulates blood throughout the body. The muscular systems in vertebrates are controlled through the nervous system although some muscles (such as the cardiac muscle) can be completely autonomous. Together with the skeletal system, it forms the musculoskeletal system, which is responsible for movement of the human body.

Muscles



There are three distinct types of muscles: skeletal muscles, cardiac or heart muscles, and smooth (non-striated) muscles. Muscles provide strength, balance, posture, movement and heat for the body to keep warm.

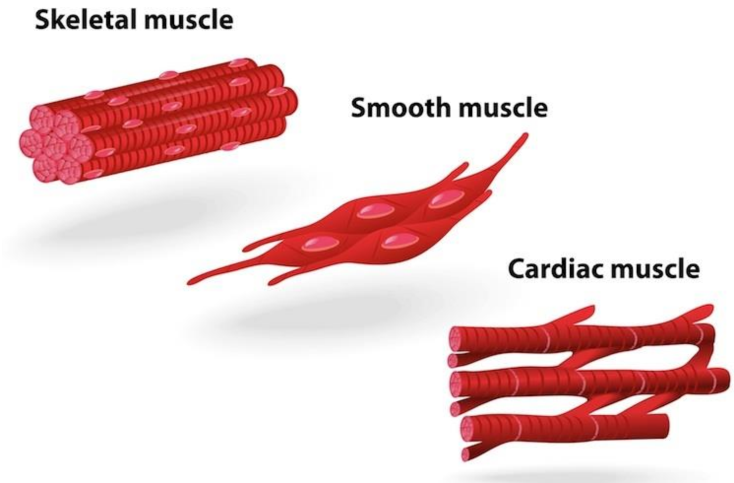


Fig: Muscle structure under the microscope

Skeletal Muscle

Skeletal muscles, like other striated muscles, are composed of myocytes, or muscle fibers, which are in turn composed of myofibrils, which are composed of sarcomeres, the basic building block of striated muscle tissue. Upon stimulation by an action potential, skeletal muscles perform a coordinated contraction by shortening each sarcomere. The best proposed model for understanding contraction is the sliding filament model of muscle contraction.

Within the sarcomere, actin and myosin fibers overlap in a contractile motion towards each other. Myosin filaments have club-shaped heads that project toward the actin filaments.

Fig: Skeletal muscle

Larger structures along the myosin filament called myosin heads are used to provide attachment points on binding sites for the actin filaments. The myosin heads move in a coordinated style; they swivel toward the center of the sarcomere, detach and then reattach to the nearest active site of the actin filament. This is called a ratchet type drive system

This process consumes large amounts of adenosine triphosphate (ATP), the energy source of the cell. ATP binds to the cross bridges between myosin heads and actin filaments. The release of energy powers the swiveling of the myosin head. When ATP is used, it becomes adenosine diphosphate (ADP), and since muscles store little ATP, they must continuously replace the discharged ADP with ATP. Muscle tissue also contains a stored supply of a fast acting recharge chemical, creatine phosphate, which when necessary can assist with the rapid regeneration of ADP into ATP.

Calcium ions are required for each cycle of the sarcomere. Calcium is released from the sarcoplasmic reticulum into the sarcomere when a muscle is stimulated to contract. This calcium uncovers the actin binding sites. When the muscle no longer needs to contract, the calcium ions are pumped from the sarcomere and back into storage in the sarcoplasmic reticulum.

There are approximately 639 skeletal muscles in the human body.

Cardiac muscle

Heart muscles are distinct from skeletal muscles because the muscle fibers are laterally connected to each other. Furthermore, just as with smooth muscles, their movement is involuntary. Heart muscles are controlled by the sinus node influenced by the autonomic nervous system.

Smooth muscle

Smooth muscles are controlled directly by the autonomic nervous system and are involuntary, meaning that they are incapable of being moved by conscious thought. Functions such as heartbeat and lungs (which are capable of being willingly controlled, be it to a limited extent) are involuntary muscles but are not smooth muscles.

Physiology

Contraction

Neuromuscular junctions are the focal point where a motor neuron attaches to a muscle. Acetylcholine, (a neurotransmitter used in skeletal muscle contraction) is released from the axon terminal of the nerve cell when an action potential reaches the microscopic junction called a synapse.

A group of chemical messengers cross the synapse and stimulate the formation of electrical changes, which are produced in the muscle cell when the acetylcholine binds to receptors on its surface. Calcium is released from its storage area in the cell's sarcoplasmic reticulum. An impulse from a nerve cell causes calcium release and brings about a single, short muscle contraction called a muscle twitch. If there is a problem at the neuromuscular junction, a very prolonged contraction may occur, such as the muscle contractions that result from tetanus. Also, a loss of function at the junction can produce paralysis.

Skeletal muscles are organized into hundreds of motor units, each of which involves a motor neuron, attached by a series of thin finger-like structures called axon terminals. These attach to and control discrete bundles of muscle fibers. A coordinated and fine tuned response to a specific circumstance will involve controlling the precise number of motor units used. While individual muscle units contract as a unit, the entire muscle can contract on a predetermined basis due to the structure of the motor unit.

Motor unit coordination, balance, and control frequently come under the direction of the cerebellum of the brain. This allows for complex muscular coordination with little conscious effort, such as when one drives a car without thinking about the process.

Tendon

A tendon is a piece of connective tissue that connects a muscle to a bone. When a muscle contracts, it pulls against the skeleton to create movement. A tendon connects this muscle to a bone, making this function possible.

Aerobic and anaerobic muscle activity

At rest, the body produces the majority of its ATP aerobically in the mitochondria without producing lactic acid or other fatiguing byproducts. During exercise, the method of ATP production varies depending on the fitness of the individual as well as the duration and intensity of exercise. At lower activity levels, when exercise continues for a long duration (several minutes or longer), energy is produced aerobically by combining oxygen with carbohydrates and fats stored in the body.

During activity that is higher in intensity, with possible duration decreasing as intensity increases, ATP production can switch to anaerobic pathways, such as the use of the creatine phosphate and the phosphagen system or anaerobic glycolysis. Aerobic ATP production is biochemically much slower and can only be used for long-duration, low-intensity exercise, but produces no fatiguing waste products that can not be removed immediately from the sarcomere and the body, and it results in a much greater number of ATP molecules per fat or carbohydrate molecule. Aerobic training allows the oxygen delivery system to be more efficient, allowing aerobic metabolism to begin quicker.

Anaerobic ATP production produces ATP much faster and allows near-maximal intensity exercise, but also produces significant amounts of lactic acid which renders high-intensity exercise unsustainable for more than several minutes. The phosphagen system is also anaerobic. It allows for the highest levels of exercise intensity, but intramuscular stores of phosphocreatine are very limited and can only provide energy for exercises lasting up to ten seconds. Recovery is very quick, with full creatine stores regenerated within five minutes.

1.2.6.4: Practical Activity

By use of a microscope oberve three pre-prepared slides containing skeletal, smooth and cardiac muscle cells. Draw well labeled diagrams to illustrate them.

1.2.6.5: Self Assessment

a). Highlight aerobic and anaerobic muscle activity

b).Describe the cardiac muscle

1.2.6.4.1:Model answers

a). Highlight aerobic and anaerobic muscle activity

At rest, the body produces the majority of its ATP aerobically in the mitochondria without producing lactic acid or other fatiguing byproducts. During exercise, the method of ATP production varies depending on the fitness of the individual as well as the duration and intensity of exercise. At lower activity levels, when exercise continues for a long duration (several minutes or longer), energy is produced aerobically by combining oxygen with carbohydrates and fats stored in the body.

During activity that is higher in intensity, with possible duration decreasing as intensity increases, ATP production can switch to anaerobic pathways, such as the use of the creatine phosphate and the phosphagen system or anaerobic glycolysis. Aerobic ATP production is biochemically much slower and can only be used for long-duration, low-intensity exercise, but produces no fatiguing waste products that cannot be removed immediately from the sarcomere and the body, and it results in a much greater number of ATP molecules per fat or carbohydrate molecule. Aerobic training allows the oxygen delivery system to be more efficient, allowing aerobic metabolism to begin quicker.

Anaerobic ATP production produces ATP much faster and allows near-maximal intensity exercise, but also produces significant amounts of lactic acid which renders high-intensity exercise unsustainable for more than several minutes. The phosphagen system is also anaerobic. It allows for the highest levels of exercise intensity, but intramuscular stores of phosphocreatine are very limited and can only provide energy for exercises lasting up to ten seconds. Recovery is very quick, with full creatine stores regenerated within five minutes.

b). Describe the cardiac muscle

Heart muscles are distinct from skeletal muscles because the muscle fibers are laterally connected to each other. Furthermore, just as with smooth muscles, their movement is involuntary. Heart muscles are controlled by the sinus node influenced by the autonomic nervous system.

1.2.6.5: References

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1.2.7: Learning outcome 7: Demonstrate knowledge of the body joints

1.2.7.1: Introduction to learning outcome

This learning outcome specifies the competencies required to identify the structure of joints, their types and functions.

1.2.7.2: Performance standards

- **Structure** of joints outlined
- **Types** of joints outlined
- **Functions** of joints are described

1.2.7.3: Information Sheet

Introduction

The point at which two or more bones meet is called a joint, or articulation. Joints are responsible for movement, such as the movement of limbs, and stability, such as the stability found in the bones of the skull.

There are two ways to classify joints: on the basis of their structure or on the basis of their function. The structural classification divides joints into bony, fibrous, cartilaginous, and synovial joints depending on the material composing the joint and the presence or absence of a cavity in the joint.

Fibrous Joints

The bones of **fibrous joints** are held together by fibrous connective tissue. There is no cavity, or space, present between the bones and so most fibrous joints do not move at all, or are only capable of minor movements. There are three types of fibrous joints: sutures, syndesmoses, and gomphoses. **Sutures** are found only in the skull and possess short fibers of connective tissue that hold the skull bones tightly in place.

Syndesmoses are joints in which the bones are connected by a band of connective tissue, allowing for more movement than in a suture. An example of a syndesmosis is the joint of the tibia and fibula in the ankle. The amount of movement in these types of joints is determined by the length of the connective tissue fibers. Gomphoses occur between teeth and their sockets;

the term refers to the way the tooth fits into the socket like a peg. The tooth is connected to the socket by a connective tissue referred to as the periodontal ligament.

Cartilaginous Joints

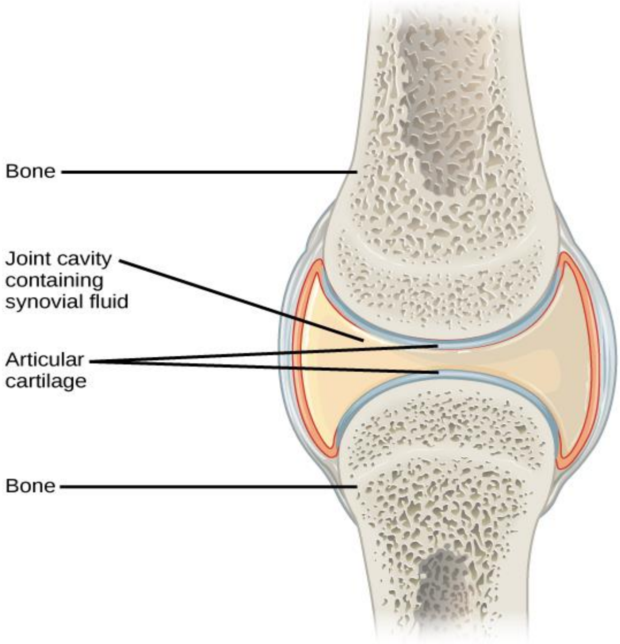

Figure 6. Synovial joints

Cartilaginous joints are joints in which the bones are connected by cartilage. There are two types of cartilaginous joints: synchondroses and symphyses. In a **synchondrosis**, the bones are joined by hyaline cartilage. Synchondroses are found in the epiphyseal plates of growing bones in children. In **symphyses**, hyaline cartilage covers the end of the bone but the connection between bones occurs through fibrocartilage. Symphyses are found at the joints between vertebrae. Either type of cartilaginous joint allows for very little movement.

Synovial Joints

Synovial joints are the only joints that have a space between the adjoining bones. This space is referred to as the synovial (or joint) cavity and is filled with synovial fluid. Synovial fluid lubricates the joint, reducing friction between the bones and allowing for greater movement. The ends of the bones are covered with articular cartilage, a hyaline cartilage, and the entire joint is surrounded by an articular capsule composed of connective tissue that allows movement of the joint while resisting dislocation. Articular capsules may also possess ligaments that hold the bones together. Synovial joints are capable of the greatest movement of the three structural joint types; however, the more mobile a joint, the weaker the joint. Knees, elbows, and shoulders are examples of synovial joints.

22 Classification of Joints on the Basis of Function

The functional classification divides joints into three categories: synarthroses, amphiarthroses, and diarthroses:

- Synarthroses are a joints that are immovable. This includes sutures, gomphoses, and synchondroses.
- Amphiarthroses are joints that allow slight movement, including syndesmoses and symphyses.
- Diarthroses are joints that allow for free movement of the joint, as in synovial joints.

23 Types of Synovial Joints

Synovial joints are further classified into six different categories on the basis of the shape and structure of the joint. The shape of the joint affects the type of movement permitted by the joint. These joints can be described as planar, hinge, pivot, condyloid, saddle, or ball-and-socket joints.

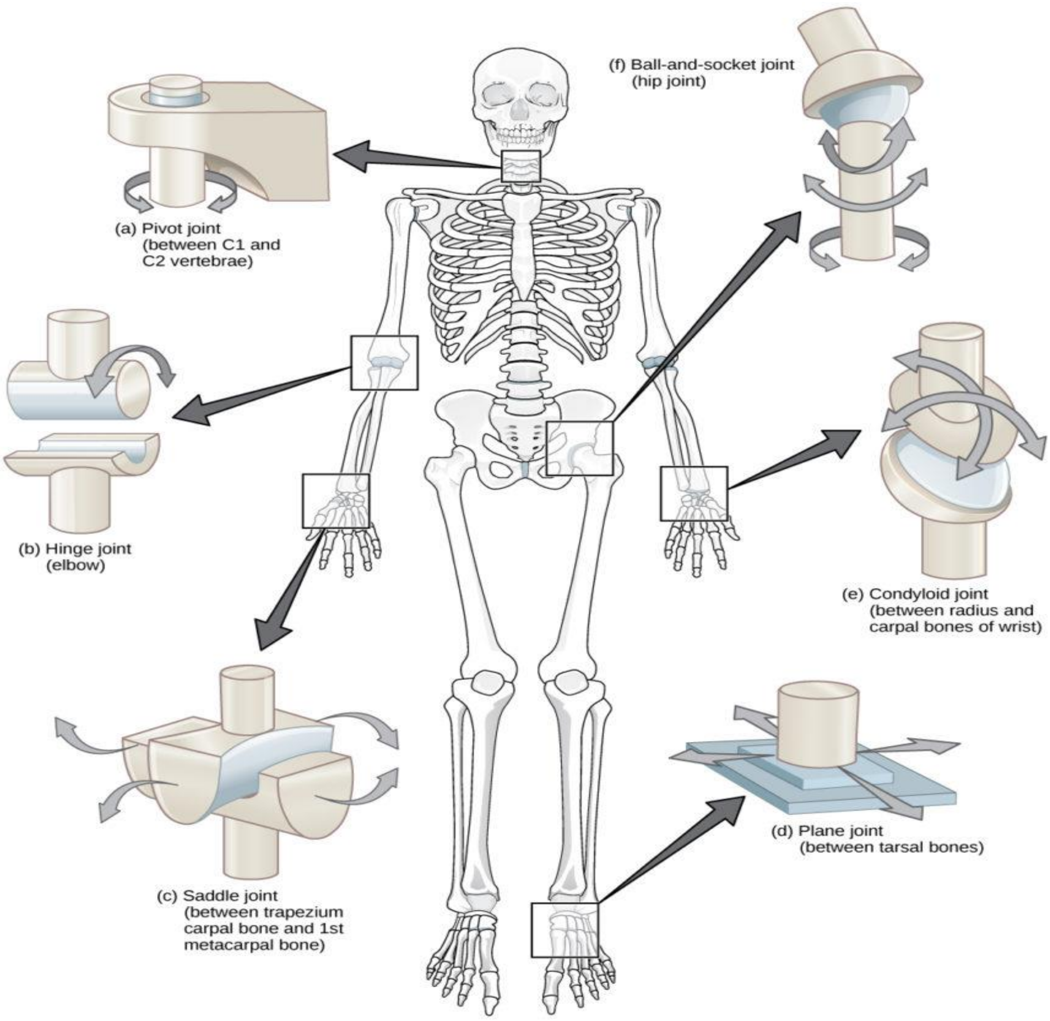

Figure 7: types of synovial joints

Planar Joints

Planar joints have bones with articulating surfaces that are flat or slightly curved faces. These joints allow for gliding movements, and so the joints are sometimes referred to as gliding joints. The range of motion is limited in these joints and does not involve rotation. Planar joints are found in the carpal bones in the hand and the tarsal bones of the foot, as well as between vertebrae.

Hinge Joints

In **hinge joints**, the slightly rounded end of one bone fits into the slightly hollow end of the other bone. In this way, one bone moves while the other remains stationary, like the hinge of a door. The elbow is an example of a hinge joint. The knee is sometimes classified as a modified hinge joint

Pivot joints

Consist of the rounded end of one bone fitting into a ring formed by the other bone. This structure allows rotational movement, as the rounded bone moves around its own axis. An example of a pivot joint is the joint of the first and second vertebrae of the neck that allows the head to move back and forth. The joint of the wrist that allows the palm of the hand to be turned up and down is also a pivot joint.

Condyloid Joints

Condyloid joints consist of an oval-shaped end of one bone fitting into a similarly oval-shaped hollow of another bone. This is also sometimes called an ellipsoidal joint. This type of joint allows angular movement along two axes, as seen in the joints of the wrist and fingers, which can move both side to side and up and down.

Saddle Joints

Saddle joints are so named because the ends of each bone resemble a saddle, with concave and convex portions that fit together. Saddle joints allow angular movements similar to condyloid joints but with a greater range of motion. An example of a saddle joint is the thumb joint, which can move back and forth and up and down, but more freely than the wrist or fingers.

Ball-and-Socket Joints

Ball-and-socket joints possess a rounded, ball-like end of one bone fitting into a cuplike socket of another bone. This organization allows the greatest range of motion, as all movement types are possible in all directions. Examples of ball-and-socket joints are the shoulder and hip joints

24 Movement at Synovial Joints

The wide range of movement allowed by synovial joints produces different types of movements. The movement of synovial joints can be classified as one of four different types: gliding, angular, rotational, or special movement.

Gliding Movement

Gliding movements occur as relatively flat bone surfaces move past each other. Gliding movements produce very little rotation or angular movement of the bones. The joints of the carpal and tarsal bones are examples of joints that produce gliding movements.

Angular Movement

Angular movements are produced when the angle between the bones of a joint changes. There are several different types of angular movements, including flexion, extension, hyperextension, abduction, adduction, and circumduction. Flexion, or bending, occurs when the angle between the bones decreases. Moving the forearm upward at the elbow or moving the wrist to move the hand toward the forearm are examples of flexion. Extension is the opposite of flexion in that the angle between the bones of a joint increases. Straightening a limb after flexion is an example of extension. Extension past the regular anatomical position is referred to as hyperextension. This includes moving the neck back to look upward, or bending the wrist so that the hand moves away from the forearm.

Abduction occurs when a bone moves away from the midline of the body. Examples of abduction are moving the arms or legs laterally to lift them straight out to the side. **Adduction** is the movement of a bone toward the midline of the body. Movement of the limbs inward after abduction is an example of adduction. **Circumduction** is the movement of a limb in a circular motion, as in moving the arm in a circular motion.

Rotational Movement

Rotational movement is the movement of a bone as it rotates around its longitudinal axis. Rotation can be toward the midline of the body, which is referred to as **medial rotation**, or away from the midline of the body, which is referred to as **lateral rotation**. Movement of the head from side to side is an example of rotation.

Special Movements

Some movements that cannot be classified as gliding, angular, or rotational are called special movements. **Inversion** involves the soles of the feet moving inward, toward the midline of the body. **Eversion** is the opposite of inversion, movement of the sole of the foot outward, away from the midline of the body. **Protraction** is the anterior movement of a bone in the horizontal plane. **Retraction** occurs as a joint moves back into position after protraction. Protraction and retraction can be seen in the movement of the mandible as the jaw is thrust outwards and then back inwards. **Elevation** is the movement of a bone upward, such as when the shoulders are shrugged, lifting the scapulae. **Depression** is the opposite of elevation—movement downward of a bone, such as after the shoulders are shrugged and the scapulae return to their normal position from an elevated position. **Dorsiflexion** is a bending at the ankle such that the toes are lifted toward the knee. **Plantar flexion** is a bending at the ankle when the heel is lifted, such as when standing on the toes. **Supination** is the movement of the radius and ulna bones of the forearm so that the palm faces forward. **Pronation** is the opposite movement, in which the palm faces backward. **Opposition** is the movement of the thumb toward the fingers of the same hand, making it possible to grasp and hold objects.

<u>1.2.7.4: Self Assessment</u>

a). Highlight four types of special movements.

b).Highlight four types of joints

<u>1.2.7.4.1: Model Answers</u>

a). Highlight four types of special movements.

. **Eversion** is the opposite of inversion, movement of the sole of the foot outward, away from the midline of the body. **Protraction** is the anterior movement of a bone in the horizontal plane. **Retraction** occurs as a joint moves back into position after protraction. Protraction and retraction can be seen in the movement of the mandible as the jaw is thrust outwards and then back inwards. **Elevation** is the movement of a bone upward, such as when the shoulders are shrugged, lifting the scapulae.

b).Highlight four types of joints.

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In **hinge joints**, the slightly rounded end of one bone fits into the slightly hollow end of the other bone. In this way, one bone moves while the other remains stationary, like the hinge of a door. The elbow is an example of a hinge joint. The knee is sometimes classified as a modified hinge joint

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1.2.8: Learning outcome 8: Demonstrate knowledge of circulatory system

1.2.8.1: Introduction to the learning outcome

This learning outcome specifies the competencies required to outline components of circulatory system, describe systemic and pulmonary circulation, describe the structure and function of blood, describe the structure and function of the heart and describe the structure and function of the blood vessels.

1.2.8.2: Performance Standards

1. Components of circulatory system are outlined
2. Systemic and pulmonary circulation are described
3. Structure and function of the blood is described
4. Structure and function of the heart is described
5. Structure and function of the blood vessels is described

1.2.8.3: Information Sheet

1. **Circulatory System -** (Cardiovascular & Lymphatic System)

Functions of the Cardiovascular System

- Distributes nutrients (e.g. glucose) and oxygen to all body cells
- Collects waste products (e.g. urea) and carbon dioxide for excretion
- Controls blood flow to the skin and extremities to increase or decrease heat loss to the environment
- Distributes hormones to target sites of action
- Aids in body defence mechanisms by delivering antibodies, platelets and leukocytes to affected areas of the body

Components of the Cardiovascular System

- The heart provides the driving force for the cardiovascular system.
- The arteries, which carry blood away from the heart, serve as distribution channels to the organs.
- The microcirculation (arterioles, capillaries and venules), which includes the capillaries, serve as the exchange region.
- The veins serve as blood reservoirs and collect the blood to return it to the heart.

The Heart as a Pump

- The heart is two pumps in series (right and left sides) that are connected by the pulmonary and systemic circulations.
- Each side of the heart is composed of two chambers: an atrium and a ventricle.
- The right atrium receives blood from tissues and organs, and the right ventricle pumps blood at low pressures through the pulmonary circulation. Blood flows from the right heart, through the pulmonary artery, to the lungs.
- The left atrium receives blood from the lungs via pulmonary veins, and the left ventricle pumps blood through the systemic circulation. Blood flows from the left heart, through the aorta, to the systemic organs except the lungs.

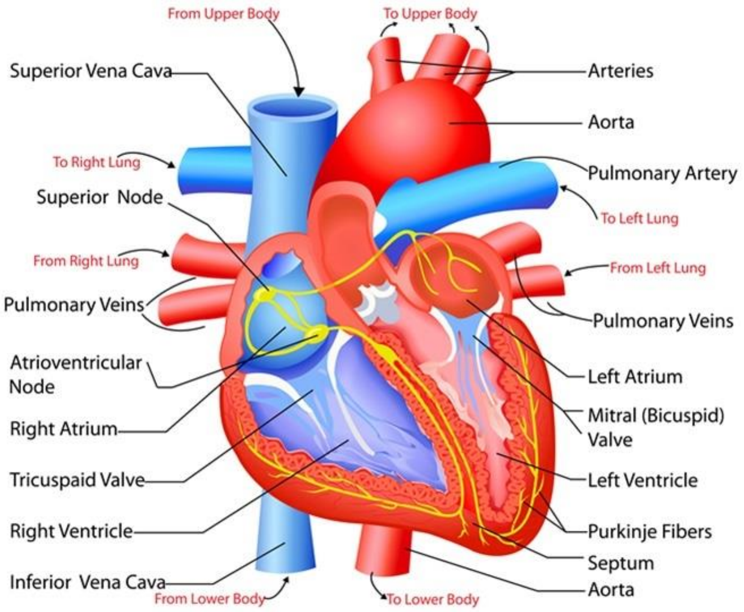

Figure 9: The human heart

Parallel Distribution to the Organs

- Although the right and left sides of the heart are connected in series, the various organs receive blood flow through parallel distribution channels.
- The **parallel arrangement** supplies organs with blood that has the **same arterial composition** (e.g. same oxygen and carbon dioxide concentrations, pH, glucose, electrolytes like sodium, potassium and calcium) and essentially the same arterial pressure.
- The **parallel arrangement** also allows for independent regulation of blood flow through different tissues as their metabolic activities change.

The systemic and pulmonary circulations

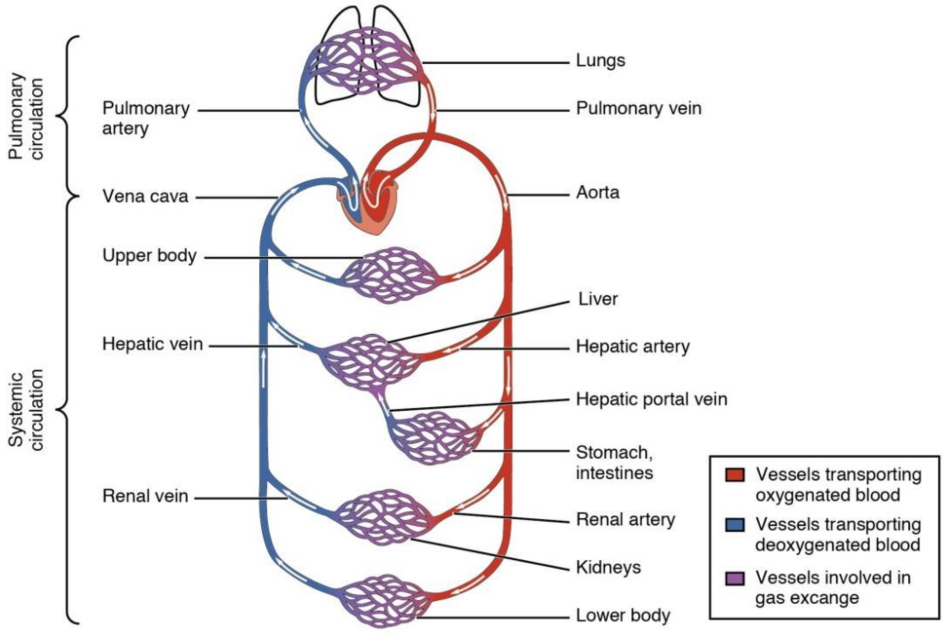

Figure 10: Parallel distribution of organs

Forces for blood flow

- For fluid flowing through a tube, flow can be represented as the average velocity of movement (cm/sec) times the cross-sectional area of the tube (cm^2).
- Blood Flow is defined as the volume of blood moving through a vessel per unit of time.
- Blood moves by bulk flow, from a region of high pressure to one of low pressure.
- Flow = ΔP (Pressure difference) divided by R (Resistance)

Flow: Blood Flow; R: Arteriolar Resistance

Capillary Exchange of Fluid

- The movement of fluids through capillaries results from the pressure (known as hydrostatic pressure) of the blood pushing against the walls of the capillaries.
- When the hydrostatic pressure inside the capillaries (capillary hydrostatic pressure) is greater than the pressure in the surrounding interstitial space (interstitial fluid hydrostatic pressure), fluid inside the capillaries is forced out into the interstitial space. This process is also known as filtration.
- When the capillary hydrostatic pressure is less than the interstitial fluid hydrostatic pressure, fluid moves back into the capillaries.
- The **reabsorption** process prevents too much fluid from leaving the capillaries.
- When fluid filters through the capillaries, the protein albumin remains behind in the decreasing volume of water. Albumin is a large molecule that normally cannot pass through the capillary membranes.

- As the concentration of albumin inside the capillaries increases, fluid begins to move back into the capillaries through osmosis.
- The osmotic (pulling) force of albumin in the intravascular space is called the plasma oncotic pressure.
- In normal conditions, capillary hydrostatic pressure exceeds plasma oncotic pressure in the arteriole end and falls below it in the venule end.
- As a result, capillary filtration occurs along the first half of the vessel and reabsorption occurs along the second half.
- Net filtration of fluid out of the capillaries usually exceeds net reabsorption of fluid into the capillaries by a slight margin, and this leads to the formation of lymph.
- **Lymph** is a combination of excess filtered fluid, filtered proteins and macromolecules (e.g. fat). **The lymphatic system** provides a route for lymph to flow from the interstitial space back into the circulatory system.

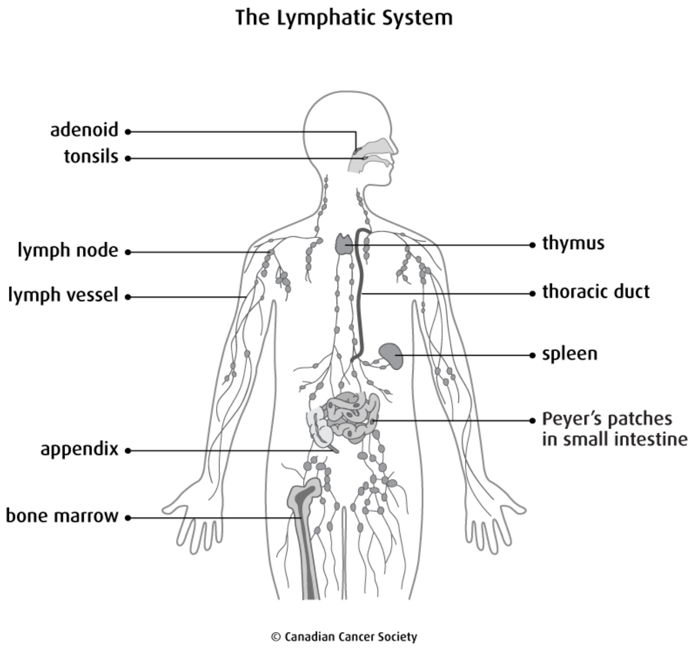

Figure 11: The lymphatic system

Lymph Vessels

- The lymph vessels of the lymphatic system return albumin and other interstitial macromolecules to the circulatory system.
- The lymph vessels possess an extensive system of one-way valves that maintain flow toward the heart.
- The lymph collected from the peripheral tissues is returned to the cardiovascular system via the thoracic duct, which empties into the left subclavian vein.
- The GI (gastrointestinal) tract and the liver produce the majority of the lymph.

Veins

- The venous system is a low-resistance, low pressure, highly distensible part of the vascular system.
- At normal pressures, the veins are about 20 times more distensible (compliant) than the arteries. In comparable vessels, the veins provide a larger cross-sectional area than do the arteries.

- Therefore, the resistance to flow and the velocity of flow are less in the veins than in the arteries.
- The veins of the dependent parts of the body are equipped with valves that prevent the backflow of venous blood. These valves also support the column of blood so that increases in capillary pressure in the dependent parts of the body (caused by hydrostatic pressure) are minimized.

Functions of the Venous System

Reservoirs

The veins serve as a fluid reservoir with about 70% of the circulating blood volume is in the veins at any one time.

Conduits

The systemic veins carry blood from the tissues to the right atrium. The pulmonary veins collect blood from the lungs and return it to the left atrium.

Factors that enhance venous return

Venoconstriction

Skeletal muscle pump

Contraction of skeletal muscle aids in venous return by compressing the veins between the contracting skeletal muscles. Competent venous valves prevent backflow during relaxation of the skeletal muscles.

Respiration

Factors that impede venous return

Peripheral pooling of blood results when an individual rises from a supine to an erect position. Peripheral pooling of blood decreases venous return (consequently lowering cardiac output and, possibly, arterial blood pressure as well), but a reflex increase of sympathetic tone to the veins and heart usually compensates for the lowered arterial blood pressure.

Positive end-expiratory pressure (PEEP) generated by mechanical ventilation can impede venous return.

25

26 Structure and Function of Blood

27 Describe the structure and function of blood in the body

Blood is important for regulation of the body’s pH, temperature, osmotic pressure, the circulation of nutrients and removal of waste, the distribution of hormones from endocrine glands, and the elimination of excess heat; it also contains components for blood clotting. Blood is made of of several components, including red blood cells, white blood cells, platelets, and the plasma, which contains coagulation factors and serum.

28 The Role of Blood in the Body

Blood, like the human blood is important for regulation of the body's systems and homeostasis. Blood helps maintain homeostasis by stabilizing pH, temperature, osmotic pressure, and by eliminating excess heat. Blood supports growth by distributing nutrients and hormones, and by removing waste. Red blood cells contain hemoglobin, which binds oxygen. These cells deliver oxygen to the cells and remove carbon dioxide.

Blood plays a protective role by transporting clotting factors and platelets to prevent blood loss after injury. Blood also transports the disease-fighting agents white blood cells to sites of infection. These cells—including neutrophils, monocytes, lymphocytes, eosinophils, and basophils—are involved in the immune response.

29 Red Blood Cells

Red blood cells, or erythrocytes (*erythro–* = "red"; *–cyte* = "cell"), are specialized cells that circulate through the body delivering oxygen to cells; they are formed from stem cells in the bone marrow. In mammals, red blood cells are small biconcave cells that at maturity do not contain a nucleus or mitochondria and are only 7–8 µm in size. In birds and non-avian reptiles, a nucleus is still maintained in red blood cells.

The red coloring of blood comes from the iron-containing protein hemoglobin. The principal job of this protein is to carry oxygen, but it also transports carbon dioxide as well. Hemoglobin is packed into red blood cells at a rate of about 250 million molecules of hemoglobin per cell. Each hemoglobin molecule binds four oxygen molecules so that each red blood cell carries one billion molecules of oxygen. There are approximately 25 trillion red blood cells in the five liters of blood in the human body, which could carry up to 25 sextillion (25×10^{21}) molecules of oxygen in the body at any time. In mammals, the lack of organelles in erythrocytes leaves more room for the hemoglobin molecules, and the lack of mitochondria also prevents use of the oxygen for metabolic respiration. Only mammals have anucleated red blood cells, and some mammals (camels, for instance) even have nucleated red blood cells. The advantage of nucleated red blood cells is that these cells can undergo mitosis. Anucleated red blood cells metabolize anaerobically (without oxygen), making use of a primitive metabolic pathway to produce ATP and increase the efficiency of oxygen transport.

The small size and large surface area of red blood cells allows for rapid diffusion of oxygen and carbon dioxide across the plasma membrane. In the lungs, carbon dioxide is released and oxygen is taken in by the blood. In the tissues, oxygen is released from the blood and carbon dioxide is bound for transport back to the lungs. Studies have found that hemoglobin also binds nitrous oxide (NO). NO is a vasodilator that relaxes the blood vessels and capillaries and may help with gas exchange and the passage of red blood cells through narrow vessels. Nitroglycerin, a heart medication for angina and heart attacks, is converted to NO to help relax the blood vessels and increase oxygen flow through the body.

A characteristic of red blood cells is their glycolipid and glycoprotein coating; these are lipids and proteins that have carbohydrate molecules attached. In humans, the surface glycoproteins and glycolipids on red blood cells vary between individuals, producing the different blood types, such as A, B, and O. Red blood cells have an average life span of 120 days, at which time they are broken down and recycled in the liver and spleen by phagocytic macrophages, a type of white blood cell.

30 White Blood Cells

White blood cells, also called leukocytes (leuko = white), make up approximately one percent by volume of the cells in blood. The role of white blood cells is very different than that of red blood cells: they are primarily involved in the immune response to identify and target pathogens, such as invading bacteria, viruses, and other foreign organisms. White blood cells are formed continually; some only live for hours or days, but some live for years.

The morphology of white blood cells differs significantly from red blood cells. They have nuclei and do not contain hemoglobin. The different types of white blood cells are identified by their microscopic appearance after histologic staining, and each has a different specialized function. The two main groups, both illustrated in are the granulocytes, which include the neutrophils, eosinophils, and basophils, and the agranulocytes, which include the monocytes and lymphocytes.

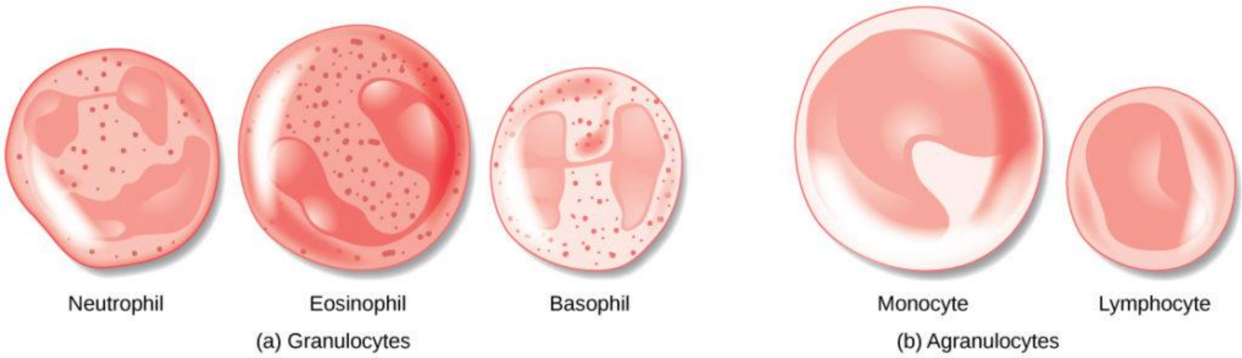

Figure 12: The white blood cells

Granulocytes contain granules in their cytoplasm; the agranulocytes are so named because of the lack of granules in their cytoplasm. Some leukocytes become macrophages that either stay at the same site or move through the blood stream and gather at sites of infection or inflammation where they are attracted by chemical signals from foreign particles and damaged cells. Lymphocytes are the primary cells of the immune system and include B cells, T cells, and natural killer cells. B cells destroy bacteria and inactivate their toxins. They also produce antibodies. T cells attack viruses, fungi, some bacteria, transplanted cells, and cancer cells. T cells attack viruses by releasing toxins that kill the viruses. Natural killer cells attack a variety of infectious microbes and certain tumor cells.

31 Components of Blood

Hemoglobin is responsible for distributing oxygen, and to a lesser extent, carbon dioxide, throughout the circulatory systems of humans, vertebrates, and many invertebrates. The blood is more than the proteins, though. Blood is actually a term used to describe the liquid that moves through the vessels and includes **plasma** (the liquid portion, which contains water, proteins, salts, lipids, and glucose) and the cells (red and white cells) and cell fragments called **platelets**. Blood plasma is actually the dominant component of blood and contains the water, proteins, electrolytes, lipids, and glucose. The cells are responsible for carrying the gases (red cells) and immune the response (white). The platelets are responsible for blood clotting. Interstitial fluid that surrounds cells is separate from the blood, but in hemolymph, they are combined. In humans, cellular components make up approximately 45 percent of the blood and the liquid plasma 55 percent. Blood is 20 percent of a person's extracellular fluid and eight percent of weight.

Platelets and Coagulation Factors

Blood must clot to heal wounds and prevent excess blood loss. Small cell fragments called platelets (thrombocytes) are attracted to the wound site where they adhere by extending many projections and releasing their contents. These contents activate other platelets and also interact with other coagulation factors, which convert fibrinogen, a water-soluble protein present in blood serum into fibrin (a non-water soluble protein), causing the blood to clot. Many of the clotting factors require vitamin K to work, and vitamin K deficiency can lead to problems with blood clotting. Many platelets converge and stick together at the wound site forming a platelet plug (also called a fibrin clot), as illustrated in Figure 4b. The plug or clot lasts for a number of days and stops the loss of blood. Platelets are formed from the disintegration of larger cells called megakaryocytes, like that shown in Figure 4a. For each megakaryocyte, 2000–3000 platelets are formed with 150,000 to 400,000 platelets present in each cubic millimeter of blood. Each platelet is disc shaped and 2–4 µm in diameter. They contain many small vesicles but do not contain a nucleus.

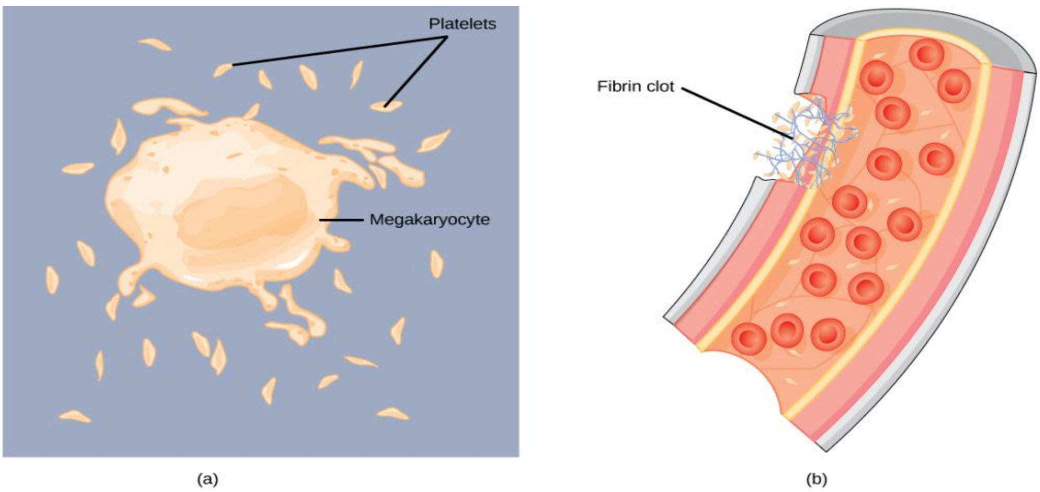

Figure 13:Platelets

Plasma and Serum

The liquid component of blood is called plasma, and it is separated by spinning or centrifuging the blood at high rotations (3000 rpm or higher). The blood cells and platelets are separated by centrifugal forces to the bottom of a specimen tube. The upper liquid layer, the plasma, consists of 90 percent water along with various substances required for maintaining the body's pH, osmotic load, and for protecting the body. The plasma also contains the coagulation factors and antibodies.

The plasma component of blood without the coagulation factors is called the **serum**. Serum is similar to interstitial fluid in which the correct composition of key ions acting as electrolytes is essential for normal functioning of muscles and nerves. Other components in the serum include proteins that assist with maintaining pH and osmotic balance while giving viscosity to the blood. The serum also contains antibodies, specialized proteins that are important for defense against viruses and bacteria. Lipids, including cholesterol, are also transported in the serum, along with various other substances including nutrients, hormones, metabolic waste, plus external substances, such as, drugs, viruses, and bacteria.

Human serum albumin is the most abundant protein in human blood plasma and is synthesized in the liver. Albumin, which constitutes about half of the blood serum protein, transports hormones and fatty acids, buffers pH, and maintains osmotic pressures. Immunoglobin is a protein antibody produced in the mucosal lining and plays an important role in antibody mediated immunity.

1.2.8.4: Self Assessment

a). Highlight factors that inhibit venous return

b).Highlight platelet and coagulation factors

1.2.8.4.1: Responses

a). Highlight factors that inhibit venous return

Peripheral pooling of blood results when an individual rises from a supine to an erect position. Peripheral pooling of blood decreases venous return (consequently lowering cardiac output and, possibly, arterial blood pressure as well), but a reflex increase of sympathetic tone to the veins and heart usually compensates for the lowered arterial blood pressure.

Positive end-expiratory pressure (PEEP) generated by mechanical ventilation can impede venous return.

b). Highlight platelet and coagulation factors

Blood must clot to heal wounds and prevent excess blood loss. Small cell fragments called platelets (thrombocytes) are attracted to the wound site where they adhere by extending many projections and releasing their contents. These contents activate other platelets and also interact with other coagulation factors, which convert fibrinogen, a water-soluble protein present in blood serum into fibrin (a non-water soluble protein), causing the blood to clot. Many of the clotting factors require vitamin K to work, and vitamin K deficiency can lead to problems with blood clotting. Many platelets converge and stick together at the wound site forming a platelet plug (also called a fibrin clot), as illustrated in Figure 4b. The plug or clot lasts for a number of days and stops the loss of blood. Platelets are formed from the disintegration of larger cells called megakaryocytes, like that shown in Figure 4a. For each megakaryocyte, 2000–3000 platelets are formed with 150,000 to 400,000 platelets present in each cubic millimeter of blood. Each platelet is disc shaped and 2–4 μm in diameter. They contain many small vesicles but do not contain a nucleus

1.2.8.5: References

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5. ^ *"The Cardiovascular System"*. *State University of New York* Downstate Medical Center. 8 March 2008.
6. ^ *"The Circu*

1.2.9:Learning outcome 9: **Demonstrate knowledge of the lymphatic system**

1.2.9.1:Introduction to the learning outcome

This learning outcome specifies the competencies required to identify components of the lymphatic system and describe their functions.

1.2.9.2: Performance standards

1.10 **Components** of lymphatic system are identified
1.11 **Functions** of the components are described

1.2.9.3: Information Sheet

Lymphatic system, a subsystem of the circulatory system in the vertebrate body that consists of a complex network of vessels, tissues, and organs. The lymphatic system helps maintain fluid balance in the body by collecting excess fluid and particulate matter from tissues and depositing them in the bloodstream. It also helps defend the body against infection by supplying disease fighting cells called lymphocytes. This article focuses on the human lymphatic system.

Lymphatic Circulation

The lymphatic system can be thought of as a drainage system needed because, as blood circulates through the body, blood plasma leaks into tissues through the thin walls of the capillaries. The portion of blood plasma that escapes is called interstitial or extracellular fluid, and it contains oxygen, glucose, amino acids, and other nutrients needed by tissue cells. Although most of this fluid seeps immediately back into the bloodstream, a percentage of it, along with the particulate matter, is left behind. The lymphatic system removes this fluid and these materials from tissues, returning them via the lymphatic vessels to the bloodstream, and thus prevents a fluid imbalance that would result in the organism's death.

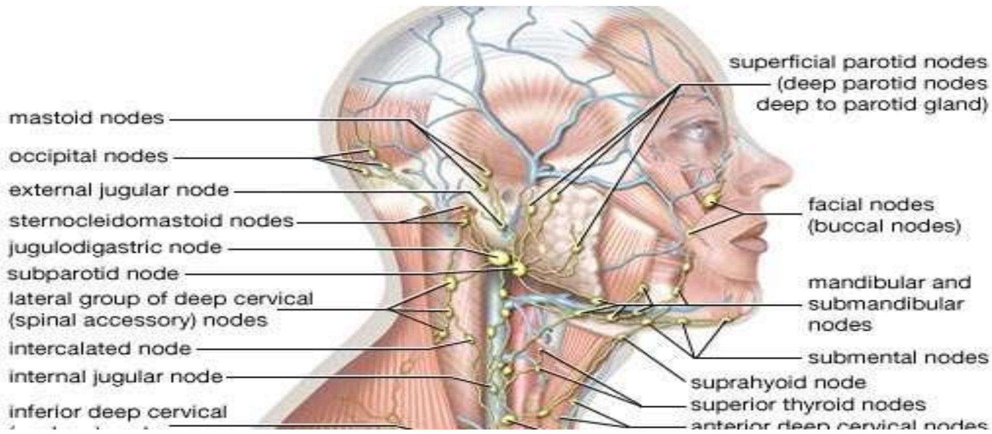

Figure: 14; Lymph nodes

The fluid and proteins within the tissues begin their journey back to the bloodstream by passing into tiny lymphatic capillaries that infuse almost every tissue of the body. Only a few regions, including the epidermis of the skin, the mucous membranes, the bone marrow, and the central nervous system, are free of lymphatic capillaries, whereas regions such as the lungs, gut, genitourinary system, and dermis of the skin are densely packed with these vessels. Once within the lymphatic system, the extracellular fluid, which is now called lymph, drains into larger vessels called the lymphatics. These vessels converge to form one of two large vessels called lymphatic trunks, which are connected to veins at the base of the neck. One of these trunks, the right lymphatic duct, drains the upper right portion of the body, returning lymph to the bloodstream via the right subclavian vein. The other trunk, the thoracic duct, drains the rest of the body into the left subclavian vein. Lymph is transported along the system of vessels by muscle contractions, and valves prevent lymph from flowing backward. The lymphatic vessels are punctuated at intervals by small masses of lymph tissue, called lymph nodes, that remove foreign materials such as infectious microorganisms from the lymph filtering through them.

Role in immunity

In addition to serving as a drainage network, the lymphatic system helps protect the body against infection by producing white blood cells called lymphocytes, which help rid the body of disease-causing microorganisms. The organs and tissues of the lymphatic system are the major sites of production, differentiation, and proliferation of two types of lymphocytes—the T lymphocytes and B lymphocytes, also called T cells and B cells. Although lymphocytes are distributed throughout the body, it is within the lymphatic system that they are most likely to encounter foreign microorganisms.

32 Lymphoid Organs

The lymphatic system is commonly divided into the primary lymphoid organs, which are the sites of B and T cell maturation, and the secondary lymphoid organs, in which further differentiation of lymphocytes occurs. Primary lymphoid organs include the thymus, bone marrow, fetal liver, and, in birds, a structure called the bursa of Fabricius. In humans the thymus and bone marrow are the key players in immune function. All lymphocytes derive from stem cells in the bone marrow. Stem cells destined to become B lymphocytes remain in the bone marrow as they mature, while prospective T cells migrate to the thymus to undergo further growth. Mature B and T lymphocytes exit the primary lymphoid organs and are transported via the bloodstream to the secondary lymphoid organs, where they become activated by contact with foreign materials, such as particulate matter and infectious agents, called antigens in this context.

33 Thymus

The thymus is located just behind the sternum in the upper part of the chest. It is a bilobed organ that consists of an outer, lymphocyte-rich cortex and an inner medulla. The differentiation of T cells occurs in the cortex of the thymus. In humans the thymus appears early in fetal development and continues to grow until puberty, after which it begins to shrink. The decline of the thymus is thought to be the reason T-cell production decreases with age.

In the cortex of the thymus, developing T cells, called thymocytes, come to distinguish between the body's own components, referred to as "self," and those substances foreign to the body, called "nonself." This occurs when the thymocytes undergo a process called positive selection, in which they are exposed to self molecules that belong to the major histocompatibility complex (MHC). Those cells capable of recognizing the body's MHC molecules are preserved, while those that cannot bind these molecules are destroyed. The thymocytes then move to the medulla of the thymus, where further differentiation occurs. There thymocytes that have the ability to attack the body's own tissues are destroyed in a process called negative selection.

Positive and negative selection destroy a great number of thymocytes; only about 5 to 10 percent survive to exit the thymus. Those that survive leave the thymus through specialized passages called efferent (outgoing) lymphatics, which drain to the blood and secondary lymphoid organs. The thymus has no afferent (incoming) lymphatics, which supports the idea that the thymus is a T-cell factory rather than a rest stop for circulating lymphocytes.

Bone marrow

In birds B cells mature in the bursa of Fabricius. (The process of B-cell maturation was elucidated in birds hence *B* for *bursa*.) In mammals the primary organ for B-lymphocyte development is the bone marrow, although the prenatal site of B-cell differentiation is the fetal liver. Unlike the thymus, the bone marrow does not atrophy at puberty, and therefore there is no concomitant decrease in the production of B lymphocytes with age.

34 Secondary lymphoid organs

Secondary lymphoid organs include the lymph nodes, spleen, and small masses of lymph tissue such as Peyer's patches, the appendix, tonsils, and selected regions of the body's mucosal surfaces (areas of the body lined with mucous membranes). The secondary lymphoid organs serve two basic functions: they are a site of further lymphocyte maturation, and they efficiently trap antigens for exposure to T and B cells.

35 Lymph nodes

The lymph nodes, or lymph glands, are small, encapsulated bean-shaped structures composed of lymphatic tissue. Thousands of lymph nodes are found throughout the body along the lymphatic routes, and they are especially prevalent in areas around the armpits (axillary nodes), groin (inguinal nodes), neck (cervical nodes), and knees (popliteal nodes). The nodes contain lymphocytes, which enter from the bloodstream via specialized vessels called the high endothelial venules. T cells congregate in the inner cortex (paracortex), and B cells are organized in germinal centres in the outer cortex. Lymph, along with antigens, drains into the node through afferent (incoming) lymphatic vessels and percolates through the lymph node, where it comes in contact with and activates lymphocytes. Activated lymphocytes, carried in the lymph, exit the node through the efferent (outgoing) vessels and eventually enter the bloodstream, which distributes them throughout the body.

36 Spleen

The spleen is found in the abdominal cavity behind the stomach. Although structurally similar to a lymph node, the spleen filters blood rather than lymph. One of its main functions is to bring blood into contact with lymphocytes. The functional tissue of the spleen is made up of two types of cells: the red pulp, which contains cells called macrophages that remove bacteria, old blood cells, and debris from the circulation; and surrounding regions of white pulp, which contain great numbers of lymphocytes. The splenic artery enters the red pulp through a web of small blood vessels, and blood-borne microorganisms are trapped in this loose collection of cells until they are gradually washed out through the splenic vein. The white pulp contains both B and T lymphocytes. T cells congregate around the tiny arterioles that enter the spleen, while B cells are located in regions called germinal centres, where the lymphocytes are exposed to antigens and induced to differentiate into antibody-secreting plasma cells.

37 Mucosa-associated tissues

Another group of important secondary lymphoid structures is the mucosa-associated lymphoid tissues. These tissues are associated with mucosal surfaces of almost any organ, but especially those of the digestive, genitourinary, and respiratory tracts, which are constantly exposed to a wide variety of potentially harmful microorganisms and therefore require their own system of antigen capture and presentation to lymphocytes. For example, Peyer's patches, which are mucosa-associated lymphoid tissues of the small intestine, sample passing antigens and expose them to underlying B and T cells. Other, less-organized regions of the gut also play a role as secondary lymphoid tissue.

1.2.9.4: Self Assessment

a). Highlight the role of the lymphatic system in immunity
b). Define the lymphatic system
c). Describe the thymus

1.2.9.4.1: Responses

a). Highlight the role of the lymphatic system in immunity

In addition to serving as a drainage network, the lymphatic system helps protect the body against infection by producing white blood cells called lymphocytes, which help rid the body of disease-causing microorganisms. The organs and tissues of the lymphatic system are the major sites of production, differentiation, and proliferation of two types of lymphocytes—the T lymphocytes and B lymphocytes, also called T cells and B cells. Although lymphocytes are distributed throughout the body, it is within the lymphatic system that they are most likely to encounter foreign microorganisms.

b). Define the lymphatic system

Lymphatic system, a subsystem of the circulatory system in the vertebrate body that consists of a complex network of vessels, tissues, and organs. The lymphatic system helps maintain fluid balance in the body by collecting excess fluid and particulate matter from tissues and depositing them in the bloodstream. It also helps defend the body against infection by supplying disease fighting cells called lymphocytes. This article focuses on the human lymphatic system.

c). Describe the thymus

The thymus is located just behind the sternum in the upper part of the chest. It is a bilobed organ that consists of an outer, lymphocyte-rich cortex and an inner medulla. The differentiation of T cells occurs in the cortex of the thymus. In humans the thymus appears early in fetal development and continues to grow until puberty, after which it begins to shrink. The decline of the thymus is thought to be the reason T-cell production decreases with age.

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Positive and negative selection destroy a great number of thymocytes; only about 5 to 10 percent survive to exit the thymus. Those that survive leave the thymus through specialized passages called efferent (outgoing) lymphatics, which drain to the blood and secondary lymphoid organs. The thymus has no afferent (incoming) lymphatics, which supports the idea that the thymus is a T-cell factory rather than a rest stop for circulating lymphocytes.

1.2.9.5: References

1. Gray's Anatomy 2018, p. 27.
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1.2.10: Learning outcome 10: Demonstrate knowledge of the endocrine system and its functions

1.2.10.1: Introduction to learning outcome

This learning outcome specifies the competencies required to identify components of the endocrine system as well as hormones and describe their functions.

1.2.10.2: Performance standards.
Components of endocrine system are identified
Hormones and their functions are described

1.2.10.3: Information Sheet

The endocrine system is a network of glands and organs located throughout the body. It's similar to the nervous system in that it plays a vital role in controlling and regulating many of the body's functions.

However, while the nervous system uses nerve impulses and neurotransmitters for communication, the endocrine system uses chemical messengers called hormones.

Endocrine System Function
The endocrine system is responsible for regulating a range of bodily functions through the release of hormones.
Hormones are secreted by the glands of the endocrine system, traveling through the bloodstream to various organs and tissues in the body. The hormones then tell these organs and tissues what to do or how to function.
Some examples of bodily functions that are controlled by the endocrine system include:

- metabolism
- growth and development
- sexual function and reproduction
- heart rate
- blood pressure
- appetite
- sleeping and waking cycles
- body temperature

Endocrine System Organs
The endocrine system is made up of a complex network of glands, which are organs that secrete substances.
The glands of the endocrine system are where hormones are produced, stored, and released. Each gland produces one or more hormones, which go on to target specific organs and tissues in the body.
The glands of the endocrine system include:

Hypothalamus. While some people don't consider it a gland, the hypothalamus produces multiple hormones that control the pituitary gland. It's also involved in regulating many functions, including sleep-wake cycles, body temperature, and appetite. It can also regulate the function of other endocrine glands.
Pituitary. The pituitary gland is located below the hypothalamus. The hormones it produces affect growth and reproduction. They can also control the function of other endocrine glands.
Pineal. This gland is found in the middle of your brain. It's important for your sleep-wake cycles.
Thyroid. The thyroid gland is located in the front part of your neck. It's very important for metabolism.
Parathyroid. Also located in the front of your neck, the parathyroid gland is important for maintaining control of calcium levels in your bones and blood.
Thymus. Located in the upper torso, the thymus is active until puberty and produces hormones important for the development of a type of white blood cell called a T cell.
Adrenal. One adrenal gland can be found on top of each kidney. These glands produce hormones important for regulating functions such as blood pressure, heart rate, and stress response.
Pancreas. The pancreas is located in your abdomen behind your stomach. Its endocrine function involves controlling blood sugar levels.

Some endocrine glands also have non-endocrine functions. For example, the ovaries and testes produce hormones, but they also have the non-endocrine function of producing eggs and sperm, respectively.

Endocrine System hormones

Hormones are the chemicals the endocrine system uses to send messages to organs and tissue throughout the body. Once released into the bloodstream, they travel to their target organ or tissue, which has receptors that recognize and react to the hormone.

Below are some examples of hormones that are produced by the endocrine system.

Hormone	**Secreting gland(s)**	**Function**
adrenaline	adrenal	increases blood pressure, heart rate, and metabolism in reaction to stress
aldosterone	adrenal	controls the body's salt and water balance
cortisol	adrenal	plays a role in stress response
dehydroepiandrosterone sulfate (DHEA)	adrenal	aids in production of body odor and growth of body hair during puberty
estrogen	ovary	works to regulate menstrual cycle, maintain pregnancy, and develop female sex characteristics; aids in sperm production

follicle stimulating hormone (FSH)	pituitary	controls the production of eggs and sperm
glucagon	pancreas	helps to increase levels of blood glucose
insulin	pancreas	helps to reduce your blood glucose levels
luteinizing hormone (LH)	pituitary	controls estrogen and testosterone production as well as ovulation
melatonin	pituitary	controls sleep and wake cycles
oxytocin	pituitary	helps with lactation, childbirth, and mother-child bonding
parathyroid hormone	parathyroid	controls calcium levels in bones and blood
progesterone	ovary	helps to prepare the body for pregnancy when an egg is fertilized
prolactin	pituitary	promotes breast-milk production
testosterone	ovary, teste, adrenal	contributes to sex drive and body density in males and females as well as development of male sex characteristics
thyroid hormone	thyroid	help to control several body functions, including the rate of metabolism and energy levels

Table 5: hormones produced by endocrine system

1.2.10.4: Self Assessment

a). Highlight glands in the endocrine system

b). State bodily functions that are controlled by the endocrine system

c).Describe endocrine system hormones

1.2.10.4.1: Responses

a). Highlight glands in the endocrine system

Hypothalamus. While some people don't consider it a gland, the hypothalamus produces multiple hormones that control the pituitary gland. It's also involved in regulating many functions, including sleep-wake cycles, body temperature, and appetite. It can also regulate the function of other endocrine glands.

Pituitary. The pituitary gland is located below the hypothalamus. The hormones it produces affect growth and reproduction. They can also control the function of other endocrine glands.

Pineal. This gland is found in the middle of your brain. It's important for your sleep-wake cycles.

Thyroid. The thyroid gland is located in the front part of your neck. It's very important for metabolism.

Parathyroid. Also located in the front of your neck, the parathyroid gland is important for maintaining control of calcium levels in your bones and blood.

Thymus. Located in the upper torso, the thymus is active until puberty and produces hormones important for the development of a type of white blood cell called a T cell.

Adrenal. One adrenal gland can be found on top of each kidney. These glands produce hormones important for regulating functions such as blood pressure, heart rate, and stress response.

Pancreas. The pancreas is located in your abdomen behind your stomach. Its endocrine function involves controlling blood sugar levels.

b). State bodily functions that are controlled by the endocrine system

- metabolism
- growth and development
- sexual function and reproduction
- heart rate
- blood pressure
- appetite
- sleeping and waking cycles
- body temperature

c).Describe endocrine system hormones

Hormones are the chemicals the endocrine system uses to send messages to organs and tissue throughout the body. Once released into the bloodstream, they travel to their target organ or tissue, which has receptors that recognize and react to the hormone.

1.2.10.5: References

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2. ^ Quesada, Ivan (2008). "Physiology of the pancreatic α-cell and glucagon secretion: role in glucose homeostasis and diabetes". Journal of Endocrinology. **199** (1): 5–19. doi:10.1677/JOE-08-0290. PMID 18669612.
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1.2.11: Learning outcome 11: Demonstrate knowledge of the nervous system and its function

1.2.11.1: Introduction to the learning outcome

This learning outcome specifies the competencies required to outline the structure and function of a neuron, describe types of neurons, *divisions of the nervous system* and their components as well as outline nervous reflexes.

1.2.11.2: Performance Standard

1.12 Structure and function of a neuron is outlined
1.13 **Types** of neurons are described
1.14 ***Divisions of the nervous system*** and their components are described
1.15 **Nervous reflexes** are outlined

1.2.11.3: Information Sheet

Functions of the Nervous System
1. Gathers information from both inside and outside the body - Sensory Function
2. Transmits information to the processing areas of the brain and spine
3. Processes the information in the brain and spine – Integration Function
4. Sends information to the muscles, glands, and organs so they can respond appropriately – Motor Function

It controls and coordinates all essential functions of the body including all other body systems
allowing the body to maintain homeostasis or its delicate balance.

The Nervous System is divided into Two Main Divisions**: Central Nervous System (CNS) and**
the Peripheral Nervous System (PNS).

Divisions of the Nervous system

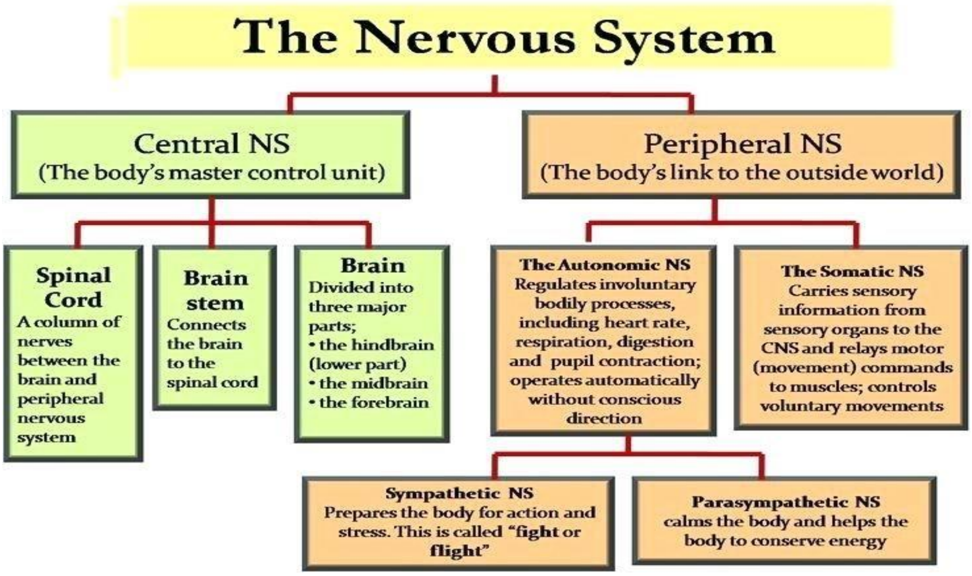

Figure 15

<u>Basic Cells of the Nervous System</u>

Neuron

• Basic functional cell of nervous system
• Transmits impulses (up to 250 mph)

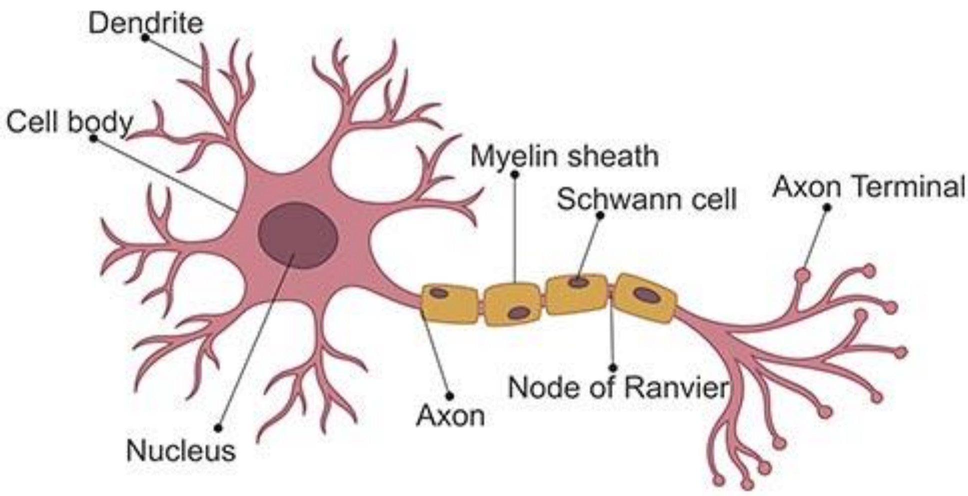

Figure 16

Parts of a Neuron
• **Dendrite** – receive stimulus and carries it impulses
toward the cell body
• **Cell Body** with nucleus – nucleus & most of
cytoplasm
• **Axon** – fiber which carries impulses away from cell body
• **Schwann Cells**- cells which produce myelin or fat layer in the Peripheral Nervous System
• **Myelin sheath** – dense lipid layer which insulates the axon – makes the axon look gray
• **Node of Ranvier** – gaps or nodes in the myelin sheath
• Impulses travel **from dendrite to cell body to axon**

Three types of Neurons

- **Sensory neurons** – bring messages to CNS
- **Motor neurons** - carry messages from CNS
- **Interneurons** – between sensory & motor neurons in the CNS

Impulses
• A **stimulus** is a change in the environment with sufficient strength to initiate a response.
• **Excitability** is the ability of a neuron to respond to the stimulus and convert it into a nerve impulse
• **All of Nothing Rule** – The stimulus is either strong enough to start and impulse or nothing happens
• Impulses are always the **same strength along a given neuron** and they are **self-propagation** – once it starts it continues to the end of the neuron in only one direction- **from dendrite to cell body to axon**
• The nerve impulse causes a movement of ions across the cell membrane of the nerve cell.

Synapse
Synapse - small gap or space between the axon of one neuron and the dendrite of another - the
neurons do not actually tough at the synapse.
It is junction between neurons which uses neurotransmitters to start the impulse in the second

neuron or an effector (muscle or gland).
The synapse insures one-way transmission of impulses.

Neurotransmitters
Neurotransmitters – Chemicals in the junction which allow impulses to be started in the second neuron.

<u>Reflex Arc</u>

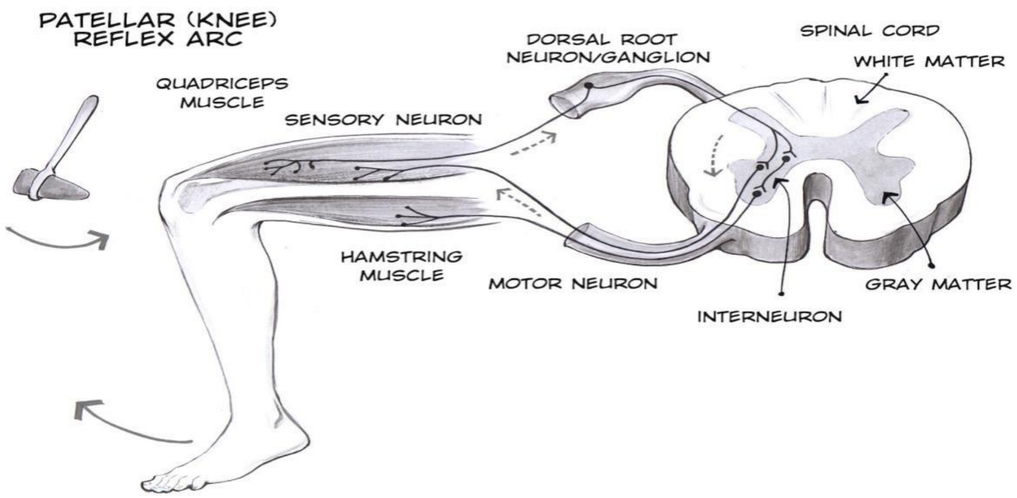

Figure 17: Components of a Reflex Arc
A. **Receptor** - reacts to a stimulus
B. **Afferent pathway (sensory neuron)** - conducts impulses to the CNS
C. **Interneuron** - consists of one or more synapses in the CNS (most are in the spine)
D. **Efferent pathway (motor neuron)** conducts impulses from CNS to effector.
E. **Effector** - muscle fibers (as in the Hamstring muscle) or glands responds by contracting or secreting a product.
Spinal reflexes - initiated and completed at the spinal cord level. Occur without the involvement of higher brain centers.

Central Nervous System
• **Brain**
- Brain stem – medulla, pons, midbrain
- Diencephalon – thalamus & hypothalamus
- Cerebellem
- Cerebrum

• Spine
- Spinal Cord

Meninges
Meninges are the three coverings around the brain & spine and help cushion, protect, and nourish the brain and spinal cord.
• Dura mater is the most outer layer, very tough

• Arachnoid mater is the middle layer and adheres to the dura mater and has weblike attachments to the innermost layer, the pia mater
• Pia mater is very thin, transparent, but tough, and covers the entire brain, following it into all its crevices (sulci) and spinal cord
• cerebrospinal fluid, which buffers, nourishes, and detoxifies the brain and spinal cord, flows through the subarachnoid space, between the arachnoid mater and the pia mater.

Regions of the Brain

Cerebellum – coordination of movement and aspects of motor learning
Cerebrum – conscious activity including perception, emotion, thought, and planning
Thalamus – Brain's switchboard – filters and then relays information to various brain regions
Medulla – vital reflexes as heart beat and respiration
Brainstem – medulla, pons, and midbrain (involuntary responses) and relays information from spine to upper brain
Hypothalamus– involved in regulating activities internal organs, monitoring information from the autonomic nervous system, controlling the pituitary gland and its hormones, and regulating sleep and appetite.

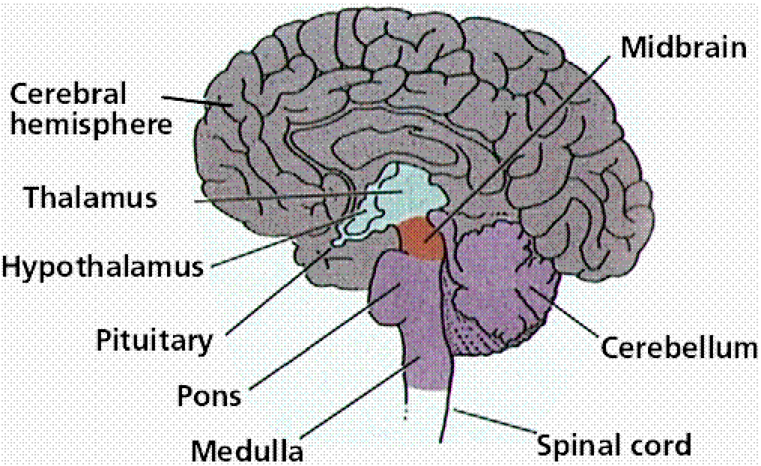

Figure 18: The brain

Cerebrum
• Is the largest portion of the brain encompasses about two-thirds of the brain mass -
• It consists of two hemispheres divided by a fissure – corpus callosum
• It includes the cerebral cortex, the medullary body, and basal ganglia

• **cerebral cortex** is the layer of the brain often referred to as gray matter because it has cell bodies and synapses but no myelin

- The cortex (thin layer of tissue) is gray because nerves in this area lack the insulation or white fatty myelin sheath that makes most other parts of the brain appear to be white.
- The cortex covers the outer portion (1.5mm to 5mm) of the cerebrum and cerebellum
- The cortex consists of folded bulges called **gyri** that create deep furrows or fissures called **sulci**
- The folds in the brain add to its surface area which increases the amount of gray matter and the quantity of information that can be processed.

Medullary body – is the white matter of the cerebrum and consists of myelinated axons

- Commisural fibers – conduct impulses between the hemispheres and for corpus callosum
- Projection fibers – conduct impulse in and out of the cerebral hemispheres
- Association fibers – conduct impulses within the hemispheres.

Basal ganglia – masses of gray matter in each hemisphere which are involved in the control of voluntary muscle movements

Lobes of the Cerebrum

• **Frontal –** motor area involved in movement and in planning & coordinating behavior
• **Parietal –** sensory processing, attention, and language
• **Temporal** – auditory perception, speech, and complex visual perceptions
• **Occipital –** visual center – plays a role in processing visual information.

Special regions

• **Broca's area –** located in the frontal lobe – important in the production of speech
• **Wernicke's area –** comprehension of language and the production of meaningful speech
• **Limbic System** – a group of brain structures (amygdala, hippocampus, septum, basal ganglia, and others) that help regulate the expression of emotions and emotional memory.

Peripheral Nervous System

Cranial nerves

• 12 pair
• Attached to undersurface of brain

Spinal nerves

• 31 pair
• Attached to spinal cord

Somatic Nervous System (voluntary)

• Relays information from skin, sense organs & skeletal muscles to CNS
• Brings responses back to skeletal muscles for voluntary responses.

Autonomic Nervous System (involuntary)

• Regulates bodies involuntary responses
• Relays information to internal organs

Two divisions

o ***Sympathetic nervous system*** – in times of stress

- Emergency response
- Fight or flight

o ***Parasympathetic nervous system*** – when body is at rest or with normal functions

- Normal everyday conditions.

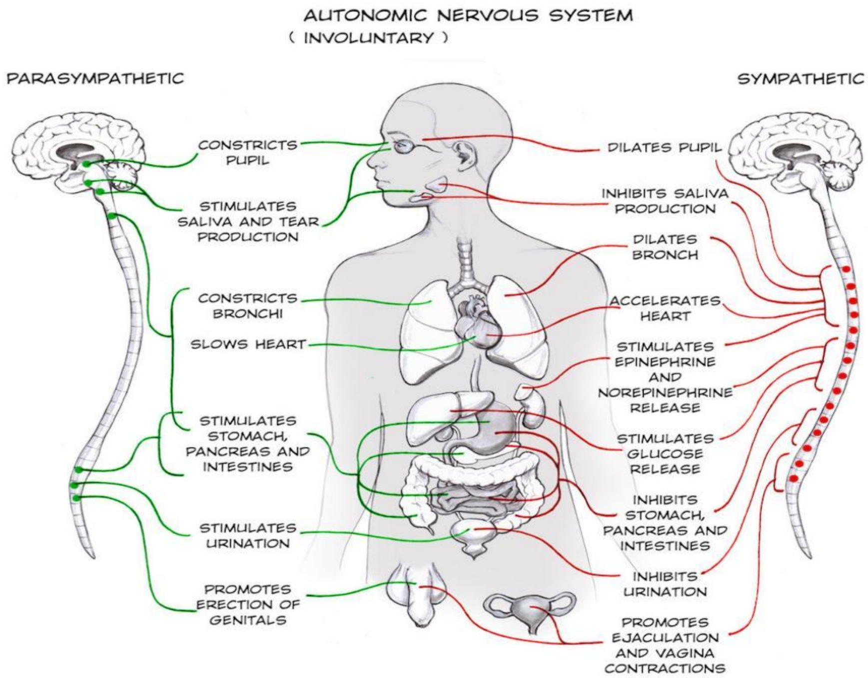

Figure 19: Autonomic nervous sytem

1.2.11.4: Self Assessment
a). Highlight the parts of a neuron.
b). Highlight regions of the brain.
c). Describe meninges

1.2.11.4.1: Responses
a). Highlight the parts of a neuron.
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• **Cell Body** with nucleus – nucleus & most of cytoplasm
• **Axon** – fiber which carries impulses away from cell body
• **Schwann Cells**- cells which produce myelin or fat layer in the Peripheral Nervous System
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b). Highlight regions of the brain.
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Cerebrum – conscious activity including perception, emotion, thought, and planning
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c). Describe meninges
Meninges are the three coverings around the brain & spine and help cushion, protect, and nourish the brain and spinal cord.

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• Arachnoid mater is the middle layer and adheres to the dura mater and has weblike attachments to the innermost layer, the pia mater
• Pia mater is very thin, transparent, but tough, and covers the entire brain, following it into all its crevices (sulci) and spinal cord
• cerebrospinal fluid, which buffers, nourishes, and detoxifies the brain and spinal cord, flows through the subarachnoid space, between the arachnoid mater and the pia mater.

1.2.11.5: References

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1.2.12: Learning outcome 12: Demonstrate knowledge of the digestive system and its function

1.2.12.1:Introduction to learning outcome

This learning outcome specifies the competencies required to identify Components of digestive system, describe structure and functions of the components as well as describe the process of digestion.

1.2.12.2: Performance standards

1.16 **Components** of digestive system are identified
1.17 Structure and functions of components are described
1.18 **Process of digestion** is described

1.2.12.3: Information sheet

The human digestive system is the collective name used to describe the alimentary canal, some accessory organs, and a variety of digestive processes that take place at different levels in the canal to prepare food eaten in the diet for absorption.

It has a general structure that is modified at different levels to provide for the processes occurring at each level.

The complex of digestive processes gradually breaks down the foods eaten until they are in a form suitable for absorption.

After absorption, nutrients are used to synthesize body constituents.

They provide the raw materials for the manufacture of new cells, hormones and enzymes, and the energy needed for these and other processes and for the disposal of waste materials.

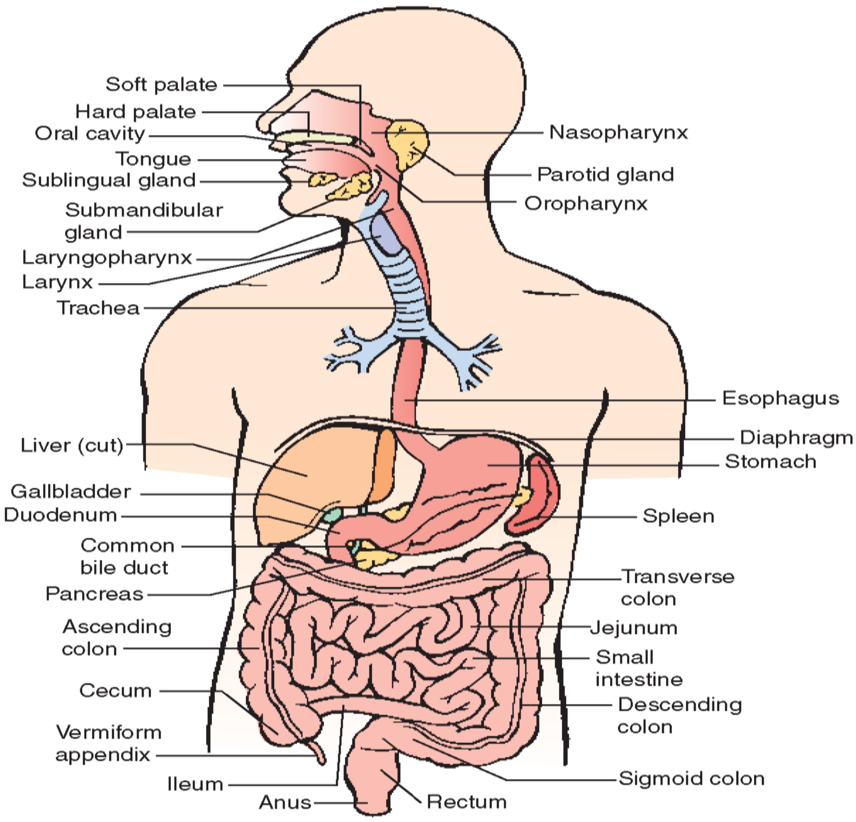

Figure: The digestive system

The alimentary canal begins at the mouth, passes through the thorax, abdomen, and pelvis and ends at the anus. It is thus a long tube through which food passes. It has various parts that are structurally remarkably similar. The parts include:

Mouth
Pharynx
Oesophagus
Stomach
Small intestine

Large intestine
Rectum and Anal canal.

Mouth

The mouth or oral cavity is bounded by muscles and bones: interiorly —by the lips, posteriorly — it is continuous with the oropharynx, laterally —by the muscles of the cheeks, superiorly —by the bony hard palate and muscular soft palate, inferiorly —by the muscular tongue and the soft tissues of the floor of the mouth.
It is lined throughout with mucous membrane, consisting of stratified squamous epithelium containing small mucus-secreting glands.
The palate forms the roof of the mouth and is divided into the anterior hard palate and the posterior soft palate. The soft palate is muscular, curves downwards from the posterior end of the hard palate, and blends with the walls of the pharynx at the sides.
The uvula is a curved fold of muscle covered with mucous membrane, hanging down from the middle of the free border of the soft palate.
It consists of the following important parts:

The Tongue

The tongue is a voluntary muscular structure that occupies the floor of the mouth.
It is attached by its base to the hyoid bone and by a fold of its mucous membrane covering, called the frenulum, to the floor of the mouth.
The superior surface consists of stratified squamous epithelium, with numerous papillae (little projections), containing nerve endings of the sense of taste, sometimes called the taste buds.
The tongue plays an important part in:

- mastication (chewing)
- deglutition (swallowing)
- speech
- taste

The Teeth

- The teeth are embedded in the alveoli or sockets of the alveolar ridges of the mandible and the maxilla.
- Each individual has two sets, the temporary or deciduous teeth, and the permanent teeth.
- At birth, the teeth of both dentitions are present in an immature form in the mandible and maxilla.
- There are 20 temporary teeth, 10 in each jaw. They begin to erupt when the child is about 6 months old, and should all be present after 24 months.
- The permanent teeth begin to replace the deciduous teeth in the 6th year of age and this dentition, consisting of 32 teeth, is usually complete by the 24th year.

Types and Functions of the teeth

The incisor and canine teeth are the cutting teeth and are used for biting off pieces of food, whereas the premolar and molar teeth, with broad, flat surfaces, are used for grinding or chewing food.

The Pharynx

- Food passes from the oral cavity into the pharynx then to the esophagus below, with which it is continuous.
- The pharynx is divided for descriptive purposes into three parts, the **nasopharynx, oropharynx, and laryngopharynx**.
- The nasopharynx is important in respiration. The oropharynx and laryngopharynx are passages common to both the respiratory and the digestive systems.

Function of Pharynx

The pharynx has roles in both the respiratory and digestive systems and can be thought of as the point where these systems diverge.

For the digestive system, its muscular walls function in the process of swallowing, and it serves as a pathway for the movement of food from the mouth to the esophagus.

- The constrictive circular muscles of the pharynx's outer layer play a big role in peristalsis. A series of contractions will help propel ingested food and drink down the intestinal tract safely.
- The inner layer's longitudinal muscles, on the other hand, will widen the pharynx laterally and lift it upward, thus allowing the swallowing of ingested food and drink.

The Oesophagus

- The esophagus is about 25 cm long and about 2 cm in diameter and lies in the median plane in the thorax in front of the vertebral column behind the trachea and the heart.
- It is continuous with the pharynx above and just below the diaphragm it joins the stomach.
- The upper and lower ends of the esophagus are closed by sphincter muscles.
- The upper cricopharyngeal sphincter prevents air from passing into the esophagus during inspiration and the aspiration of oesophageal contents.
- The cardiac or lower oesophageal sphincter prevents the reflux of acid gastric contents into the esophagus.

Functions of Oesophagus

- The esophagus serves to pass food and liquids from the mouth down to the stomach. This is accomplished by periodic contractions (peristalsis).
- The esophagus is an important connection to the digestive system through the thoracic cavity, which protects the heart and lungs.
- Two sphincters on either side of the esophagus separate food into small units known as a bolus.

The Stomach

- The stomach is a J-shaped dilated portion of the alimentary tract situated in the epigastric, umbilical, and left hypochondriac regions of the abdominal cavity.
- The stomach is continuous with the esophagus at the cardiac sphincter and with the duodenum at the pyloric sphincter.
- It has two curvatures. The lesser curvature is short, lies on the posterior surface of the stomach, and is the downwards continuation of the posterior wall of the esophagus. Just before the pyloric sphincter, it curves upwards to complete the J shape.
- Where the esophagus joins the stomach the anterior region angles acutely upwards, curves downwards forming the greater curvature then slightly upwards towards the pyloric sphincter.
- The stomach is divided into three regions: the fundus, the body, and the antrum.

- At the distal end of the pyloric antrum is the pyloric sphincter, guarding the opening between the stomach and the duodenum.
- Stomach size varies with the volume of food it contains, which may be 1.5 liters or more in an adult.
- In the stomach, gastric muscle contraction consists of a churning movement that breaks down the bolus and mixes it with gastric juice and peristaltic waves that propel the stomach contents towards the pylorus.
- About 2 liters of gastric juice are secreted daily by special secretory glands in the mucosa.
- It consists of Water, mineral salts, mucus secreted by goblet cells in the glands and on the stomach surface, hydrochloric acid, Intrinsic factor, inactive enzyme precursors, etc.

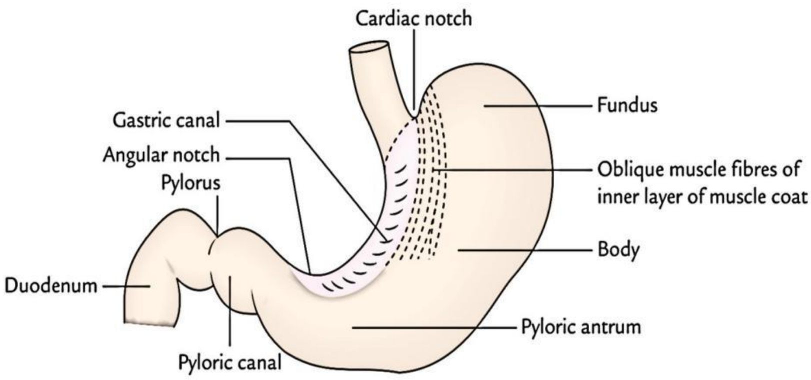

Figure 21: The stomach

Functions of the Stomach

- Temporary storage allowing time for the digestive enzymes, pepsins, to act.
- Chemical digestion — pepsins convert proteins to polypeptides.
- Mechanical breakdown — the three smooth muscle layers enable the stomach to act as a churn, gastric juice is added and the contents are liquefied to chime.
- Performs limited absorption of water, alcohol and some lipid-soluble drugs
- Non-specific defense against microbes — provided by hydrochloric acid in gastric juice.
- Preparation of iron for absorption further along the track — the acid environment of the stomach solubilizes iron salts, which is required before iron can be absorbed
- Production of intrinsic factor needed for absorption of vitamin B12 in the terminal ileum
- Regulation of the passage of gastric contents into the duodenum. When the chyme is sufficiently acidified and liquefied, the pyloric antrum forces small jets of gastric contents through the pyloric sphincter into the duodenum.

The Small Intestine

- The small intestine is continuous with the stomach at the pyloric sphincter and leads into the large intestine at the ileocaecal valve.
- It is a little over 5 meters long and lies in the abdominal cavity surrounded by the large intestine.
- In the small intestine, the chemical digestion of food is completed and most of the absorption of nutrients takes place.
- The small intestine comprises three main sections continuous with each other:

1. **The duodenum**: It is about 25 cm long and curves around the head of the pancreas. Secretions from the gall bladder and pancreas are released into the duodenum through a

common structure, the hepatopancreatic ampulla, and the opening into the duodenum is guarded by the hepatopancreatic sphincter (of Oddi).

2. **The jejunum**: It is the middle section of the small intestine and is about 2 meters long.
3. **The ileum**, or terminal section, is about 3 meters long and ends at the ileocaecal valve, which controls the flow of material from the ileum to the caecum, the first part of the large intestine, and prevents regurgitation.

- The surface area of the small intestine mucosa is greatly increased by permanent circular folds, villi, and microvilli.
- The villi are tiny finger-like projections of the mucosal layer into the intestinal lumen, about 0.5 to 1 mm long.
- Their walls consist of columnar epithelial cells, or enterocytes, with tiny microvilli (1 µm long) on their free border.

Functions of the small intestine

- The small intestine is the part of the intestines where 90% of the digestion and absorption of food occurs, the other 10% taking place in the stomach and large intestine.
- The main function of the small intestine is the absorption of nutrients and minerals from food.

The Large Intestine

- It is about 1.5 meters long, beginning at the caecum in the right iliac fossa and terminating at the rectum and anal canal deep in the pelvis.
- Its lumen is larger than that of the small intestine. It forms an arch around the coiled-up small intestine.
- The colon is divided into the caecum, ascending colon, transverse colon, descending colon, sigmoid colon rectum, and anal canal.

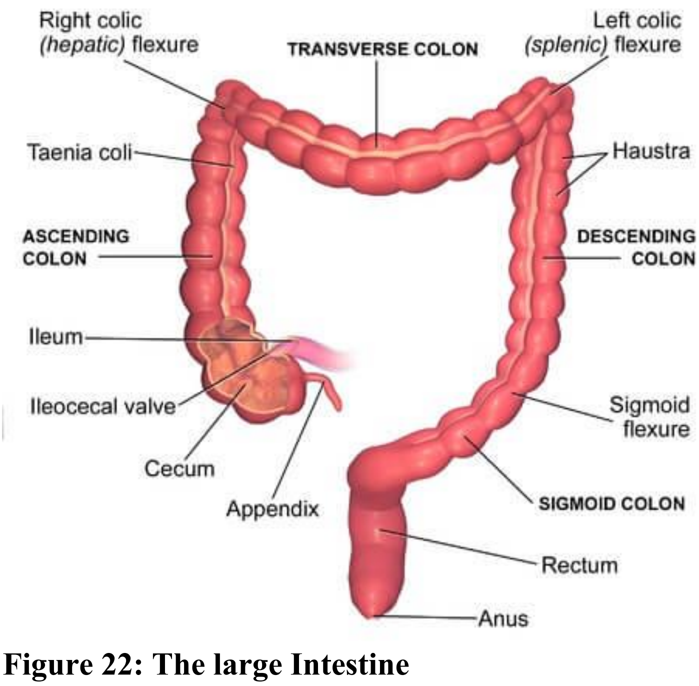

Figure 22: The large Intestine

The caecum

- This is the first part of the colon. It is a dilated region which has a blind end inferiorly and is continuous with the ascending colon superiorly.
- Just below the junction of the two, the ileocaecal valve opens from the ileum.
- The vermiform appendix is a fine tube, closed at one end, which leads from the caecum. It is usually about 13 cm long and has the same structure as the walls of the colon but contains more lymphoid tissue.

The ascending colon

- This passes upwards from the caecum to the level of the liver where it curves acutely to the left at the hepatic flexure to become the transverse colon.

The transverse colon

- This is a loop of the colon which extends across the abdominal cavity in front of the duodenum and the stomach to the area of the spleen where it forms the splenic flexure and curves acutely downwards to become the descending colon.

The descending colon

- This passes down the left side of the abdominal cavity then curves towards the midline. After it enters the true pelvis it is known as the sigmoid colon.

The sigmoid colon

- This part describes an S-shaped curve in the pelvis then continues downwards to become the rectum.

The Rectum and the anal canal

- It is a slightly dilated section of the colon about 13 cm long. It leads from the sigmoid colon and terminates in the anal canal.
- The anal canal is a short passage about 3.8 cm long in the adult and leads from the rectum to the exterior.
- Two sphincter muscles control the anus; the internal sphincter, consisting of smooth muscle fibers, is under the control of the autonomic nervous system and the external sphincter, formed by skeletal muscle, is under voluntary control.

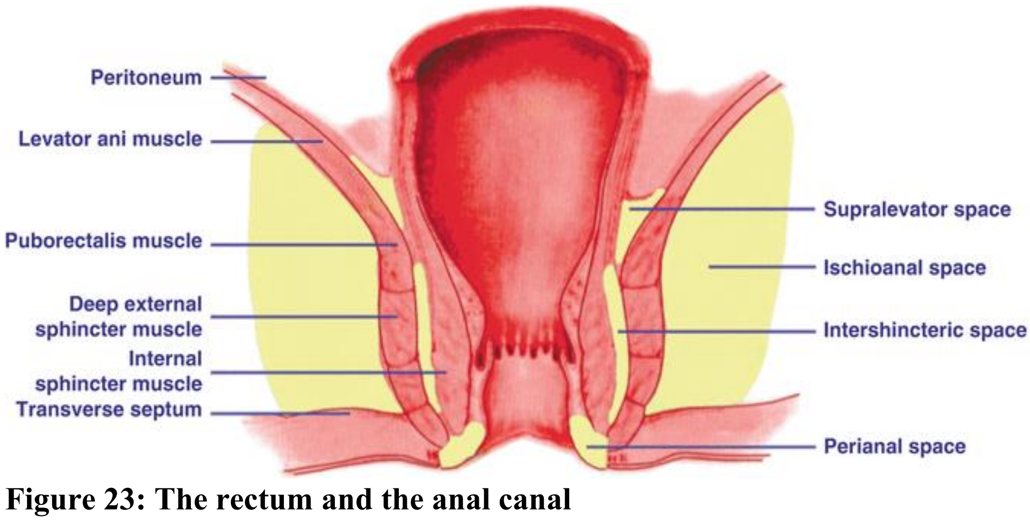

Figure 23: The rectum and the anal canal

Functions of the large intestine, rectum and anal canal

Absorption

- The contents of the ileum which pass through the ileocaecal valve into the caecum are fluid, even though some water has been absorbed in the small intestine.
- In the large intestine absorption of water continues until the familiar semisolid consistency of feces is achieved.

- Mineral salts, vitamins, and some drugs are also absorbed into the blood capillaries from the large intestine.

Microbial activity

- The large intestine is heavily colonized by certain types of bacteria, which synthesize vitamin K and folic acid. They include *Escherichia coli, Enterobacter aerogenes, Streptococcus faecalis,* and *Clostridium perfringens* (welchii).

Defaecation

- Usually, the rectum is empty, but when a mass movement forces the contents of the sigmoid colon into the rectum the nerve endings in its walls are stimulated by a stretch.
- Defaecation involves involuntary contraction of the muscle of the rectum and relaxation of the internal anal sphincter.
- Contraction of the abdominal muscles and lowering of the diaphragm increases the intra-abdominal pressure (Valsalva's maneuver) and so assists the process of defaecation.

1.2.12.4: Self Assessment

a). Highlight parts of the large intestine.
b). Highlight functions of the stomach

1.2.12.4.1: Responses

a). Highlight parts of the large intestine

The caecum

- This is the first part of the colon. It is a dilated region which has a blind end inferiorly and is continuous with the ascending colon superiorly.
- Just below the junction of the two, the ileocaecal valve opens from the ileum.
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The sigmoid colon

- This part describes an S-shaped curve in the pelvis then continues downwards to become the rectum.

b). Highlight functions of the stomach

- Temporary storage allowing time for the digestive enzymes, pepsins, to act.
- Chemical digestion — pepsins convert proteins to polypeptides.
- Mechanical breakdown — the three smooth muscle layers enable the stomach to act as a churn, gastric juice is added and the contents are liquefied to chime.
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- Preparation of iron for absorption further along the track — the acid environment of the stomach solubilizes iron salts, which is required before iron can be absorbed
- Production of intrinsic factor needed for absorption of vitamin B12 in the terminal ileum
- Regulation of the passage of gastric contents into the duodenum. When the chyme is sufficiently acidified and liquefied, the pyloric antrum forces small jets of gastric contents through the pyloric sphincter into the duodenum.

1.2.13: Learning outcome 13: Demonstrate knowledge of the urinary system and its function

1.2.13.1: Introduction to learning outcome

This learning outcome specifies the competencies required to identify Components of the urinary system, describe structure and functions of components and describe process of urine formation.

1.2.13.2: Performance Standard

1.19 **Components** of urinary system are identified
1.20 Structure and functions of components are described
1.21 **Process of urine formation** is described

1.2.13.3:Information Sheet

Thousands of metabolic processes in myriad body cells produce hundreds of waste products. The urinary system removes them by filtering and cleansing the blood as it passes through the kidneys. Another vital function is the regulation of the volume, acidity, salinity, concentration, and chemical composition of blood, lymph, and other body fluids. Under hormonal control, the kidneys continually monitor what they release into the urine to maintain a healthy chemical balance.

The urinary system is composed of a pair of kidneys, a pair of ureters, a bladder, and a urethra. These components together carry out the urinary system's function of regulating the volume and composition of body fluids, removing waste products from the blood, and expelling the waste and excess water from the body in the form of urine. The two kidneys are reddish organs resembling beans in shape that are situated on either side of the abdomen just above the waist and towards the back of the body. The kidneys contain microscopic filtering units that remove waste, unwanted minerals, and excess water from the blood as urine. Each kidney is connected to the bladder by a long tube called a ureter, which transports urine away.

The bladder is a hollow, muscular organ situated centrally in the pelvis; it stores urine until it is convenient to release it. At a certain volume, stretch receptors in its wall transmit nervous impulses that initiate a conscious desire to urinate. The urethra then conducts urine from the bladder to the outside.

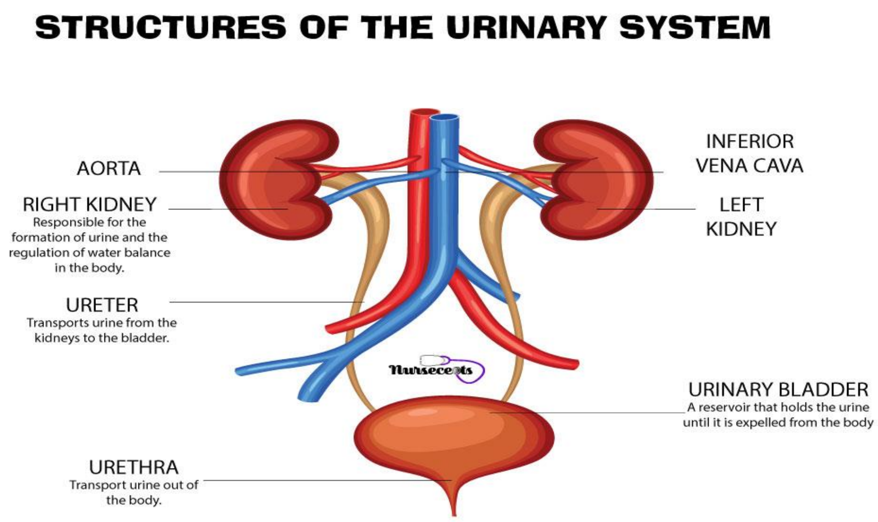

Figure 24: The urinary system

THE KIDNEYS
The kidneys sit at the back of the abdominal wall and at the start of the urinary system. These organs are constantly at work:

- Nephrons, tiny structures in the renal pyramids, filter gallons of blood each day.
- The kidneys reabsorb vital substances, remove unwanted ones, and return the filtered blood back to the body.
- As if they weren't busy enough, the kidneys also create urine to remove all the waste.

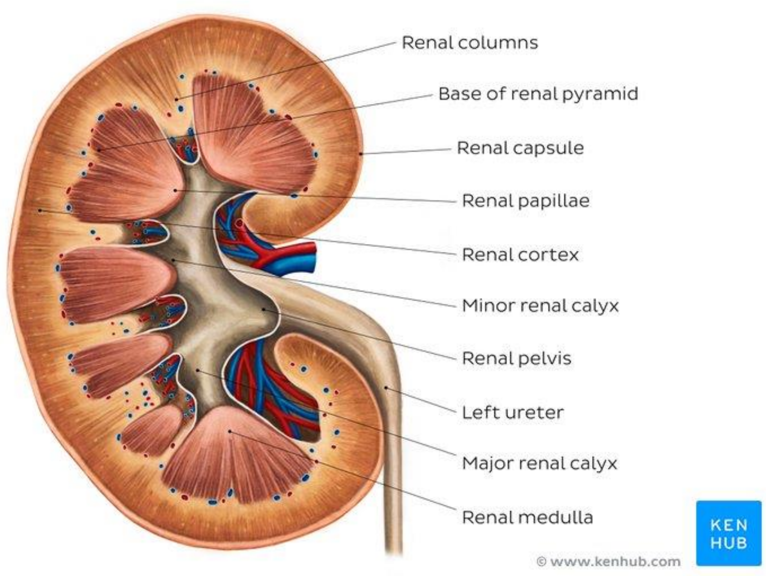

Figure 25: The kidney

The kidneys are located behind the peritoneum, and so are called retroperitoneal organs. They sit in the back of the abdomen between the levels of the T12 and L03 vertebrae. The right kidney is slightly lower than the left kidney to accommodate the liver. Both kidneys are bean-shaped and about the size of an adult fist.

Blood enters the kidneys through renal arteries. These arteries branch into tiny capillaries that interact with urinary structures inside the kidneys (namely the nephrons). Here the blood is filtered. Waste is removed and vital substances are reabsorbed back into the bloodstream. The filtered blood leaves through the renal veins.

Each kidney contains over 1 million tiny structures called nephrons. The nephrons are located partly in the cortex and partly inside the renal pyramids, where the nephron tubules make up most of the pyramid mass. Nephrons perform the primary function of the kidneys: regulating the concentration of water and other substances in the body. They filter the blood, reabsorb what the body needs, and excrete the rest as urine.

The kidneys filter unwanted substances from the blood and produce urine to excrete them.
There are 3 main steps of urine formation:

- Glomerular filtration,
- Reabsorption
- Secretion.

These processes ensure that only waste and excess water are removed from the body.

Each nephron has a glomerulus, the site of blood filtration. The glomerulus is a network of capillaries surrounded by a cuplike structure, the glomerular capsule (or Bowman's capsule). As blood flows through the glomerulus, blood pressure pushes water and solutes from the capillaries into the capsule through a filtration membrane. This glomerular filtration begins the urine formation process.

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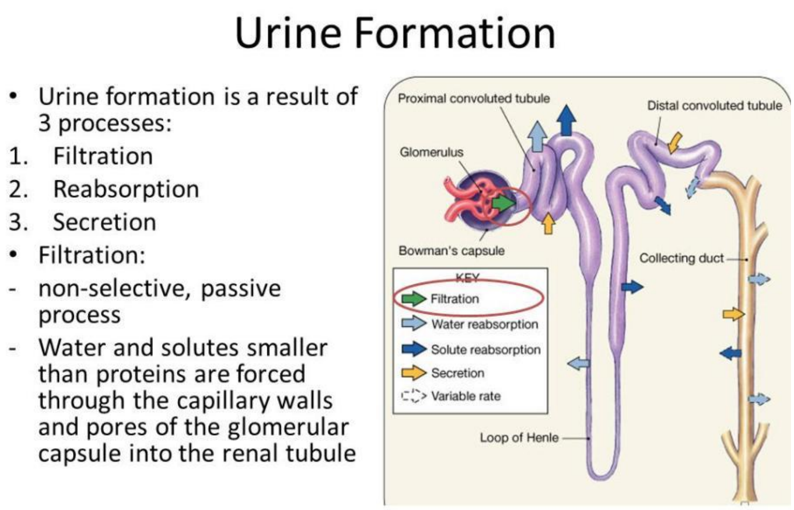

Figure 26: Urine formation

The kidneys filter unwanted substances from the blood and produce urine to excrete them. There are 3 main steps of urine formation:

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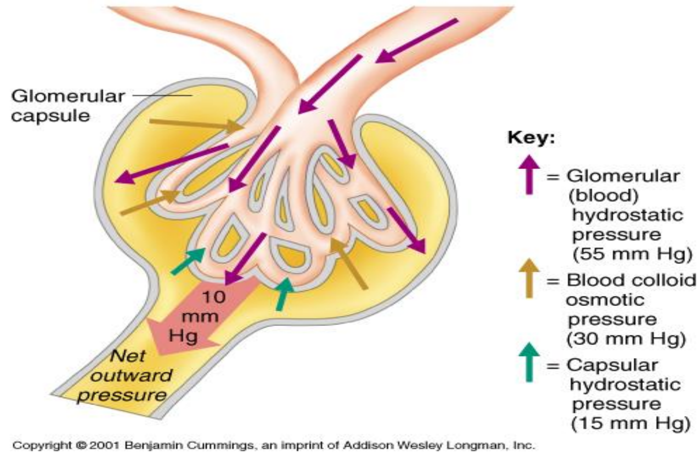

Figure 27: Glomerular filtration

The nephrons of the kidneys process blood and create urine through a process of filtration, reabsorption, and secretion. Urine is about 95% water and 5% waste products. Nitrogenous wastes excreted in urine include urea, creatinine, ammonia, and uric acid. Ions such as sodium, potassium, hydrogen, and calcium are also excreted.

Urine produced in the kidneys travels down the ureters into the urinary bladder. The bladder expands like an elastic sac to hold more urine. As it reaches capacity, the process of micturition, or urination, begins. Involuntary muscle movements send signals to the nervous system, putting the decision to urinate under conscious control.

The internal urethral sphincter and the external urethral sphincter both provide muscle control for the flow of urine. The internal sphincter is involuntary. It surrounds the opening of the bladder to the urethra and relaxes to allow urine to pass. The external sphincter is voluntary. It surrounds the urethra outside the bladder and must be relaxed for urination to occur.

Smooth muscle stretch initiates the micturition reflex by activating stretch receptors in the bladder wall. This autonomic reflex causes the detrusor muscle to contract and the internal urethral sphincter muscle to relax, allowing urine to flow into the urethra. The stretch receptors also send a message to the thalamus and the cerebral cortex, giving voluntary control over the external urethral sphincter. We usually gain this control of urination between the ages of 2 and 3, as our brains develop.

1.2.13.4: Self Assessment

a). State the steps involved in urine formation**.**

b). Describe the kidney

1.2.13.4.1: Responses

a). State the steps involved in urine formation

- Glomerular filtration,
- Reabsorption
- Secretion

b). Describe the kidney

The kidneys are located behind the peritoneum, and so are called retroperitoneal organs. They sit in the back of the abdomen between the levels of the T12 and L03 vertebrae. The right kidney is slightly lower than the left kidney to accommodate the liver. Both kidneys are bean-shaped and about the size of an adult fist.

Blood enters the kidneys through renal arteries. These arteries branch into tiny capillaries that interact with urinary structures inside the kidneys (namely the nephrons). Here the blood is filtered. Waste is removed and vital substances are reabsorbed back into the bloodstream. The filtered blood leaves through the renal veins.

Each kidney contains over 1 million tiny structures called nephrons. The nephrons are located partly in the cortex and partly inside the renal pyramids, where the nephron tubules make up most of the pyramid mass. Nephrons perform the primary function of the kidneys: regulating the concentration of water and other substances in the body. They filter the blood, reabsorb what the body needs, and excrete the rest as urine.

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1.2.14: Learning outcome 14: Demonstrate knowledge of the respiratory system and its function

1.2.14.1: Introduction to learning outcome

This learning outcome specifies the competencies required to identify Components of the respiratory system, describe the structure and function of components and describe the respiration process.

1.2.14.2:Performance standard

1.22 **Components** of respiratory system are identified
1.23 Structure and functions of components are described
1.24 Respiration process is described

1.2.14.3:Information sheet

In biology, respiration has two meanings: - at the cellular level, it refers to the chemical reactions that take place in the mitochondria, which require oxygen, and are the principle source of energy for eukaryotic cells; - at the level of the whole organism, it refers to the process of taking in oxygen from the environment and giving carbon dioxide back to it.

Cells need oxygen to generate ATP in their mitochondria
• Of all organs, your liver has the greatest need: 81 litres of oxygen a day, while your brain requires 76 litres.
• A top marathoner uses about 500 litres of oxygen in the course of a race.
• The main physiological difference between a sedentary person and a runner is the number of mitochondria per cell, which increases with training.

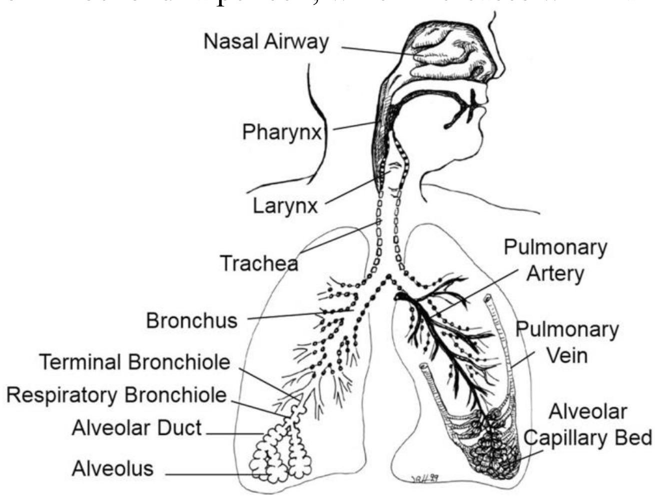

Figure28: The respiratory system

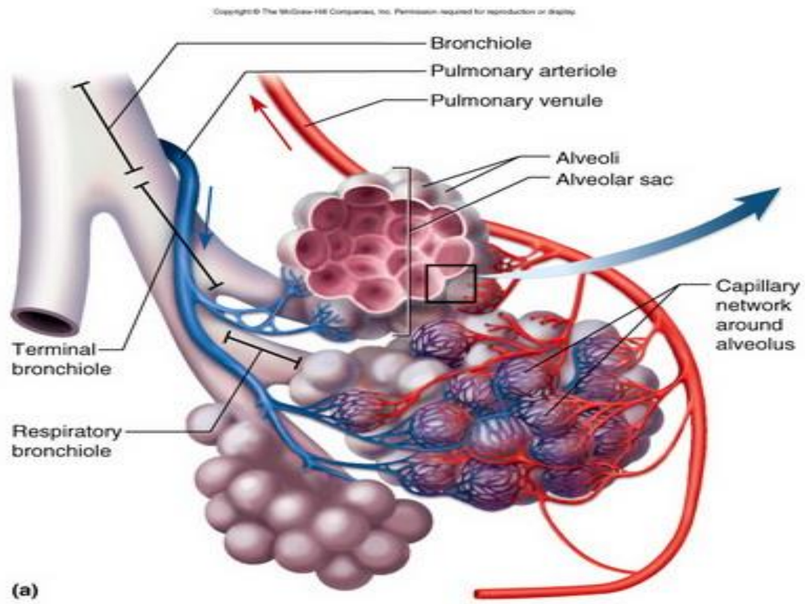

Figure 29: The alveoli

A pair of human lungs has some 300 million alveoli, where gas exchange takes place, providing a respiratory surface of about 70 square meters.

The human respiratory system

- Air enters through the nose or mouth and passes into the pharynx, past the larynx, and down the trachea, bronchi and bronchioles to the alveoli in the lungs.
- Gas exchange takes place in the alveoli. The barrier between the air in an alveolus and the blood in its capillaries is only 0.5 micrometer.
- Oxygen and carbon dioxide diffuse into and out of the bloodstream through the capillaries surrounding the walls of the alveoli.

The trachea, bronchi and bronchioles, which serve mainly to transport air by bulk flow, are lined with epithelial cells.
• These include both mucus-secreting and ciliated cells.
• The mucus coats the epithelium of the respiratory system and traps foreign particles.
• The cilia beat continuously, pushing mucus and foreign particles toward the pharynx, to be expelled.

Mechanics of respiration
• Air flows into or out of the lungs when the air pressure within the alveoli differs from the pressure of the external air atmospheric pressure).
• The pressure in the lungs is varied by changes in the volume of the thoracic cavity, caused by the contraction and relaxation of the diaphragm and of intercostal ("between the-ribs") muscles.

Transport and exchange of gases
• Oxygen is almost insoluble in blood plasma.

• The blood transports oxygen thanks to respiratory pigments which, in vertebrates, are called hemoglobin. Each hemoglobin molecule can bind four molecules of oxygen.
• In the capillaries of the alveoli, where the partial pressure of oxygen is high, most of the hemoglobin is combined with oxygen.
• In the tissues, where the partial pressure is lower, oxygen is released from the hemoglobin molecules into the plasma and diffuses into the tissues.

Carbon dioxide is more soluble than oxygen and some of it is dissolved in the blood.
• However, most CO_2 reacts with water to form carbonic acid, a weak acid that dissociates to form bicarbonate (HCO_3^-) and hydrogen (H^+) ions:
• $CO_2 + H_2O$ □ H_2CO_3 □$HCO_3 + H^+$
Carbon Water Carbonic Bicarbonate Hydrogen
dioxide acid ion ion
• The reaction can go in either direction, depending on the partial pressure of carbon dioxide in the blood.

Control of Respiration
• The rate and depth of respiration are controlled by respiratory neurons in the brainstem which activate motor neurons in the spinal cord that cause the diaphragm and intercostal muscles to contract.
• In addition, chemoreceptor cells, located in the carotid arteries, signal the respiratory neurons when the concentration of oxygen decreases.
• Centres in the brain and chemoreceptors simultaneously monitor the concentration of dissolved carbon dioxide and hydrogen.
• The system is highly sensitive to the slightest changes in the chemical composition of the blood (particularly hydrogen ion, which reflects the concentration of carbon dioxide).

1.2.14.4: Self Assessment
a). Describe the mechanics of respiration
b). Describe the movement of air from the atmosphere to the alveoli.

1.2.14.4.1: Responses
a). Describe the mechanics of respiration
• Air flows into or out of the lungs when the air pressure within the alveoli differs from the pressure of the external air atmospheric pressure).
• The pressure in the lungs is varied by changes in the volume of the thoracic cavity, caused by the contraction and relaxation of the diaphragm and of intercostal ("between the-ribs") muscles
b). Describe the movement of air from the atmosphere to the alveoli.
- Air enters through the nose or mouth and passes into the pharynx, past the larynx, and down the trachea, bronchi and bronchioles to the alveoli in the lungs.
- Gas exchange takes place in the alveoli. The barrier between the air in an alveolus and the blood in its capillaries is only 0.5 micrometer.
- Oxygen and carbon dioxide diffuse into and out of the bloodstream through the capillaries surrounding the walls of the alveoli.

1.2.14.5: References

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1.2.15:Learning outcome 15: Demonstrate knowledge of the special senses and their functions

1.2.15.1:Introduction to learning outcome

This learning outcome specifies the competencies required to identify Sensory organs are identified, outline structures of special organs and describe their functions.

1.2.15.2:Performance standard

1.25 Sensory organs are identified
1.26 Structures of special organs are outlined
1.27 Functions of special organs are described

1.2.15.3:Information sheet

Sense organs

Sensory Receptors - receive input, generate receptor potentials and with enough summation, generate action potentials in the neurons they are part of or synapse with

5 Types of Sensory Receptors - based on the type of stimuli they detect:
1. **Mechanoreceptors** - pressure receptors, stretch receptors, and specialized mechanoreceptors involved in movement and balance.
2. **Thermoreceptors** - skin and viscera, respond to both external and internal temperature
3. **Pain receptors** - stimulated by lack of O_2, chemicals released from damaged cells and inflammatory cells
4. **Chemoreceptors** - detect changes in levels of O_2, CO_2, and H^+ ions (pH) as well as chemicals that stimulate taste and smell receptors

5. **Photoreceptors** - stimulated by light

Distribution of Receptors in the body:
Special Senses
• mediated by relatively complex sense organs of the head, innervated by cranial nerves
• vision, hearing, equilibrium, taste and smell
General (somesthetic, somatosensory)
• receptors widely distributed in skin, muscles, tendons, joints, and viscera
• they detect touch, pressure, stretch, heat, cold and pain, blood pressure

Special Senses
Sensation and perception
• Vision – Eye
• Hearing – Ear
• Equilibrium – Ear
• Taste – Taste receptors
• Smell – Olfactory system

General Senses
• Skin – Hot, cold, pressure, pain
• Muscles, joints, and tendons – proprioceptors- stretch receptors respond to stretch or compression
• Pain Receptors – somatic or visceral

Eye - Vision
Processes
• Light energy is transduced into neural activity
• Neural activity is processed by the brain
• ***Note:*** For an analogy, you can imagine taking a picture with a camera. The eye is the camera, the retina, which is a specialized part of the brain at the back of the eye, is the film, and the parts of the brain that process visual information is the photoshop.

Human visual systems permit light reflected off distant objects to be:
• Localized relative to the individual within his or her environment
• Identified based on size, shape, color, and past experience
• Perceived to be moving (or not)
• Detected in a wide variety of lighting conditions

Sequence of events
• Light entering the eye is focused on the retina
• Retina converts light energy into neuronal activity
• Axons of the retinal neurons are bundled to form the optic nerves
• Visual information is distributed to several brain structures that perform different functions.

Eye **–** the organ used to sense light
Three layers –
1. ***Outer layer*** consists of sclera and cornea
2. ***Middle layer*** consists of choroid, ciliary body and iris
3. ***Inner layer*** consists of retina

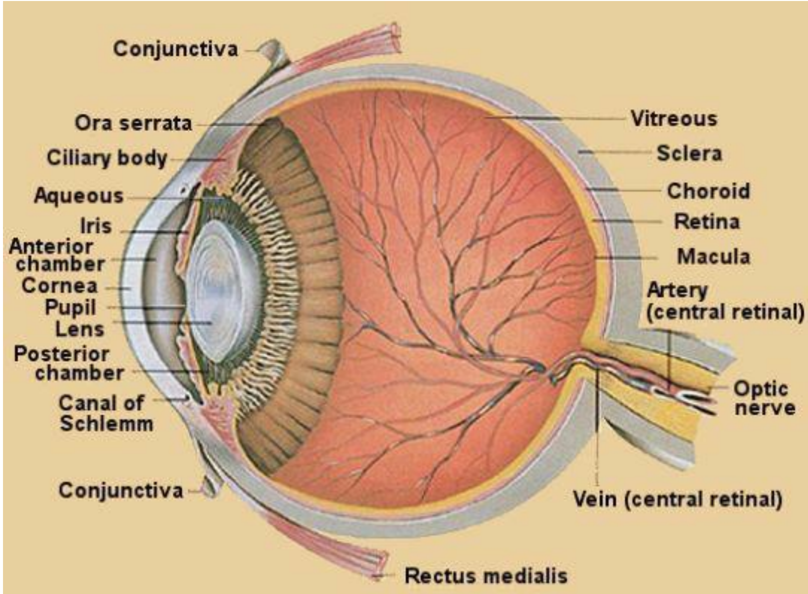

Figure 30: The eye

Extraocular muscles--attached to the eye and skull and allow movement

Anatomy of the Eye
Gross anatomy
Functions of the major parts of the eye:
• **Sclera or Scleroid Layer – (white of eye)** the outermost layer that forms the eyeball- a tough protective layer of connective tissue that helps maintain the shape of the eye and provides an attachment for the muscles that move the eye
• **Conjunctiva**--membrane inside the eyelid attached to the sclera
• **Cornea** - the transparent surface covering the iris and pupil- a clear, dome-shaped part of the sclera covering the front of the eye through which light enters the eye
• **Anterior Chamber** – a small chamber between the cornea and the pupil
• **Aqueous Humor** - fluid behind the cornea - the clear fluid that fills that anterior chamber of the eye and helps to maintain the shape of the cornea providing most of the nutrients for the lens and the cornea and involved in waste management in the front of the eye
• **Choroid Layer** - middle layer of the eye containing may blood vessels
• **Ciliary Body** - the ciliary body is a circular band of muscle that is connected and sits immediately behind the iris- produces aqueous humor, changes shape of lens for focusing, and
• **Iris** - circular muscle that controls the diameter of the pupil - the pigmented front portion of the choroid layer and contains the blood vessels - it determines the eye color and it controls the amount of light that enters the eye by changing the size of the pupil (an albino only has the blood vessels – not pigment so it appears red or pink because of the blood vessels)
• **Lens** - a crystalline structure located just behind the iris - it focuses light onto the retina
• **Pupil** - the opening in the center of the iris- it changes size as the amount of light changes (the more light, the smaller the hole) and it allows light to reach the retina
• **Vitreous** - a thick, transparent liquid that fills the center of the eye - it is mostly water and gives the eye its form and shape (also called the **vitreous humor**)
• **Retina** - axons of the retina leaving the eye - sensory tissue that lines the back of the eye. It contains millions of photoreceptors (**rods for black & white and cones for color**) that convert light rays into electrical impulses that are relayed to the brain via the optic nerve
• **Optic nerve** - the nerve that transmits electrical impulses from the retina to the brain

Opthalmoscopic appearance (Retina as seen through the pupil)
• **Note:** in photographs, the red appearance of the eye is actually the retina photographed. Double flash camera causes the pupil to constrict.
• **Optic disk (blind spot)--**no vision is possible
o Blood vessels originate here. The vessels
shadow the retina
o Optic nerve fibers exit here
o No photoreceptors
• **Macula**--area of the retina responsible for central vision (vs. peripheral)
• **Fovea**--center of the retina (where most of the cones are)

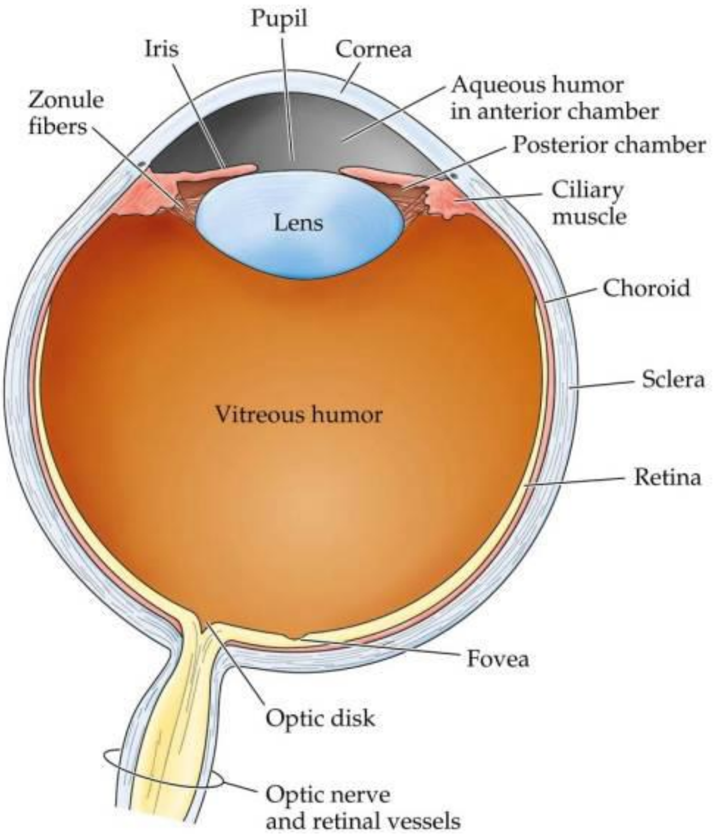

Figure 31: The eye

Common eye defects include
• **Myopia** or nearsightedness where the eyeball is too long or the cornea is too steep;
• **Hyperopia** or far sightedness where the eyeball is short or lens cannot become round enough:
• **Presbyopia** where the muscles controlling the bulging of the lens become weak as we age;
• **Cataracts** where the lens becomes fogged;
• **Nyctalopia** or night blindness where vision is impaired in dim light and in the dark due to pigment rhodospin in the rods not functioning properly External features of the eye.

Cross sectional anatomy
• **Lens--**transparent surface that contributes to the formation of images (w/i 9 meters)
• **Ciliary muscles**--change the shape of the lens and allow focusing

• **Vitreous humor**--more viscous than the aqueous humor - Lies between the lens and the retina and provides spherical shape
• **Retina -** inner most layer of cells at the back of the eye - Transduces light energy into neural activity

EAR – HEARING

Outer Ear & ear canal – brings sound into eardrum
Eardrum – vibrates to amplify sound & separates inner and middle ear
Middle ear has 3 small bones or **Ossicles** = anvil, stirrup, stapes – amplify sound (small bones) which vibrate sound.
Eustachian tube – connects middle ear to throat and equalizes pressure on eardrum
Cochlea – in inner ear – has receptors for sound & sends signals to brain via Auditory Nerve
Process of hearing:
• Sound waves enter your outer ear and travel through your ear canal to the middle ear.
• The ear canal channels the waves to your eardrum, a thin, sensitive membrane stretched tightly over the entrance to your middle ear.
• The waves cause your eardrum to vibrate.
• It passes these vibrations on to the hammer, one of three tiny bones in your ear. The hammer vibrating causes the anvil, the small bone touching the hammer, to vibrate. The anvil passes these vibrations to the stirrup, another small bone which touches the anvil. From the stirrup, the vibrations pass into the inner ear.
• The stirrup touches a liquid filled sack and the vibrations travel into the cochlea, which is shaped like a shell.
• Inside the cochlea, a vestibular system formed by three semicircular canals that are approximately at right angles to each other and which are responsible for the sense of balance and spatial orientation.
It has chambers filled with a viscous fluid and small particles (**otoliths**) containing calcium carbonate. The movement of these particles over small hair cells in the inner ear sends signals to the brain that are interpreted as motion and acceleration. The brain processes the information from the ear and lets us distinguish between different types of sounds.\

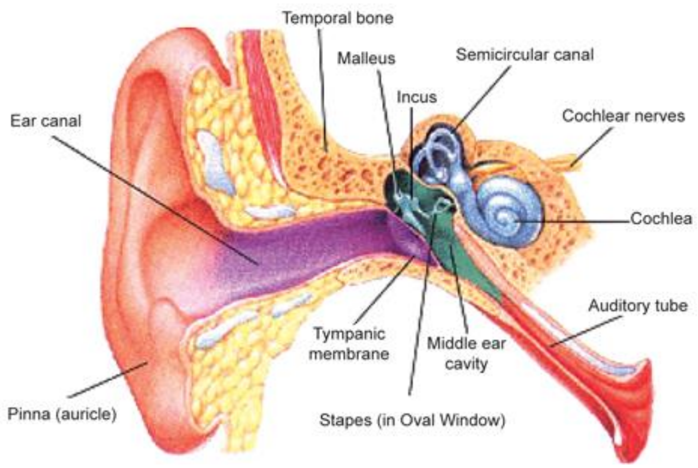

Figure 32: The ear

Ear – Equilibrium

Equilibrium
• Equilibrium is a response to movements of the head - Example: a cat landing on its feet if dropped from upside down
• Vestibular Apparatus: the equilibrium receptors of the inner ear
• Divided into static and dynamic equilibrium
Static Equilibrium
• When the body is not moving
• **Maculae**: receptors within the membrane sacs of the vestibule that report on the position of the head with respect to the pull of gravity when the body is not moving.
• Each macula is a patch of receptor cells with their "hairs" embedded in the otolithic membrane
• **Otolithic Membrane**: a jelly-like substance containing otoliths
• **Otoliths**: tiny stones made of calcium salts that roll in response to changes in the pull of gravity. When otoliths move, they pull on the gel and this bends the hairs. Activated hair cells send impulses along the vestibular nerve
• **Vestibular Nerve:** (Cranial Nerve VIII) transmits
signals to the cerebellum
Dynamic Equilibrium
• Receptors in the semicircular canals respond to angular or rotary movements of the head.
• Semicircular canals are oriented in the three planes of space
• **Crista Ampullaris**: receptor region that consists of a tuft of hair cells covered with a gelatinous cap called the **cupula**
• When the head moves in an angular direction:
o The endomlymph lags behind
o As the cupula drags against the stationaryendolymph, the cupula bends
o This stimulates hair cells to transmit signals to the vestibular nerve
• When you are moving at a constant rate, receptors stop sending impulses
• You no longer have the sensation of motion until you change speed or direction
Vision plays a significant role in balance. Approximately twenty percent of the nerve fibers from the eyes interact with the vestibular system.

Taste and Smell – Chemical Receptors
Taste buds
• The mouth contains around 10,000 taste buds, most of which are located on and around the tiny bumps on your tongue.
• Every taste bud detects **five primary tastes**:
o Sour
o Sweet
o Bitter
o Salty
o Umami - salts of certain acids (for example monosodium glutamate or MSG)
• Each of your taste buds contains 50-100 specialised receptor cells.
• Sticking out of every single one of these receptor cells is a tiny taste hair that checks out the food chemicals in your saliva.
• When these taste hairs are stimulated, they send nerve impulses to your brain.
• Each taste hair responds best to one of the five basic tastes.
Smell Receptors or Olfactory receptors
• Humans able to detect thousands of different smells
• Olfactory receptors occupy a stamp-sized area in the roof of the nasal cavity, the hollow space inside the nose

• Tiny hairs, made of nerve fibers, dangle from all your olfactory receptors. They are covered with a layer of mucus.
• If a smell, formed by chemicals in the air, dissolves in this mucus, the hairs absorb it and excite your olfactory receptors.
• A few molecules are enough to activate these extremely sensitive receptors.
• Olfactory Hairs easily fatigued so you do not notice smells
• Linked to memories - when your olfactory receptors are stimulated, they transmit impulses to your brain and the pathway is directly connected to the limbic system - the part of your brain that deals with emotions so you usually either like or dislike a smell
• Smells leave long-lasting impressions and are strongly linked to your memories
• **Much of what we associate as taste also involves smell – that is why hot foods "taste" different than "cold" foods.**

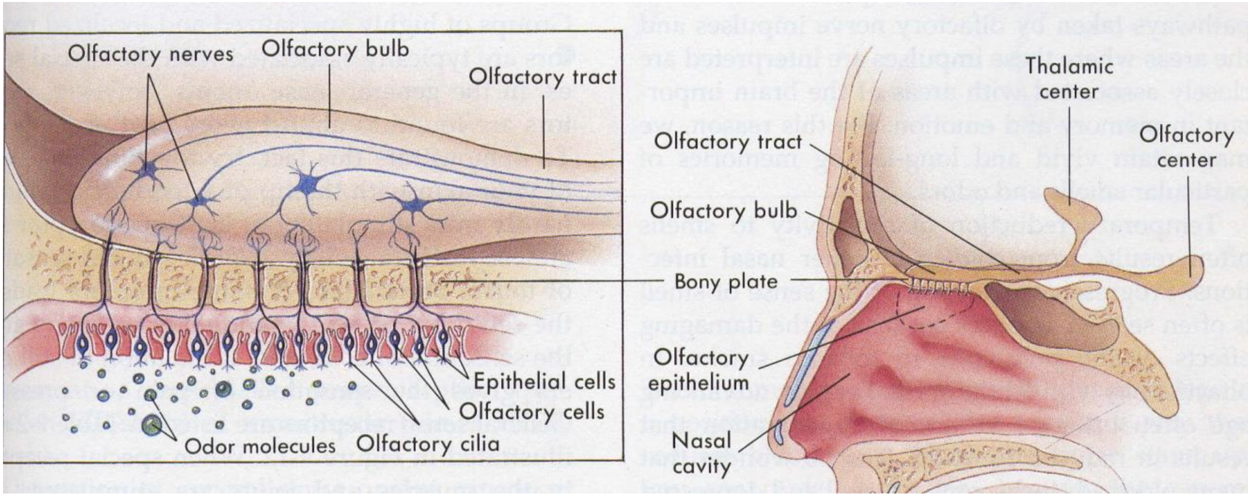

Figure 33: Smell receptors

1.2.15.4: Self Assessment
a). Highlight five types of sensory receptors
b). Describe the process of hearing
c). Highlight five common eye defects

1.2.15.4.1: Responses
a**). Highlight five types of sensory receptors**
1. **Mechanoreceptors** - pressure receptors, stretch receptors, and specialized mechanoreceptors involved in movement and balance.
2. **Thermo receptors** - skin and viscera, respond to both external and internal temperature
3. **Pain receptors** - stimulated by lack of O2, chemicals released from damaged cells and inflammatory cells
4. **Chemoreceptors** - detect changes in levels of O2, CO2, and H+ ions (pH) as well as chemicals that stimulate taste and smell receptors
5. **Photoreceptors** - stimulated by light

b). Describe the process of hearing
Sound waves enter your outer ear and travel through your ear canal to the middle ear.
• The ear canal channels the waves to your eardrum, a thin, sensitive membrane stretched tightly over the entrance to your middle ear.

• The waves cause your eardrum to vibrate.
• It passes these vibrations on to the hammer, one of three tiny bones in your ear. The hammer vibrating causes the anvil, the small bone touching the hammer, to vibrate. The anvil passes these vibrations to the stirrup, another small bone which touches the anvil. From the stirrup, the vibrations pass into the inner ear.
• The stirrup touches a liquid filled sack and the vibrations travel into the cochlea, which is shaped like a shell.
• Inside the cochlea, a vestibular system formed by three semicircular canals that are approximately at right angles to each other and which are responsible for the sense of balance and spatial orientation.
It has chambers filled with a viscous fluid and small particles (**otoliths**) containing calcium carbonate. The movement of these particles over small hair cells in the inner ear sends signals to the brain that are interpreted as motion and acceleration. The brain processes the information from the ear and lets us distinguish between different types of sounds

c). Highlight five common eye defects
• **Myopia** or nearsightedness where the eyeball is too long or the cornea is too steep;
• **Hyperopia** or far sightedness where the eyeball is short or lens cannot become round enough:
• **Presbyopia** where the muscles controlling the bulging of the lens become weak as we age;
• **Cataracts** where the lens becomes fogged;
• **Nyctalopia** or night blindness where vision is impaired in dim light and in the dark due to pigment rhodospin in the rods not functioning properly External features of the eye.

1.2.15.5: References

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1.2.16: Learning outcome 16: Demonstrate knowledge of the reproductive system and its functions

1.2.16.1: Introduction to learning outcome

This learning outcome specifies the competencies required to identify Components of the human reproductive system, describe the structure and components as well as the fertilization process.

1.2.16.2: Performance standard

1.28 **Components** of the ***human reproductive system*** are identified
1.29 Structure and functions of components are described
1.30 Fertilization process is described

1.2.16.3: Information sheet

Function of the reproductive system

Sexual reproduction requires a male and a female of the same species to copulate and combine their genes in order to produce a new individual who is genetically different from his parents. Sexual reproduction relies on meiosis to shuffle the genes , so that new combinations of genes occur in each generation , allowing some of the offspring of survive in the constantly – changing environment . The male reproductive system produces, sustains, and delivers sperm cells (spermatozoa) to the female reproductive tract. The female reproductive system produces, sustains , and allows egg cells (oocytes) to be fertilized by sperm . It also supports the development of an offspring (gestation) and gives birth to a new individual (parturition) .

Male Reproductive System

- **Testis**: Sex organ that produces sperm in a process called spermatogenesis, and male sex hormones (testosterone).
- Developed in a male fetus near the kidneys, and descend to the scrotum about 2 months before birth.
- Each testis is enclosed by a layer of fibrous connective tissue called tunica alumina.
- Each testis contains about 250 functional units called lobules; each lobule contains about 4 seminiferous tubules where spermatogenesis occurs.
- All somniferous tubules in a testis converge and form a channel called rate testis.

Testis

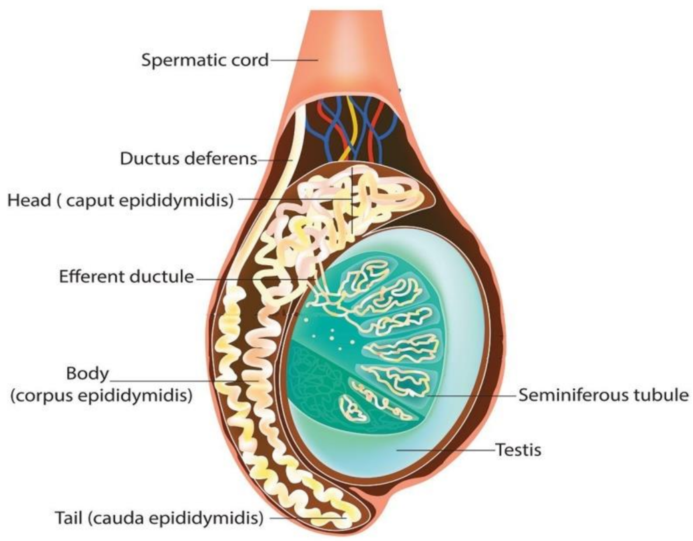


Figure 33: The Testis

Scrotum: A pouch – like cutaneous extension that contains the two testes. Located outside of pelvic cavity to prevent overheating of testes [internal temperature of scrotum is always about 3 ˚F below body temperature] .

Epididymis: An expanded tubule from the rate testis where sperm is stored (for about 3 days) , matured and become fully functional. Contains cilia on its columnar epithelium that help move sperm toward vas deferens during ejaculation. .
Vas deferens: A tubule (about 10 inches long) that connects epididymis to the urethra for transporting sperm during ejaculation. It contains smooth muscle that undergoes rapid peristalsis during ejaculation.

Accessory sex glands

Seminal vesicles: secrete an alkaline solution that makes up 60% of the semen volume; this seminal fluid contains fructose (nutrient for the sperm) and prostaglandins (substances that stimulate uterine contraction during sexual excitation).
Prostate gland: secretes a slightly acidic, milky white fluid that makes up about 30% of semen volume ; this fluid helps neutralize the pH of semen and vaginal secretion.
Bulb urethral gland: secretes a clear lubricating fluid that aids in sexual intercourse.

Reproductive organs of the male

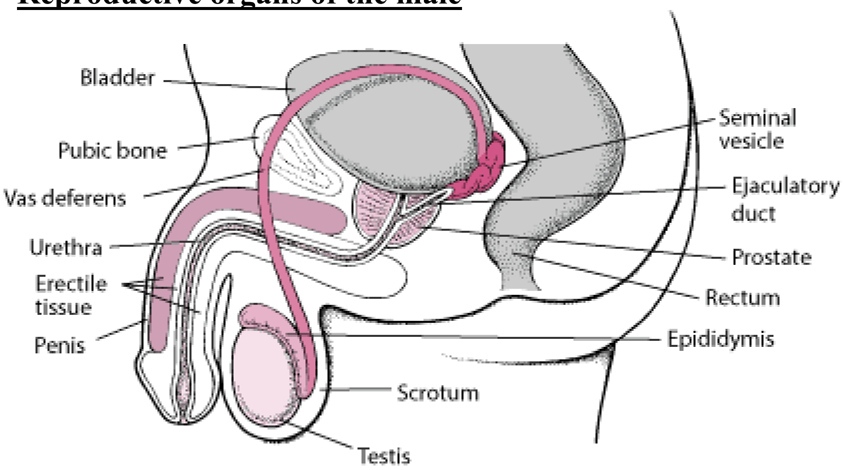


Figure 34

Urethra: a tubule located inside the penis for urine excretion and semen ejaculation .Contains smooth muscle that performs rapid peristalsis during ejaculation .
Penis: a copulatory organ that is responsible for delivering the sperm to the female reproductive tract. Contains 2 erectile tissues called corpus cavernosa and corpus spongiosum, where the latter one enlarges and forms the glans penis due to increased blood flow during sexual excitation.
During sexual excitement, parasympathetic nerves cause vasodilatation in the penis, allowing erectile tissues to swell and erect the penis.
During ejaculation, sympathetic nerves cause vas deferens, urethra and erectile tissues to contract, forcefully expelling semen (a mixture of sex gland fluids and about 300 million sperm) outward.

Seminiferous Tubules

About 1,000 seminiferous tubules in each testis conduct spermatogenesis.
Between the tubules are specialized glandular cells called interstitial cells (or leydig's cells) which produce testosterone.
Inside the tubules are specialized cells called sertoli's cells which support and nourish the sperm.

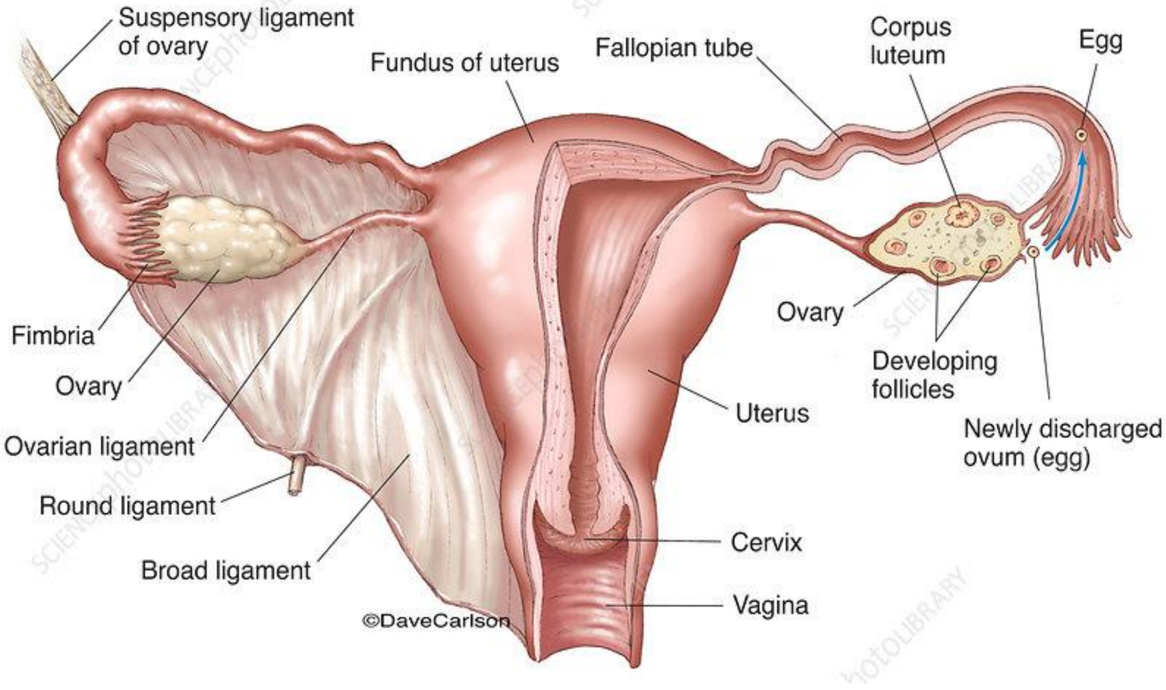
Spermatogenesis

Spermatogonia (containing 46 chromosomes) undergo DNA replication and produce primary spermatocytes (with 46 pairs of chromosomes) .[some spermatozoid undergo mitosis to maintain a large population , so that spermatogenesis can be continuous for many decades].
Primary spermatocytes undergo "crossing - over" to shuffle their genes ,and undergo meiosis I which results in secondary spermatocytes (each containing 46 unique chromosomes) .
Secondary spermatocytes undergo meiosis II which produces spermatids (with 23 unique chromosomes).
Spermatids now transform themselves into spermatozoa (also containing 23 unique chromosomes) in a final event called spermatogenesis.

Female reproductive system

Ovary:
Primary sex organ that produces egg cells in a process called oogenesis, and also produces female sex hormones such as estrogens and progesterone.
Developed near the kidneys during fetal development, and toward the end of pregnancy descend into the pelvic cavity.
Consists of ovarian cortex where the ovarian cycle occurs and ovarian medulla where scar tissues and connective tissue are located.
Enclosed by a layer of cubical cells called germinal epithelium.
Bound to the uterine tubes and uterus by ovarian ligaments.

Figure 35: Internal reproductive organs of a female



Structure of an ovary

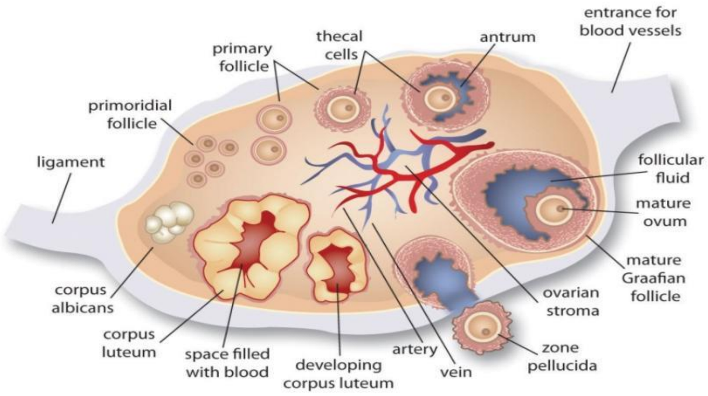


Figure 36

Uterine tube (or fallopian tube): consists of firmbriae, finger – like appendages that collect the ovum from the ovary during ovulation.

Infundibulum channels the ovum from the firmbriae into the uterine tube.

Ampulla is the curvature of the uterine tube where most fertilization occurs.

Inner wall of uterine tube is made of ciliated mucosa, where the cilia propel the ovum toward the uterus.

Uterus

A pear – shaped cavity formed by the union of the two uterine tubes .

Composed of 3 layers of tissue – perimetrium (fibrous connective tissue) , myometrium (smooth muscle), and endometrium (epithelial and connective tissues) .

After fertilization, embryo adheres to the endometrial layer for further development – an event called implantation.

To prepare for implantation and development, endometrium is stimulated by estrogens to thicken and becomes vascularzed – a process called the menstrual cycle.

Myometrium, under the stimulation of oxytocin, contracts during labor to expel the fetus into the vagina.

The base of uterus is closed by a narrow passageway called cervix to prevent the entry of foreign substances.

Vagina: an elastic channel inferior to the cervix that serves as the "birth canal" during parturition. Also serves as the copulatory receptacle , where it receives the penis during sexual intercourse. In addition to the acids secretion from cervix, it also coveys uterine secretions (i.e. menstrual flow).

Oogenesis

In the ovarian cortex, a process called oogenesis (formation of egg) occurs to develop a mature ovum. Before birth , several million cells called primordial oocytes exist in the ovaries – most of them spontaneously degenerate .

At birth, only 1 million primordial oocytes are left ; and by puberty (age 10-11) ,only 400,000 remain in the ovaries .

From puberty to menopause, some of these primordial oocytes (containing 46 chromosomes) undergo DNA replication and become primary oocytes (with 46 pairs of chromosomes).

Primary oocytes will then undergo "crossing - over" to shuffle their genes, and meiosis I will occur to divide the cells into secondary oocytes (containing 46 unique chromosomes) and the first polar bodies (also containing 46 unique chromosomes ; but will be degenerated).

Oogenesis now is arrested where Events of oogenesis the ovary discharges a mature secondary oocyte into the uterine tube (in a process called ovulation) .
Meiosis II is reactivated when this secondary oocyte is fertilized by a sperm (if no fertilization occurs, secondary oocyte is discarded along with the menstrual flow), instantly dividing the 46 chromosomes into 23 (inside the second polar body) and another 23 will be united with the 23 chromosomes released from the sperm.

Mechanism of erection, lubrication, and orgasm in human female

1. Sexual stimulation.
2. Arteries in the erectile tissue dilate, vagina expands and elongates
3. Engorged and swollen vagina increases friction from movement of penis.
4. Parasympathetic nerves impulses from sacral portion of the spinal nerve is enhanced.
5. Sexual stimulation intensifies.
6. Vestibule glands secrete mucus to lubricate.
7. Orgasm: rhythmic contraction of muscles of the perineum, muscular walls of uterus, and uterine tube.

Hormonal control of reproductive function

Hormones from the hypothalamus,
Anterior Pituitary gland and ovaries, play important roles in the control of sex cell maturation, and development and maintenance of female secondary sex characteristics.
Female sex hormones:
A female body remains reproductively immature until about 10 years of age when gonadotropin secretion increases.
The most important female sex hormones are estrogen and progesterone.
Estrogen is responsible for the development and maintenance of most female secondary sex characteristics.
Progesterone causes change in the uterus.

Hormonal control of secondary sex characteristic
The hypothalamus releases GnRH, which stimulates the Ant. Pituitary gland.
The Ant. pituitary gland secretes FSH and LH.
FSH stimulates the maturation of a follicle.
Granulose cells of the follicle produce and secrete estrogen; LH stimulated certain cells to secrete estrogen precursor molecules.
Estrogen is responsible for the development and maintenance of most female secondary sex characteristics.
Concentration of Androgens affects other secondary sex characteristics, including skeletal growth and growth of hair.
Progesterone, secreted by the ovaries, affect cyclical changes in the uterus and mammary glands.

Ovarian cycle
A series of event in the ovarian cortex in order to produce a mature ovum and sex hormones. Lasts for about 28 days, where from day 1 to 13 the mature ovum is developed and estrogens are released, on day 14 ovulation occurs to discharge the ovum , and from day 15 to 28 scar tissues are formed and progesterone is released . On day 1, hypothalamus secretes Latinizing hormone releasing hormone (LHRH) to the anterior pituitary gland, which in turn secretes follicle – stimulating hormone (FSH) to the ovaries. Upon receiving FSH, about 20-25 primary follicles develop into secondary follicles. (Primary oocytes located inside primary follicles undergo meiosis I and become secondary oocoytes, contained in secondary follicles).

Follicular cells in secondary follicles begin to secrete estrogens (for communicating with hypothalamus and anterior pituitary and for developing the endometrium) .
With continuous stimulation of FSH and some Latinizing hormone (LH) ,secondary follicles continue to grow larger and develop multiple layers of follicular cells (while the secondary oocytes within are unchanged).
By day 13 , only 1 secondary follicle will fully mature and become the graafian follicle (or mature follicle) which secretes a large amount of estrogens to the hypothalamus – anterior pituitary system for signaling ovulation (using a positive feedback mechanism).
On day 14 , large amounts of LH ("LH surge") will be secreted by anterior pituitary , inducing ovulation where the graafian follicle ruptures and releases the secondary oocyte into uterine tube .

From days 15 to 25 , graafian follicle degenerates and becomes corpus hemorrhagicum ("a bleeding body") then corpus luteum ("a yellow body"; containing lutein cells that secrete progesterone and some estrogens to continuum stimulating the development of endometrium).
By day 26 , if no fertilization occurs to the secondary oocyte, resulting in a lack of human chorionic Gonadotropin hormone (HCG) from the embryo , corpus luteum degenerates into corpus albicans . [if fertilization did occur , HCG will continuously simulate corpus luteum for 2-3 months , allowing high levels of estrogens and progesterone to maintain pregnancy in the first trimester] .
When corpus luteum degenerates, the declining levels of estrogens and progesterone will signal the hypothalamus – anterior pituitary system to initiate another ovarian cycle.

Fertilization
Fertilization, union of a sperm nucleus, of paternal origin, with an egg nucleus, of maternal origin, to form the primary nucleus of an embryo. In all organisms the essence of fertilization is, in fact, the fusion of the hereditary material of two different sex cells, or gametes, each of which carries half the number of chromosomes typical of the species. The most primitive form of fertilization, found in microorganisms and protozoans, consists of an exchange of genetic material between two cells.
The first significant event in fertilization is the fusion of the membranes of the two gametes, resulting in the formation of a channel that allows the passage of material from one cell to the other. Fertilization in advanced plants is preceded by pollination, during which pollen is transferred to, and establishes contact with, the female gamete or macrospore. Fusion in advanced animals is usually followed by penetration of the egg by a single spermatozoon.

The result of fertilization is a cell (zygote) capable of undergoing cell division to form a new individual.

The fusion of two gametes initiates several reactions in the egg. One of these causes a change in the egg membrane(s), so that the attachment of and penetration by more than one spermatozoon cannot occur. In species in which more than one spermatozoon normally enters an egg (polyspermy), only one spermatozoal nucleus actually merges with the egg nucleus. The most important result of fertilization is egg activation, which allows the egg to undergo cell division. Activation, however, does not necessarily require the intervention of a spermatozoon; during parthenogenesis, in which fertilization does not occur, activation of an egg may be accomplished through the intervention of physical and chemical agents. Invertebrates such as aphids, bees, and rotifers normally reproduce by parthenogenesis.

38 Maturation Of The Egg

Maturation is the final step in the production of functional eggs (oogenesis) that can associate with a spermatozoon and develop a reaction that prevents the entry of more than one spermatozoon. In addition, the cytoplasm of a mature egg can support the changes that lead to fusion of spermatozoal and egg nuclei and initiate embryonic development.

39 Egg surface

Certain components of an egg's surface, especially the cortical granules, are associated with a mature condition. Cortical granules of sea urchin eggs, aligned beneath the plasma membrane (thin, soft, pliable layer) of mature eggs, have a diameter of 0.8–1.0 micron (0.0008–0.001 millimetre) and are surrounded by a membrane similar in structure to the plasma membrane surrounding the egg. Cortical granules are formed in a cell component known as a Golgi complex, from which they migrate to the surface of the maturing egg.

The surface of a sea urchin egg has the ability to affect the passage of light unequally in different directions; this property, called birefringence, is an indication that the molecules comprising the surface layers are arranged in a definite way. Since birefringence appears as an egg matures, it is likely that the properties of a mature egg membrane are associated with specific molecular arrangements. A mature egg is able to support the formation of a zygote nucleus; i.e., the result of fusion of spermatozoal and egg nuclei. In most eggs the process of reduction of chromosomal number (meiosis) is not completed prior to fertilization. In such cases the fertilizing spermatozoon remains beneath the egg surface until meiosis in the egg has been completed, after which changes and movements that lead to fusion and the formation of a zygote occur.

40 Egg coats

The surfaces of most animal eggs are surrounded by envelopes, which may be soft gelatinous coats (as in echinoderms and some amphibians) or thick membranes (as in fishes, insects, and mammals). In order to reach the egg surface, therefore, spermatozoa must penetrate these envelopes; indeed, spermatozoa contain enzymes (organic catalysts) that break them down. In some cases (e.g., fishes and insects) there is a channel, or micropyle, in the envelope, through which a spermatozoon can reach the egg.

The jelly coats of echinoderm and amphibian eggs consist of complex carbohydrates called sulfated mucopolysaccharides. The envelope of a mammalian egg is more complex. The egg is surrounded by a thick coat composed of a carbohydrate protein complex called zona pellucida. The zona is surrounded by an outer envelope, the corona radiata, which is many cell layers thick and formed by follicle cells adhering to the oocyte before it leaves the ovarian follicle.

Although it once was postulated that the jelly coat of an echinoderm egg contains a substance (fertilizin) thought to have an important role not only in the establishment of sperm-egg interaction but also in egg activation, fertilizin now has been shown identical with jelly-coat material, rather than a substance continuously secreted from it. Yet there is evidence that the egg envelopes do play a role in fertilization; i.e., contact with the egg coat elicits the acrosome reaction (described below) in spermatozoa.

Events Of Fertilization

41 Sperm-egg association

The acrosome reaction of spermatozoa is a prerequisite for the association between a spermatozoon and an egg, which occurs through fusion of their plasma membranes. After a spermatozoon comes in contact with an egg, the acrosome, which is a prominence at the anterior tip of the spermatozoa, undergoes a series of well-defined structural changes. A structure within the acrosome, called the acrosomal vesicle, bursts, and the plasma membrane surrounding the spermatozoon fuses at the acrosomal tip with the membrane surrounding the acrosomal vesicle to form an opening. As the opening is formed, the acrosomal granule, which is enclosed within the acrosomal vesicle, disappears. The dissolution of the granule releases a substance called a lysin, which breaks down the egg's vitelline coat, allowing passage of the spermatozoon to the egg. The acrosomal membrane region opposite the opening adheres to the nuclear envelope of the spermatozoon and forms a shallow out pocketing, which rapidly elongates into a thin tube, the acrosomal tubule that extends to the egg surface and fuses with the egg plasma membrane. The tubule thus formed establishes continuity between the egg and the spermatozoon and provides a way for the spermatozoal nucleus to reach the interior of the egg. Other spermatozoal structures that may be carried within the egg include the midpiece and part of the tail; the spermatozoal plasma membrane and the acrosomal membrane, however, do not reach the interior of the egg. In fact, whole spermatozoa injected into unfertilized eggs cannot elicit the activation reaction or merge with the egg nucleus. As the spermatozoal nucleus is drawn within the egg, the spermatozoal plasma membrane breaks down; at the end of the process, the continuity of the egg plasma membrane is re-established. This description of the process of sperm-egg association, first documented for the acorn worm *Saccoglossus* (phylum Enteropneusta), generally applies to most eggs studied thus far.

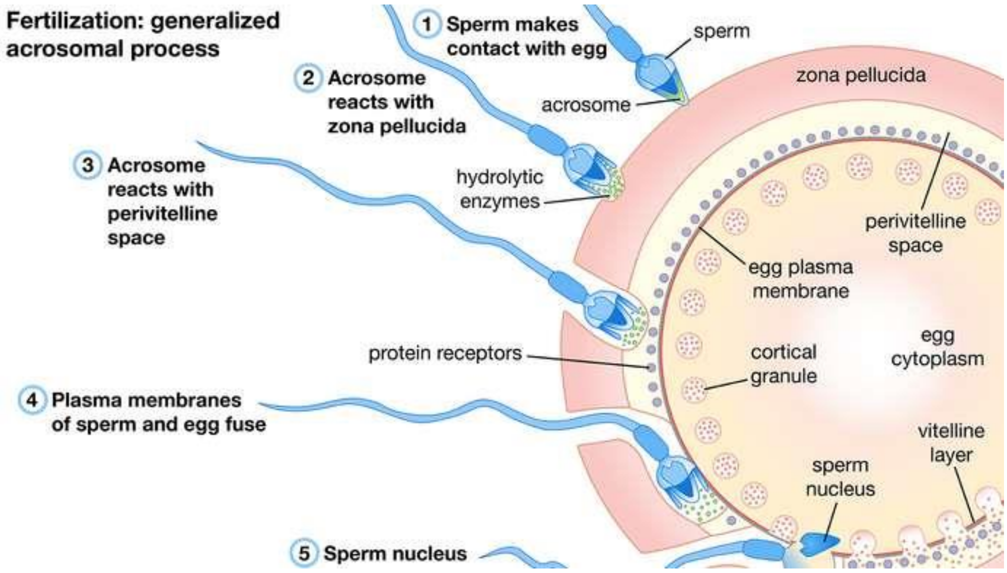


Figure 37:
During their passage through the female genital tract of mammals, spermatozoa undergo physiological change, called capacitation, which is a prerequisite for their participation in fertilization; they are able to undergo the acrosome reaction, traverse the egg envelopes, and reach the interior of the egg. Dispersal of cells in the outer egg envelope (corona radiata) is caused by the action of an enzyme (hyaluronidase) that breaks down a substance (hyaluronic acid) binding corona radiata cells together. The enzyme may be contained in the acrosome and released as a result of the acrosome reaction, during passage of the spermatozoon through the corona radiata. The reaction is well advanced by the time a spermatozoon contacts the thick coat surrounding the egg itself (zona pellucida). The pathway of a spermatozoon through the zona pellucida appears to be an oblique slit created by the oscillation of the sperm's head near the point of its attachment to the zona pellucida.

Association of a mammalian spermatozoon with the egg surface occurs along the lateral surface of the spermatozoon, rather than at the tip as in other animals, so that the spermatozoon lies flat on the egg surface. Several points of fusion occur between the plasma membranes of the two gametes (i.e., the breakdown of membranes occurs by formation of numerous small vesicles).

42 Specificity of sperm-egg interaction

Fertilization is strictly species-specific, and the egg's coating, the zona pellucida, plays an important role in the binding process between sperm and egg. In general, the biochemistry of the zona pellucida of one species differs from that of another, and thus it only matches up and binds with sperm of the appropriate species. For example, among the echinoderms, solutions of the jelly coat clump, or agglutinate, only spermatozoa of their own species. In both echinoderms and amphibians, however, slight damage to an egg surface makes fertilization possible with spermatozoa of different species (heterologous fertilization); this procedure has been used to obtain certain hybrid larvae. In addition, binding between sperm and egg of different species may occur when the zona pellucida of the egg is removed.

The eggs of ascidians, or sea squirts, members of the chordate subphylum Tunicata, are covered with a thick membrane called a chorion. The space between the chorion and the egg is filled with cells called test cells. The gametes of ascidians, which have both male and female reproductive organs in one animal, mature at the same time, yet self-fertilization does not occur. If the chorion and the test cells are removed, however, not only is fertilization with spermatozoa of different species possible, but self-fertilization also can occur.

43 Prevention of polyspermy

Most animal eggs are monospermic; i.e., only one spermatozoon is admitted into an egg. In some eggs, protection against the penetration of the egg by more than one spermatozoon (polyspermy) is due to some property of the egg surface; in others, however, the egg envelopes are responsible. The ability of some eggs to develop a polyspermy-preventing reaction depends on a molecular rearrangement of the egg surface that occurs during egg maturation (oogenesis). Although immature sea urchin eggs have the ability to associate with spermatozoa, they also allow multiple penetration; i.e., they are unable to develop a polyspermy-preventing reaction. Since the mature eggs of most animals are fertilized before completion of meiosis and are able to develop a polyspermy-preventing reaction, specific properties of the egg surface must have differentiated by the time meiosis stops, which is when the egg is ready to be fertilized.

In some mammalian eggs defense against polyspermy depends on properties of the zona pellucida; i.e., when a spermatozoon has started to move through the zona, it does not allow the penetration of additional spermatozoa (zona reaction). In other mammals, however, the zona reaction either does not take place or is weak, as indicated by the presence of numerous spermatozoa in the space between the zona and egg surface. In such cases the polyspermy-preventing reaction resides in the egg surface. Although the eggs of some kinds of animals (e.g., some amphibians, birds, reptiles, and sharks) are naturally polyspermic, only one spermatozoal nucleus fuses with an egg nucleus to form a zygote nucleus; all of the other spermatozoa degenerate.

44 Formation of the fertilization membrane

The most spectacular changes that follow fertilization occur at the egg surface. The best known example, that of the sea urchin egg, is described below. An immediate response to fertilization is the raising of a membrane, called a vitelline membrane, from the egg surface. In the beginning the membrane is very thin; soon, however, it thickens, develops a well-organized molecular structure, and is called the fertilization membrane. At the same time an extensive rearrangement of the molecular structure of the egg surface occurs. The events leading to formation of the fertilization membrane require about one minute.

At the point on the outer surface of the sea urchin egg at which a spermatozoan attaches, the thin vitelline membrane becomes detached. As a result the membranes of the cortical granules come into contact with the inner aspect of the egg's plasma membrane and fuse with it, the granules open, and their contents are extruded into the perivitelline space; i.e., the space between the egg surface and the raised vitelline membrane. Part of the contents of the granules merge with the vitelline membrane to form the fertilization membrane; if fusion of the contents of the cortical granules with the vitelline membrane is prevented, the membrane remains thin and soft. Another material that also derives from the cortical granules covers the surface of the egg to form a transparent layer, called the hyaline layer, which plays an important role in holding together the cells (blastomeres) formed during division, or cleavage, of the egg. The plasma membrane surrounding a fertilized egg, therefore, is a mosaic structure containing patches of the original plasma membrane of the unfertilized egg and areas derived from membranes of the cortical granules. The events leading to the formation of the fertilization membrane are accompanied by a change of the electric charge across the plasma membrane, referred to as the fertilization potential, and a concurrent outflow of potassium ions (charged particles); both of these phenomena are similar to those that occur in a stimulated nerve fibre. Another effect of fertilization on the plasma membrane of the egg is a several-fold increase in its permeability to various molecules. This change may be the result of the activation of some surface-located membrane transport mechanism.

45 Formation of the zygote nucleus

After its entry into the egg cytoplasm, the spermatozoal nucleus, now called the male pronucleus, begins to swell, and its chromosomal material disperses and becomes similar in appearance to that of the female pronucleus. Although the membranous envelope surrounding the male pronucleus rapidly disintegrates in the egg, a new envelope promptly forms around it. The male pronucleus, which rotates 180° and moves towards the egg nucleus, initially is accompanied by two structures (centrioles) that function in cell division. After the male and female pronuclei have come into contact, the spermatozoal centrioles give rise to the first cleavage spindle, which precedes division of the fertilized egg. In some cases fusion of the two pronuclei may occur by a process of membrane fusion; in this process, two adjoining membranes fuse at the point of contact to give rise to the continuous nuclear envelope that surrounds the zygote nucleus.

1.2.16.4: Self Assessment

a). Describe formation of the zygote nucleus

b). Highlight the specificity of sperm egg intergration

c). Describe the ovarian cycle

1.2.16.4.1: Responses

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c). Describe the ovarian cycle

A series of event in the ovarian cortex in order to produce a mature ovum and sex hormones. Lasts for about 28 days, where from day 1 to 13 the mature ovum is developed and estrogens are released, on day 14 ovulation occurs to discharge the ovum , and from day 15 to 28 scar tissues are formed and progesterone is released . On day 1, hypothalamus secretes Latinizing hormone releasing hormone (LHRH) to the anterior pituitary gland, which in turn secretes follicle – stimulating hormone (FSH) to the ovaries. Upon receiving FSH, about 20-25 primary follicles develop into secondary follicles. (Primary oocytes located inside primary follicles undergo meiosis I and become secondary oocoytes, contained in secondary follicles).

Follicular cells in secondary follicles begin to secrete estrogens (for communicating with hypothalamus and anterior pituitary and for developing the endometrium) .

With continuous stimulation of FSH and some Latinizing hormone (LH) ,secondary follicles continue to grow larger and develop multiple layers of follicular cells (while the secondary oocytes within are unchanged).

By day 13 , only 1 secondary follicle will fully mature and become the graafian follicle (or mature follicle) which secretes a large amount of estrogens to the hypothalamus – anterior pituitary system for signaling ovulation (using a positive feedback mechanism).

On day 14 , large amounts of LH ("LH surge") will be secreted by anterior pituitary , inducing ovulation where the graafian follicle ruptures and releases the secondary oocyte into uterine tube .

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When corpus luteum degenerates, the declining levels of estrogens and progesterone will signal the hypothalamus – anterior pituitary system to initiate another ovarian cycle.

1.2.16.5: References

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